STRUCTURE THEORY OF p-ADIC REDUCTIVE GROUPS (2026 SPRING @ NTU)

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Contents

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1. Week 1: Course overview (motivation of Bruhat–Tits theory)

1.1. **Representations of p-adic reductive groups.** Let F be a non-archimedean local field, i.e., a finite extension of the p-adic number field $\mathbb{Q}_p$ or $\mathbb{F}_p(\!(t)\!)$. Let G be a connected reductive group over F. For example, one can choose G to be GL_n, SL_n, SO_n, Sp_{2n}, etc...

We consider the group $G(F)$ of F-rational points of G. When $G = \mathrm{GL}_n$, the group $G(F)$ is equipped with a topology induced from $\mathrm{GL}_n(F) \hookrightarrow M_n(F) \cong F^{\oplus n^2}$. In general, by choosing an embedding $G \hookrightarrow \mathrm{GL}_n$, we can give $G(F)$ the topology induced from $\mathrm{GL}_n(F)$. With respect to this topology, $G(F)$ becomes a non-discrete topological group which is locally compact and totally disconnected (i.e., every element forms a connected component). We call a topological group obtained in this way a *p-adic reductive group*[1].

[1]Note that this terminology is a bit confusing because the base field k may not be a p-adic field (i.e., of characteristic zero).

Definition 1.1. Let π be a representation of $G(F)$ on a $\mathbb{C}$-vector space V.

(1) We say that π is *smooth* if every $v \in V$ is contained in V^K (subspace of K-fixed vectors) for some open compact subgroup K of $G(F)$.
(2) We say that π is *admissible* if π is smooth and V^K is finite dimensional for any open compact subgroup K of $G(F)$.

Remark 1.2. It is known that an irreducible representation of a p-adic reductive group is smooth if and only if admissible.

We are interested in

$$\mathrm{Rep}(G(F)) := (\text{category of smooth representations of } G(F)),$$
$$\mathrm{Irr}(G(F)) := \{\text{irreducible smooth representations of } G(F)\}/\sim.$$

Why? The possible explanations are the following, for example:

- By so-called *Harish-Chandra's Lefschetz principle*, $\mathrm{Rep}(G(F))$ is expected to be "parallel" to $\mathrm{Rep}(G(\mathbb{R}))$ for a real reductive group $G(\mathbb{R})$ (although p-adic and real reductive groups look totally different!). Representation theory of real reductive groups has a long history; thus it is not strange at all to study its analogue for p-adic reductive groups.
- Irreducible smooth representations of G arise as local components of *automorphic representations*, which is a modern generalization of the classical notion of modular forms. Hence it is technically very important to study $\mathrm{Irr}(G(F))$ for understanding modular forms systematically.
- By the so-called *local Langlands conjecture*, $\mathrm{Irr}(G(F))$ is expected to be related to *Galois representations* of k. It is quite interesting to study how "number theory of F" is encoded in representation theory of $G(F)$ (or vice versa) from the viewpoint of the local Langlands correspondence.
- In a certain sense, $\mathrm{Rep}(G(F))$ should be something "generalizing"[2] $\mathrm{Rep}(\mathbb{G}(\mathbb{F}_q))$ for a finite group of Lie type $\mathbb{G}(\mathbb{F}_q)$. It is quite interesting to see how various techniques or phenomena in representation theory of finite groups of Lie type (for example, the theory of Deligne-Lusztig) can be generalized to that of p-adic reductive groups.

(Actually, all of these are deeply related to each other. So maybe I should say I can only come up with just one reason.)

As we can see from the definition, it must be crucially important to analyze open compact subgroups of $G(F)$ for understanding $\mathrm{Rep}(G(F))$.

1.2. **Hecke algebra and types.** (See, for example, [BH06, BK98] for details of the content of this subsection.)

As before, we consider a p-adic reductive group $G(F)$; to make the notation lighter, we simply denote it by G hereafter. We first considier the space of "test functions":

$$C_c^\infty(G) := \{f\colon G \to \mathbb{C} \mid f \text{ is locally constant and compactly supported}\}.$$

By fixing a Haar measure dg on G, we can introduce a $\mathbb{C}$-algebra structure on $C_c^\infty(G)$. To be more precise, for any $f_1, f_2 \in C_c^\infty(G)$, we define $f_1 * f_2 \in C_c^\infty(G)$ by

$$(f_1 * f_2)(g) := \int_G f_1(gh) \cdot f_2(h^{-1})\, dg.$$

[2] I don't know the meaning of this.

This operation $*$ is called the *convolution product* for test functions. We can check that $C_c^\infty(G)$ is an associative $\mathbb{C}$-algebra with respect to $*$. But please be careful that $C_c^\infty(G)$ is non-commutative and even non-unital in general. We call this algebra the *Hecke algebra of* G and often denote by $\mathcal{H}(G)$.

Suppose that (π, V) is a smooth representation of G. Then we may regard (π, V) as an $\mathcal{H}(G)$-module in the following manner. For any $f \in \mathcal{H}(G)$, we define an operator "$\pi(f)$" on V by

$$v \mapsto \pi(f)(v) := \int_G f(g)\pi(g)v\,dg.$$

This gives a well-defined action of $\mathcal{H}(G)$ on V. Moreover, the smoothness of (π, V) implies that $\mathcal{H}(G) \cdot V = V$; note that this does not necessarily hold for general $\mathcal{H}(G)$-modules because $\mathcal{H}(G)$ may not have the unit element! In general, we say that an $\mathcal{H}(G)$-module V is *smooth* if $\mathcal{H}(G) \cdot V = V$.

The importance of the Hecke algebra is encapsulated in the following fact (but it is not extremely difficult to prove this fact; see, e.g., [BH06, §4.2]):

Fact 1.3. *The procedure above gives a categorical equivalence*

$$\mathrm{Rep}(G) \xrightarrow{\cong} \mathrm{Mod}(\mathcal{H}(G)),$$

where the latter denotes the category of smooth $\mathcal{H}(G)$*-modules.*

By Fact 1.3, in order to study $\mathrm{Rep}(G)$, it is enough to "understand $\mathcal{H}(G)$ completely". But of course this change of viewpoint does not immediately make the problem so easier because the Hecke algebra $\mathcal{H}(G)$ is too huge to analyze.

The key idea is to note the *idempotents* of $\mathcal{H}(G)$. For example, let K be an open compact subgroup of G. Then we can easily check that the normalized characteristic function of K

$$e_K := \frac{1}{dg(K)} \cdot \mathbb{1}_K \in \mathcal{H}(G)$$

is an idempotent, i.e., $e_K * e_K = e_K$. This implies that the subset

$$\mathcal{H}(G, K) := e_K * \mathcal{H}(G) * e_K \subset \mathcal{H}(G)$$

is a subalgebra with unit e_K. (Another description of $\mathcal{H}(G, K)$ is: $\mathcal{H}(G, K) = \{f \in \mathcal{H}(G) \mid f \text{ is bi-}K\text{-invariant}\}$.)

For a smooth representation (π, V) of G (hence regareded also as an $\mathcal{H}(G)$-module), we consider the action of the subring $\mathcal{H}(G, K)$ of $\mathcal{H}(G)$. The point is that $\pi(e_K)$ gives the projector $V \to V^K$, hence the action of $\mathcal{H}(G, K)$ on V factors through the projector $V \to V^K$. If we let $\mathrm{Rep}(G, K)$ be the subcategory

$$\mathrm{Rep}(G, K) := ((\pi, V) \in \mathrm{Rep}(G) \text{ such that } V^K \text{ generates } V)$$

of $\mathrm{Rep}(G)$, then we get a functor

$$\mathrm{Rep}(G, K) \to \mathrm{Mod}(\mathcal{H}(G, K))\colon (\pi, V) \mapsto V^K,$$

where the action of $\mathcal{H}(G, K)$ on V^K is the one induced by that of $\mathcal{H}(G)$ on V. This functor induces a bijection between the set $\mathrm{Irr}(G, K)$ of irreducible representations of G having nonzero K-fixed vectors and the set of simple $\mathcal{H}(G, K)$-modules. Since $\mathrm{Irr}(G)$ is 'covered' by $\mathrm{Irr}(G, K)$ over all open compact subgroups K of G, in principle, we can determine $\mathrm{Irr}(G)$ by studying $\mathrm{Irr}(G, K)$ through $\mathcal{H}(G, K)$ for all K.

This procedure can be generalized as follows. We fix an open compact subgroup K of G and its irreducible smooth representation ρ (which must be automatically finite-dimensional). We define a test function $e_\rho \in \mathcal{H}(G)$ by

$$e_\rho(g) := \frac{\dim \rho}{dg(K)} \cdot \begin{cases} \Theta_\rho(g^{-1}) & \text{if } g \in K, \\ 0 & \text{othrewise,} \end{cases}$$

where $\Theta_\rho \colon K \to \mathbb{C}$ denotes the trace character of ρ. Then e_ρ is an idempotent of $\mathcal{H}(G)$, hence gives rises to a subalgebra with unit e_ρ

$$\mathcal{H}(G,\rho) := e_\rho * \mathcal{H}(G) * e_\rho \subset \mathcal{H}(G).$$

Similarly to before, we consider the following subcategory of $\mathrm{Rep}(G)$:

$$\mathrm{Rep}(G,\rho) := ((\pi, V) \in \mathrm{Rep}(G) \text{ such that } V^\rho \text{ generates } V),$$

where V^ρ denotes the ρ-isotypic part of $(\pi, V)|_K$. Then we get a functor

$$\mathrm{Rep}(G,\rho) \to \mathrm{Mod}(\mathcal{H}(G,\rho)) \colon (\pi, V) \mapsto V^\rho.$$

If this functor gives a categorical equivalence, we can analyze the subcategory $\mathrm{Rep}(G,\rho)$ of $\mathrm{Rep}(G)$ through the module category for a $\mathbb{C}$-algebra $\mathcal{H}(G,\rho)$, which is still non-commutative in general, but much smaller than $\mathcal{H}(G)$ (and has the unit!). In fact, this functor is not always an equivalence. But this observation naturally leads us to the following definition.

Definition 1.4. We say that (K,ρ) is a *type* if the functor

$$\mathrm{Rep}(G,\rho) \to \mathrm{Mod}(\mathcal{H}(G,\rho)) \colon (\pi, V) \mapsto V^\rho.$$

gives an equivalence of categories.

Example 1.5. Let $I \subset G$ be an *Iwahori subgroup* of G. For example, when $G = \mathrm{GL}_n(\mathbb{Q}_p)$, you can take I to be the subgroup of $\mathrm{GL}_n(\mathbb{Z}_p)$ consisting of matrices whose modulo-p-reductions are upper-triangular. In this case, $(I, \mathbb{1})$ is known to be a type. In other words, $\mathrm{Rep}(G, I)$ is equivalent to $\mathrm{Mod}(\mathcal{H}(G, I))$ by the above association. The $\mathbb{C}$-algebra $\mathcal{H}(G, I)$ is sometimes (or often) called the *Iwahori–Hecke algebra*.[3]

From these observations, we may expect that the study of $\mathrm{Rep}(G)$ could be eventually reduced to the study of $\mathcal{H}(G,\rho)$ for all pairs (G). To sophisticate this idea, we need to introduce the theory of *Bernstein decomposition*, which we discuss next.

1.3. **Induction and compact induction.** Recall that, when a representation of a subgroup is given, we can construct a representation of a bigger group using it; this is called the *induction* of representations. In the context of representation theory of p-adic groups, we typically consider two kinds of inductions; the *(smooth) induction* and the *compact induction*. Let us briefly review them here (see [BH06] for more details).

For any closed subgroup $H \subset G$ and its smooth representation (ρ, W), we define the *(smooth) induction* $(\mathrm{Ind}_H^G \rho, \mathrm{Ind}_H^G W)$, which is a smooth representation of G, by

$$\mathrm{Ind}_H^G W := \{f \colon G \to W \mid \exists K \text{ s.t. } f(hgk) = \sigma(h)(f(g)) \text{ for any } h \in H,\, k \in K\}$$

[3]There is a more general and abstract notion of the "Iwahori–Hecke algebra". The Iwahori–Hecke algebra in our context is a special case of more general "Iwahori–Hecke algebras".

(K is some open compact subgroup of G depending on f), where G acts on $\operatorname{Ind}_H^G W$ via right-translation of W-valued functions. This gives a functor

$$\operatorname{Ind}_H^G \colon \operatorname{Rep}(H) \to \operatorname{Rep}(G).$$

When H is open, we define the *compact induction* ($\text{c-Ind}_H^G \rho, \text{c-Ind}_H^G W$) in the same way, but additionally requiring that the support of $f \colon G \to W$ is compact-modulo-H. This also gives a functor

$$\text{c-Ind}_H^G \colon \operatorname{Rep}(H) \to \operatorname{Rep}(G).$$

On the other hand, we can also define the restriction functor in the obvious manner:

$$\operatorname{Res}_H^G \colon \operatorname{Rep}(G) \to \operatorname{Rep}(H).$$

In fact, these are adjunctions to each other

$$\text{c-Ind}_H^G \dashv \operatorname{Res}_H^G \dashv \operatorname{Ind}_H^G,$$

both of which are referred to as the *Frobenius reciprocity.*

In the context of representation theory of p-adic reductive groups, the following variant of an induction is particularly important.

Definition 1.6. Let P be a parabolic subgroup of G with a Levi subgroup M (defined over F; again write M for $M(F)$ by abuse of notation). Let π_M be a smooth representation of M. We define the *parabolic induction* of π_M to G by

$$i_P^G(\pi_M) := \operatorname{Ind}_P^G(\pi_M \otimes \delta_P),$$

where π_M is regarded as a representation of P through the natural quotient (by the unipotent radical) map $P \twoheadrightarrow M$ and δ_P denotes the "modulus character" of P.

Definition 1.7. We say that an irreducible smooth representation (π, V) of G is *supercuspidal* if there does not exist any pair (M, π_M) of a proper Levi subgroup $M \subsetneq G$ and an irreducible smooth representation π_M of M such that π is realized as a subquotient of the parabolic induction $i_P^G(\pi_M)$.

Fact 1.8. *For any irreducible smooth representation (π, V) of G, there uniquely (up to G-conjugation) exists a pair (M, π_M) of a Levi subgroup $M \subset G$ and an irreducible supercuspidal representation π_M of M such that π is realized as a subquotient of $i_P^G(\pi_M)$.*

Definition 1.9. We call the pair (M, π_M) associated to π as in the previous fact the *cuspidal support* of π. Also, we call such a pair *cuspidal pair* of G.

We may naively understand that supercuspidal representations are "building blocks" of $\operatorname{Rep}(G)$.

The following lemma is useful to construct supercuspidal representations:

Lemma 1.10. *Let K be an open compact-mod-center subgroup of G. Let ρ be an irreducible smooth representation of K. If the compact induction $\text{c-Ind}_K^G \rho$ is of finite length, then its all irreducible constituents are supercuspidal.*

The converse is also expected to be true, but it has been a long-standing "folklore conjecture":

Conjecture 1.11. *Any irreducible supercuspidal representation of G is realized in the compact induction of an irreducible smooth representation of an open compact-mod-center subgroup.*

1.4. **Bernstein decomposition.** We next explain the Bernstein decomposition of $\mathrm{Rep}(G)$; the references of this subsection are, e.g., [Ber84, Roc09].

We consider an equivalence relation on the set of cuspidal pairs of G as follows:

> we say cuspidal pairs (M, π_M) and $(M', \pi_{M'})$ are *inertially equivalent* if there exists an unramified character $\chi\colon M \to \mathbb{C}^\times$ such that $(M, \pi_M \otimes \chi)$ and $(M', \pi_{M'})$ are G-conjugate.

Here, loosely speaking, an "unramified character" means that χ is trivial on "$M(\mathcal{O}_F)$".

Let us write $\mathfrak{C}(G)$ for the set of inertially-equivalence classes of cuspidal pairs of G. Then the uniqueness fact on the cuspidal supports especially implies the following decomposition of $\mathrm{Irr}(G)$:

$$\mathrm{Irr}(G) = \bigsqcup_{\mathfrak{s} \in \mathfrak{C}(G)} \mathrm{Irr}_{\mathfrak{s}}(G),$$

where $\mathrm{Irr}_{\mathfrak{s}}(G)$ denotes the set of isomorphism classes of irreducible smooth representations of G whose cuspidal supports lie in $\mathfrak{s}$.

In fact, this decomposition can be upgraded to a decomposition of $\mathrm{Rep}(G)$. Let $\mathrm{Rep}_{\mathfrak{s}}(G)$ be the subcategory of $\mathrm{Rep}(G)$ consisting of smooth representations whose irreducible subquotients are all in $\mathrm{Irr}_{\mathfrak{s}}(G)$.

Theorem 1.12 (Bernstein decomposition)**.** *We have a decomposition*

$$\mathrm{Rep}(G) \cong \prod_{\mathfrak{s} \in \mathfrak{C}(G)} \mathrm{Rep}_{\mathfrak{s}}(G).$$

This categorical decomposition is in fact the "finest" one. Therefore, to understand $\mathrm{Rep}(G)$, it is enough to investigate each block $\mathrm{Rep}_{\mathfrak{s}}(G)$.

Recall that, by definition, a pair (K, ρ) is called a type when the functor

$$\mathrm{Rep}(G, \rho) \cong \mathrm{Mod}(\mathcal{H}(G, \rho))\colon V \mapsto V^{\rho}$$

is an equivalence of categories. In fact, types are closely related to the Bernstein decomposition as follows:

Fact 1.13 ([BK98, Theorem 4.3])**.** *Let K be an open compact subgroup of G and ρ an irreducible smooth representation of K. The pair (K, ρ) is a type if and only if*

$$\mathrm{Rep}(G, \rho) \cong \prod_{\mathfrak{s} \in \mathfrak{S}} \mathrm{Rep}_{\mathfrak{s}}(G)$$

for a finite subset $\mathfrak{S} \subset \mathfrak{C}(G)$.

The finite subset $\mathfrak{S}$ as in this fact is uniquely determined by a given type (K, ρ); let us say that (K, ρ) is an $\mathfrak{S}$*-type* in this situation. When $\mathfrak{S}$ is a singleton $\{\mathfrak{s}\}$, we simply say (K, ρ) is an $\mathfrak{s}$*-type*.

If (K, ρ) is an $\mathfrak{s}$-type, we get

$$\mathrm{Rep}_{\mathfrak{s}}(G) \cong \mathrm{Mod}(\mathcal{H}(G, \rho)).$$

Thus the natural question we come up with here is:

can we always find an $\mathfrak{s}$-type for each $\mathfrak{s} \in \mathfrak{C}(G)$?

If this is true, then, combined with the Bernstein decomposition, the study of $\mathrm{Rep}(G)$ ultimately comes down to studying the structure of Hecke algebras $\mathcal{H}(G, \rho)$ of an $\mathfrak{s}$-type (K, ρ) for each $\mathfrak{s} \in \mathfrak{C}(G)$.

However, the answer has not been known at present. For example, let us consider an inertially-equivalence class $\mathfrak{s} = [(G, \pi)]$ whose Levi part is G itself. If we can find an $\mathfrak{s}$-type (K, ρ), then it especially means that $\pi \in \mathrm{Rep}_{\mathfrak{s}}(G) = \mathrm{Rep}(G, \rho)$, hence, in particular, $\pi^\rho \neq 0$. In other words, we have $\mathrm{Hom}_K(\rho, \pi) \neq 0$. By Frobenius reciprocity, this is equivalent to $\mathrm{Hom}_G(\text{c-Ind}_K^G \rho, \pi) \neq 0$. This means that π is realized in the compact induction of (K, ρ).[4]

So, much remains unknown, but what we want to do is in some sense clear.

(1) For each $\mathfrak{s}$, we want to construct an $\mathfrak{s}$-type (K, ρ).
(2) For each type (K, ρ), we want to determine the structure of $\mathcal{H}(G, \rho)$.

In fact, much progress has been made on these problems so far; I cannot explain anything here. But, what is clear in any case is that it must be absolutely indispensable to deeply study the open compact subgroups of G. Especially;

- Which kind of ("nice") open compact subgroups does G have?
- Which kind of group-theoretic/representation-theoretic properties do they have?
- How can we classify them?

Bruhat–Tits theory gives answers to these questions.

1.5. **Flavor of Bruhat–Tits theory.** Let us begin with the following very basic example. Let G be GL_2 over $\mathbb{Q}_p$. Then $\mathrm{GL}_2(\mathbb{Z}_p)$ is a maximal open compact subgroup of $G(\mathbb{Q}_p) = \mathrm{GL}_2(\mathbb{Q}_p)$. By taking the reduction modulo p, we get a homomorphism from $\mathrm{GL}_2(\mathbb{Q}_p)$ to $\mathrm{GL}_2(\mathbb{F}_p)$:

$$\mathrm{GL}_2(\mathbb{Z}_p) \twoheadrightarrow \mathrm{GL}_2(\mathbb{F}_p)\colon \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} \overline{a} & \overline{b} \\ \overline{c} & \overline{d} \end{pmatrix}.$$

The point here is that the target group $\mathrm{GL}_2(\mathbb{F}_p)$ is a *finite group of Lie type* in the sense[5] that it is regarded as the group of $\mathbb{F}_p$-rational points of a connected reductive group over $\mathbb{F}_p$. In particular, we can define the notion of *cuspidality* for representations of $\mathrm{GL}_2(\mathbb{F}_p)$ in the same way as above. Let us take an irreducible cuspidal representation κ of $\mathrm{GL}_2(\mathbb{F}_p)$. (Note that such a representation exists, but it is not so obvious; see [BH06, §6].) By pulling back κ along the above surjection, we get an irreducible smooth representation of $\mathrm{GL}_2(\mathbb{Z}_p)$ (let's again write κ). Let Z be the center of $\mathrm{GL}_2(\mathbb{Q}_p)$, i.e., the subgroup of non-zero scalar matrices. We extend the representation κ from $\mathrm{GL}_2(\mathbb{Z}_p)$ to $Z \cdot \mathrm{GL}_2(\mathbb{Z}_p)$ (write $\tilde{\kappa}$) and take its compact induction to $\mathrm{GL}_2(\mathbb{Q}_p)$:

$$\pi_{\tilde{\kappa}} := \text{c-Ind}_{Z\cdot\mathrm{GL}_2(\mathbb{Z}_p)}^{\mathrm{GL}_2(\mathbb{Q}_p)} \tilde{\kappa}.$$

Fact 1.14. (1) *The representation $\pi_{\tilde{\kappa}}$ is irreducible and supercuspidal.*
(2) *The pair $(\mathrm{GL}_2(\mathbb{Z}_p), \kappa)$ is an $\mathfrak{s}$-type for $\mathfrak{s} = [(\mathrm{GL}_2(\mathbb{Q}_p), \pi_{\tilde{\kappa}})]$.*

This construction does not exhaust all supercuspidal representations of $\mathrm{GL}_2(\mathbb{Q}_p)$ at all. However, at least we can see that some part of them are found by looking at the reduction of the maximal open compact subgroup $\mathrm{GL}_2(\mathbb{Z}_p)$ and studying its representation theory (especially, classifying cuspidal representations of $\mathrm{GL}_2(\mathbb{F}_p)$).

[4] Precisely speaking, we have to be careful about the center (to apply the Frobenius reciprocity, we first have to extend ρ to a compact-mod-center subgroup using the central character of π).

[5] the definition of a "finite group of Lie type" is actually broader than the one explained here.

We next recall that I claimed above that the pair $(I, \mathbb{1})$ is also a type, where I is the Iwahori subgroup of $\mathrm{GL}_2(\mathbb{Q}_p)$ given by

$$I := \begin{pmatrix} \mathbb{Z}_p^\times & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times \end{pmatrix}.$$

Note that this group is contained in $\mathrm{GL}_2(\mathbb{Z}_p)$, hence we can think about the modulo p reduction of I as well. The image of the modulo p reduction map $I \to \mathrm{GL}_2(\mathbb{F}_p)$ is not just a random subgroup; it is given by the upper-triangular Borel subgroup of $\mathrm{GL}_2(\mathbb{F}_p)$. From this example, we can guess that it must be also important to look at subgroups of maximal open compact subgroups whose reductions have some special group-theoretic properties from the viewpoint of theory of finite groups of Lie type (for example, being parabolic subgroups). The maximal open compact subgroup $\mathrm{GL}_2(\mathbb{Z}_p)$ of $\mathrm{GL}_2(\mathbb{Q}_p)$ is an example of a *parahoric subgroup* (especially, *special (hyperspecial) subgroup*) in the sense of Bruhat–Tits theory. The Iwahori subgroup I is also an example of a *minimal* parahoric subgroup.

The key object in Bruhat–Tit theory is a topological space called the *Bruhat–Tits building*. The Bruhat–Tits building classifies all parahoric subgroups in the sense that each point gives rise to a parahoric subgroup of $G(F)$.

Suppose that a parahoric subgroup is given (although we do not yet know even its definition at this point). Firstly, the above example of $\mathrm{GL}_2(\mathbb{Z}_p)$ suggests that we should be able to consider the "reduction" of the parahoric subgroup and also that the resulting finite group of Lie type should be related to the original one (just as $\mathrm{GL}_2(\mathbb{F}_p)$ is obtained from $\mathrm{GL}_2(\mathbb{Z}_p)$). However, what does the "reduction" exactly mean? How we can consider the above intuitive operation of taking modulo p for general p-adic reductive groups? To formulate it precisely, we need to think about the integral models of the algebraic group G over $\mathcal{O}_F$ seriously; once we can realize a parahoric subgroup of $G(F)$ as the set of $\mathcal{O}_F$-rational points of an integral model $\mathcal{G}$ of G, we can define the reduction of the parahoric subgroup to be the special fiber of $\mathcal{G}$ (its $\mathbb{F}_q$-rational points).

Secondly, as mentioned above, the above construction enables us to find only some part of supercuspidal representations. Recall that $\mathrm{GL}_2(\mathbb{Z}_p)$ has the following standard filtration ("congruence filtration"):

$$\mathrm{GL}_2(\mathbb{Z}_p) \supset I_2 + pM_2(\mathbb{Z}_p) \supset I_2 + p^2M_2(\mathbb{Z}_p) \supset \cdots.$$

The second subgroup $I_2+pM_2(\mathbb{Z}_p)$ is nothing but the kernel of $\mathrm{GL}_2(\mathbb{Z}_p) \twoheadrightarrow \mathrm{GL}_2(\mathbb{F}_p)$. In fact, for any $r \geq 1$, there are types arising from irreducible representations of $\mathrm{GL}_2(\mathbb{Z}_p)$ which are trivial on $I_2 + p^{r+1}M_2(\mathbb{Z}_p)$ but not on $I_2 + p^rM_2(\mathbb{Z}_p)$. This observation suggests that it is also important to understand further smaller open compact subgroups contained in parahoric subgroups conceptually. The congruence filtration above is an example of the *Moy–Prasad filtration*, which is defined for any parahoric subgroup. Bruhat–Tits theory enables us to define more general open compact subgroups (at least containing Moy–Prasad filtrations).

1.6. **References for this course.** Bruhat–Tits theory was established in [BT72, BT84]. In this course, I will completely follow the book [KP23] of Kaletha–Prasad, which was published recently. This means that this course note would be a low quality imitation of a proper (far from being equal) subset of [KP23]. So I recommend that you look at this note just for checking what is covered in my course and also which kind of exercises (needed to be solved for getting a course credit) are given.

2. Week 2: Outline and Examples of Bruhat–Tits theory ([KP23, §3])

2.1. Affine spaces.

Let k be a field. Let V be a k-vector space.

Definition 2.1. An *affine space over* V is a non-empty set A equipped with a simply transitive action of V (as an additive group).[6]

- We define $\dim(A) := \dim(V)$.
- When we simply say an *affine space over* k, it means a pair (V, A) of V and an affine space A over V.
- For any $x, y \in A$, we write "$x - y$" for the unique element of V satisfies $x = y + (x - y)$ (here, the symbol "$+$" denotes the action of V on A).

Definition 2.2. Let A be an affine space over V. An *affine subspace of* A is a non-empty subset $B \subset A$ such that $W := \{x - y \in V \mid x, y \in B\}$ is a k-subspace of V (hence B is an affine space over W). We call W the *derivative of* B and write $W = \nabla B$.

Definition 2.3. Let (V, A) and (V', A') be affine spaces over k. We call a map $F\colon A \to A'$ an *affine map* if there exists a k-linear map $f\colon V \to V'$ satisfying $F(x + v) = F(x) + f(v)$ for any $x \in A$ and $v \in V$ (note that such a map f is necessarily unique). We call f the *derivative of* F and write $f = \nabla F$.

For an affine space A over V, we let $\mathrm{Aff}(A)$ denote the set of affine automorphisms of A, which forms a group with respect to the composition. We can check that we have the following natural short exact sequence:

$$0 \to V \xrightarrow{T_{(-)}} \mathrm{Aff}(A) \xrightarrow{\nabla(-)} \mathrm{Aut}(V) \to 1.$$

Here, the map $T_{(-)}\colon V \to \mathrm{Aff}(A)\colon v \mapsto T_v$ is given by $T_v(x) := x + v$.

Definition 2.4. Let A be an affine space over V. Let Γ be a group acting on A. We say that this Γ-action is *affine* if all automorphisms given by Γ are affine, i.e., we have a homomorphism $\Gamma \to \mathrm{Aff}(A)$.

Definition 2.5. Let A be an affine space over V.

1. An *affine functional on* A is an affine map $\psi\colon A \to k$. We write A^* for the set of affine functionals on A. We shortly write $\dot{\psi} := \nabla\psi \in V^* := \mathrm{Hom}_k(V, k)$.
2. For an affine functional ψ on A, we let H_ψ denote the zero locus of ψ, i.e., $H_\psi = A^{\psi=0} = \{x \in A \mid \psi(x) = 0\}$.

Note that A^* has a structure of a k-vector space with respect to the pointwise addition and k-multiplication of affine functionals. The derivative map $\nabla\colon A^* \to V^*$ is then a k-linear map which is surjective. Its kernel is given by the set of constant maps, hence we get a short exact sequence

$$0 \to k \to A^* \to V^* \to 0.$$

By taking the k-duals of this sequence, we obtain

$$0 \to V \to A^{**} \to k \to 0.$$

Concretely, the first map is described as follows: $V \to A^{**}\colon v \mapsto \langle -, v\rangle$, where $\langle -, v\rangle \in A^{**}$ is $\langle \psi, v\rangle := \langle \dot{\psi}, v\rangle$.

[6]We also call such A a V-*torsor*. But this terminology is more general in the sense that it can be used for any space equipped with a simply-transitive action of a group.

2.2. **Affine root systems ([KP23, §1.3]).** Let V be a finite dimensional $\mathbb{R}$-vector space. Let A be an affine space over V. We consider the short exact sequence discusses above:

$$0 \to V \to A^{**} \to \mathbb{R} \to 0.$$

Definition 2.6. An *affine root system* is a subset $\Psi \subset A^*$ satisfying the following:

(1) The subset Ψ spans A^* and does not contain constant functionals. Moreover, $\Phi := \nabla\Psi \subset V^*$ is a finite set.
(2) For any $\psi \in \Psi$, there exists $\dot{\psi}^\vee \in V$ satisfying $\langle \dot{\psi}, \dot{\psi}^\vee \rangle = 2$. Moreover, the *reflection* $r_{\psi,\dot{\psi}^\vee} \colon A^* \to A^*$ defined by

$$r_{\psi,\dot{\psi}^\vee}(x) := x - \dot{x}(\dot{\psi}^\vee)\psi \in A^*$$

preserves the set Ψ.
(3) For any $\psi, \eta \in \Psi$, we have $\langle \dot{\psi}, \dot{\eta}^\vee \rangle \in \mathbb{Z}$.
(4) For any $a \in \Phi$, $\{\psi \in \Psi \mid \dot{\psi} = a\}$ does not have an accumulation point.

- We call Ψ the set of *affine roots*.
- We call $\dim(V)$ the *rank* of Ψ.
- We write $W(\Psi)$ for the group of k-linear automorphisms on A^* generated by $\{r_{\psi,\dot{\psi}^\vee} \mid \psi \in \Psi\}$. The group $W(\Psi)$ is called the *affine Weyl group* of Ψ.

Fact 2.7. (1) *The set $\Psi \subset V^*$ of derivatives of Ψ forms a root system (we call Ψ the "derivative root system" of Ψ).*
(2) *The set $\Psi^{\mathrm{nd}} := \{\psi \in \Psi \mid \psi/2 \notin \Psi\}$ of "nondivisible affine roots" forms an affine root system.*
(3) *The set $\Psi^{\mathrm{nm}} := \{\psi \in \Psi \mid 2\psi \notin \Psi\}$ of "nonmultipliable affine roots" forms an affine root system.*

Definition 2.8. We say that an affine root system Ψ is *reduced* if $\Psi = \Psi^{\mathrm{nd}} = \Psi^{\mathrm{nm}}$.

2.3. **Rough outline of Bruhat–Tits theory.** In the following, we let k be a non-archimedean local field.[7] Let $\omega \colon k \to \mathbb{Z}$ be the normalized valuation of k. Let $\mathfrak{o}$ be the ring of integers of k and $\mathfrak{m}$ the maximal ideal of $\mathfrak{o}$. We write $\mathfrak{f} := \mathfrak{o}/\mathfrak{m}$. (In this course, all the notation follow those of [KP23].)

Let G be a connected reductive group over k.

2.3.1. *Bruhat–Tits building.* We can associate to G a topological (contractible, complete metric) space $\mathcal{B}(G)$ called the *(reduced) Bruhat–Tits building* of G, which is equipped with an action of $G(k)$. The Bruhat–Tits building $\mathcal{B}(G)$ has a polysimplicial structure (i.e., a product of several simplicial structures; we will examine this notion soon) which is preserved by the action of $G(k)$. Each polisimplice is called a facet; a maximal facet is called a *chamber* (a.k.a. alcove) and and a minimal one is called a *vertex*.

2.3.2. *Apartments.* The Bruhat–Tits building $\mathcal{B}(G)$ is equipped with a family of polysimplicial subcomplexes called *apartments*. Apartments of $\mathcal{B}(G)$ bijectively correspond to maximal k-split tori of G. If two maximal k-split tori are conjugate by $g \in G(k)$, then the associated apartments are also conjugate (transferred) by the action of $g \in G(k)$ on $\mathcal{B}(G)$. In particular, since all maximal k-split tori of G are $G(k)$-conjugate, all apartments are conjugate to each other.

[7]In fact, theory works for any Henselian discrete valuation field with perfect residue field.

2.3.3. *Affine root system.* Each apartment $\mathcal{A}$ is an affine space over $\mathbb{R}^n$, where n is the rank of a maximal k-split torus of G. The polysimplicial structure of $\mathcal{A}$ is given by a family of hyperplanes of $\mathcal{A}$, i.e., chambers of $\mathcal{A}$ are the connected components of the complement of the union of these hyperplanes in $\mathcal{A}$. These hyperplanes are defined to be the zero loci of a set of affine functionals on $\mathcal{A}$, which forms an *affine root system* associated to G. The root system obtained by taking the derivative (linear part) of this affine root system is the relative root system of G with respect to the chosen maximal k-split torus.

2.3.4. *Various open subgroups of $G(k)$.* Recall that, from the viewpoint of representation theory or harmonic analysis on p-adic reductive groups, we are interested in open compact subgroups of $G(k)$. An important role of the Bruhat–Tits building $\mathcal{B}(G)$ is to classify certain classes of open compact subgroups of $G(k)$. Each point x of $\mathcal{B}(G)$ gives rise to several open compact subgroups. The hierarchy is as follows:

$$G(k) \supset G(k)^1 \supset G(k)^1_x = G(k)_{\tilde{x}} \supset G(k)^0_x = G(k)_{x,0} \supset G(k)_{x,r} \supset \cdots$$

Let us explain each of this chain briefly.

(1) $G(k)_x$ (point stabilizer):
The simplest way of producing a subgroup using a given point $x \in \mathcal{B}(G)$ is to take its stabilizer group with respect to the action of $G(k)$ on $\mathcal{B}(G)$. The stabilizer group $G(k)_x$ is open. However, since $Z_G(k)$ acts on $\mathcal{B}(k)$ (where Z_G denotes the center of G) acts trivially, $G(k)_x$ always contains $Z_G(k)$. This means that $G(k)_x$ cannot be compact unless the center Z_G is anisotropic (i.e., the split part of Z_G is trivial). In fact, $G(k)_x$ is always compact modulo $Z_G(k)$. Thus, the crux is how to cut this subgroup so as to make it compact, while preserving openness.

(2) $G(k)^1_x$ (cut $G(k)_x$ using $G(k)^1$):
Let ω the normalized valuation of k and $X^*(G) := \mathrm{Hom}_k(G, \mathbb{G}_\mathrm{m})$. We put

$$G(k)^1 := \{g \in G(k) \mid \omega(\chi(g)) = 0 \text{ for all } \chi \in X^*(G)\}.$$

Then we see that any compact subgroup of $G(k)$ must be contained in $G(k)^1$ (try to show this; the exercise below). Hence any open compact subgroup of $G(k)_x$ must be contained in $G(k)^1_x := G(k)_x \cap G(k)^1$. A key fact in Bruhat–Tits theory is that $G(k)^1_x$ is open compact subgroup. In fact, we also have the following fact:

Fact 2.9. *Any compact subgroup of $G(k)$ is contained in $G(k)^1_x$ for some $x \in \mathcal{B}(G)$.*

(3) $G(k)^1_x$ (stabilizer taken in the enlarged building):
The group $G(k)^1_x$ induced above has an alternative interpretation. We define the *enlarged building* of G by

$$\widetilde{\mathcal{B}}(G) := \mathcal{B}(G) \times (X_*(A_G) \otimes_\mathbb{Z} \mathbb{R}),$$

where A_G is the maximal central split torus of G. Here, $X_*(A_G) \otimes_\mathbb{Z} \mathbb{R}$ is equipped with an action of $G(k)$ such that the kernel is exactly equal to $G(k)^1$. Therefore, if we choose a lift $\tilde{x} \in \widetilde{\mathcal{B}}(G)$ of $x \in \mathcal{B}(G)$ and consider the stabilizer group of $\tilde{x}$, we obtain $G(k)_{\tilde{x}} = G(k)^1_x$.

(4) $G(k)^0_x$ (parahoric subgroup):

For a connected reductive group G over k, Kottwitz [Kot97] defined a homomorphism (called the *Kottwitz homomorphism*)

$$\kappa_G\colon G(k) \to (\pi_1(G)_{I_k})^{\mathrm{Gal}(k^{\mathrm{ur}}/k)},$$

where k^{ur} is the maximal unramified extension of k and I_k is the inertia subgroup of $\mathrm{Gal}(k^{\mathrm{sep}}/k)$. We let $G(k)^0$ be the kernel of the Kottwitz homomorphism. We put $G(k)^0_x := G(k)^0 \cap G(k)_x$. This group is open compact and contained in $G(k)^1_x$; we call $G(k)^0_x$ the *parahoric subgroup* of $G(k)$ associated to the point $x \in \mathcal{B}(G)$. In fact, the parahoric subgroup depends only on the facet $\mathcal{F} \subset \mathcal{B}(G)$ containing x, hence we often write $G(k)^0_{\mathcal{F}}$ instead of $G(k)^0_x$.

(5) $G(k)_{x,r}$ (Moy–Prasad filtration):
The parahoric subgroup $G(k)^0_x$ has a descending filtration $\{G(k)_{x,r}\}_{r\in\mathbb{R}_{\geq 0}}$ of open compact subgroups labeled by $\mathbb{R}_{\geq 0}$ such that $G(k)_{x,0} = G(k)^0_x$. This is called the *Moy–Prasad filtration* of $G(k)^0_x$. In contrast to the parahoric subgroup, which depends only on the facet $\mathcal{F}$ containing x, the Moy–Prasad filtration depends on the point x itself.

Exercise 2.10. Prove that any compact subgroup of $G(k)$ is contained in $G(k)^1$.

Another important aspect of Bruhat–Tits theory is to realize all of the above subgroups also as the groups of $\mathfrak{o}$-valued points of smooth integral models of G. For example, the intuitive operation of taking the "modulo-p-reduction" of various open compact subgroup can be justified as taking the special fibers of those integral models. Bruhat–Tits constructed a smooth model $\mathcal{G}^1_x$ of G over $\mathfrak{f}$ such that $\mathcal{G}^1_x(\mathfrak{o}) = G(k)^1_x$. The special fiber of $\mathcal{G}^1_x$ is disconnected in general, hence it makes sense to look at the relative identity component $\mathcal{G}^0_x$ of $\mathcal{G}^1_x$. In fact, the parahoric subgroup $G(k)^0_x$ is nothing but $\mathcal{G}^0_x(\mathfrak{o})$. Similarly, we can also construct a smooth integral model $\mathcal{G}_{x,r}$ such that $\mathcal{G}_{x,r}(\mathfrak{o}) = G(k)_{x,r}$. The structure of the reductive quotient of the special fiber of $\mathcal{G}^0_x$ can be completely described in terms of affine root system and the facet $\mathcal{F}$ containing x. In this way, we obtain a path connecting finite groups of Lie type and p-adic reductive groups.

2.4. **The case of SL_2.** Let us first consider the case of $G = \mathrm{SL}_2$:

$$\mathrm{SL}_2(k) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; a,b,c,d \in k,\ ad-bc=1 \right\}.$$

2.4.1. *Filtrations on maximal torus and root subgroups.* Let T be the diagonal maximal torus of G. Let $a \in X^*(T) := \mathrm{Hom}(T, \mathbb{G}_{\mathrm{m}})$ and $a^\vee \in X_*(T) := \mathrm{Hom}(\mathbb{G}_{\mathrm{m}}, T)$ be the standard positive (i.e., with respect to the upper triangular Borel subgroup) root and coroot, respectively. More concretely, we have

$$a(\mathrm{diag}(x, x^{-1})) = x^2, \quad a^\vee(x) = \mathrm{diag}(x, x^{-1}).$$

Recall that $G = \mathrm{SL}_2$ is generated by T and root subgroups $U_{\pm a}$; more concretely, U_a and U_{-a} are the subgroups of upper-triangular and lower triangular unipotent matrices, respectively. The basic idea of producing various open compact subgroups of $G(k)$ is firstly defining descending filtrations on $T(k)$ and $U_{\pm a}(k)$ and combining them.

We let

$$T(k)_0 := a^\vee(\mathfrak{m}^\times), \quad T(k)_n := a^\vee(1+\mathfrak{m}^n) \quad (n>0).$$

By letting $u_{\pm a}\colon \mathbb{G}_{\mathrm{a}} \cong U_{\pm a}$ be

$$u_a(x) := \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \quad u_{-a}(x) := \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix},$$

we also get filtrations $\{u_{\pm a}(\mathfrak{m}^n)\}_{n\in\mathbb{Z}}$ of $U_{\pm a}(k)$.

2.4.2. *Apartment.* We define

$$\mathcal{A} := X_*(T) \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}a^\vee$$

and call $\mathcal{A}$ the *standard apartment* of $G = \mathrm{SL}_2$. To each $x \in \mathcal{A}$, we associate an open compact subgroup $\mathcal{P}_x$ of $G(k)$ by

$$\mathcal{P}_x := \langle T(k)_0,\, u_a(\mathfrak{m}^{\lfloor\langle a,x\rangle\rfloor}),\, u_{-a}(\mathfrak{m}^{\lfloor\langle -a,x\rangle\rfloor})\rangle.$$

Here, the outer $\langle\cdots\rangle$ means the group generated by $(\cdots)$ and the inner (on the exponents) $\langle -,-\rangle$ means the pairing between $X^*(T)$ and $X_*(T)$.

Example 2.11. (1) Let $x := 0$. Then we have $\mathcal{P}_x = \langle T(k)_0, u_a(\mathfrak{o}), u_{-a}(\mathfrak{o})\rangle$. In the matrix form, this is

$$\begin{pmatrix} \mathfrak{o} & \mathfrak{o} \\ \mathfrak{o} & \mathfrak{o} \end{pmatrix}.$$

(2) Let $x := \frac{1}{2}a^\vee$. Then we have $\mathcal{P}_x = \langle T(k)_0, u_a(\mathfrak{m}^{-1}), u_{-a}(\mathfrak{m})\rangle$ (Note that $\langle a, a^\vee\rangle = 2$). In the matrix form, this is

$$\begin{pmatrix} \mathfrak{o} & \mathfrak{m}^{-1} \\ \mathfrak{m} & \mathfrak{o} \end{pmatrix}.$$

(3) Let $x := ra^\vee$, where $0 < r < \frac{1}{2}$. Then we have $\mathcal{P}_x = \langle T(k)_0, u_a(\mathfrak{o}), u_{-a}(\mathfrak{m})\rangle$. In the matrix form, this is

$$\begin{pmatrix} \mathfrak{o} & \mathfrak{o} \\ \mathfrak{m} & \mathfrak{o} \end{pmatrix}.$$

In fact, $\mathcal{P}_0$ and $\mathcal{P}_{\frac{1}{2}a^\vee}$ completely represent the $G(k)$-conjugacy classes of maximal open compact subgroup of $G(k)$ while $\mathcal{P}_{ra^\vee}$ for $0 < r < \frac{1}{2}$ represents the unique conjugacy class of Iwahori subgroups. This reflects the fact that the interval $(1, \frac{1}{2})a^\vee$ is an chamber of $\mathcal{A}$ and the points 0 and $\frac{1}{2}a^\vee$ are vertices attached to the chamber.

2.4.3. *Affine roots.* The apartment $\mathcal{A}$ gives a parametrization of certain open compact subgroups of $\mathrm{SL}_2(k)$ by the above discussion. As the previous example suggest, the points of $\mathcal{A}$ which produces 'maximal' open compact subgroups are exactly given by $\frac{1}{2}\mathbb{Z}a^\vee = \{x \in \mathcal{A} \mid \langle a, x\rangle \in \mathbb{Z}\} \subset \mathcal{A}$.

We interpret this set from a different viewpoint. Let us consider the set of affine functionals on $\mathcal{A}$ given by

$$\Psi := \{b + j \mid b \in \Phi,\, j \in \mathbb{Z}\}.$$

More precisely, each $b + j$ is regarded as a function $\mathcal{A} \to \mathbb{R}$ by

$$(b + j)(x) := \langle b, x\rangle + j.$$

In fact, Ψ forms an affine root system; then $x \in \mathcal{A}$ belongs to $\frac{1}{2}\mathbb{Z}a^\vee$ if and only if it some affine root takes 0 at x. So the set of zeros of affine roots parametrized the points of the apartment whose open compact subgroups are maximal.

Consequently, we can find a very obvious 1-dimensional simplicial structures on $\mathcal{A}$. The set of vertices is given by $\frac{1}{2}\mathbb{Z}a^\vee \subset \mathcal{A}$. The set of edges (1-dimensional facets) are given by the intervals divided by vertices. So the maximal facets in this situation are connected components of the complement in $\mathcal{A}$ of zero loci of affine roots. As we saw in the previous example, the group $\mathcal{P}_x$ associated to a point x in any 1-dimensional interval depends only on the interval; independent of the choice of the point in the interval.

2.4.4. *Affine Weyl group.* As claimed above without proof, different faces $\mathcal{F}_1$ and $\mathcal{F}_2$ of $\mathcal{A}$ (vertices or edges) may associate open compact subgroups which are $G(k)$-conjugate. This phenomena can be understood in terms of the *affine Weyl group.*

We first consider $\mathbb{Z}a^\vee = X_*(T) \subset \mathcal{A}$. This abelian group acts on $\mathcal{A}$ via translation. Note that this action preserves the simplicial structure of $\mathcal{A}$. Moreover, for any facet $\mathcal{F} \subset \mathcal{A}$, the open comact subgroups associated to $\mathcal{F}$ and $\mathcal{F}+1$ are conjugate by $a^\vee(\varpi^{-1})$.

We next consider the Weyl group of the root system Φ, which is just given by $\mathbb{Z}/2\mathbb{Z}$. This acts on $\mathcal{A} = X_*(T) \otimes_{\mathbb{Z}} \mathbb{R}$ via inversion, hence a facet $\mathcal{F}$ is mapped to $-\mathcal{F}$. The corresponding open compact subgroups are conjugate by

$$w_0 := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

These actions of two groups can be combined together into the action of $\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$ on $\mathcal{A}$. In fact, this is nothing but the affine Weyl group $W(\Psi)$.

We can also think of $W(\Psi)$ as the group as follows. Let N be the normalizer group of T in G. The group $T(k)$ acts on $\mathcal{A}$ via translation;

$$t \cdot x := x - \omega(t).$$

This action can be extended to an action of $N(k)$ by letting $w_0 \in N(k)$ act on $\mathcal{A}$ via inversion. Since the kernel of this action is $T(k)_0$, we get an identificaion

$$N(k)/T(k)_0 \cong W(\Psi) = \mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}.$$

2.4.5. *Building.* Now we define the Bruhat–Tits building $\mathcal{B}$ for SL_2. What we want to construct here is a simplicial complex $\mathcal{B}$ having the following properties:

- $\mathcal{B}$ has a $G(k)$-action.
- $\mathcal{B}$ contains $\mathcal{A}$.
- For any $x \in \mathcal{A}$, its stabilizer in $G(k)$ is given by $\mathcal{P}_x$.

For this, we consider the equivalence relation on $G(k) \times \mathcal{A}$ given by

$$(g, x) \sim (h, y) \iff \text{There exists } n \in N(k) \text{ s.t. } y = nx \text{ and } g^{-1}hn \in \mathcal{P}_x.$$

If we put $\mathcal{B} := (G(k) \times \mathcal{A})/{\sim}$, then $\mathcal{B}$ satisfies the above desired properties. The action of $G(k)$ on $\mathcal{B}$ is given by $g \cdot (h, y) := (gh, y)$. We define a map from $\mathcal{A}$ to $\mathcal{B}$ by

$$\mathcal{A} \to \mathcal{B} \colon x \mapsto (1, x).$$

In fact, this map is injective by the following:

Fact 2.12. *The stabilizer group of $x \in \mathcal{A}$ in $N(k)$ equals $N(k) \cap \mathcal{P}_x$.*

Hence we may regard $\mathcal{A}$ as a subset of $\mathcal{B}$. We call a subset of $\mathcal{B}$ of the form $g\mathcal{A}$ for some $g \in G(k)$ an *apartment* of $\mathcal{B}$. The building $\mathcal{B}$ has a simplicial structure.

Exercise 2.13.
(1) Prove the above fact (the stabilizer group of $x \in \mathcal{A}$ in $N(k)$ equals $N(k) \cap \mathcal{P}_x$).
(2) Deduce the injectivity of $\mathcal{A} \hookrightarrow \mathcal{B}$ from this fact.

2.5. **The case of** SU_3**.** We next consider an example of a non-split (but quasi-split) connected reductive group. Let l/k be a quadratic Galois extension. Then the *special unitary group* SU_3 in three variables associated to l/k is defined to be a connected reductive group over k whose set of R-rational points is given by

$$\mathrm{SU}_3(R) = \{g \in \mathrm{SL}_3(l \otimes_k R) \mid {}^t\bar{g}Jg = J\},$$

where J is the anti-diagonal matrix whose anti-diagonal entries are $(1, -1, 1)$ and $\bar{g}$ denotes the Galois conjugate of g (i.e., apply $\sigma \otimes \mathrm{id}$ to entries of g, where σ is the nontrivial element of $\mathrm{Gal}(l/k)$). Thus, especially, we have

$$\mathrm{SU}_3(k) = \{g \in \mathrm{SL}_3(l) \mid {}^t\bar{g}Jg = J\}.$$

We write $G := \mathrm{SU}_3$.

2.5.1. *Structure of* SU_3. Let T be the subgroup of diagonal matrices in SU_3. Then this is a maximal k-torus of G. Concretely, the group of its k-rational points is given by

$$T(k) = \{\mathrm{diag}(t, \bar{t}/t, \bar{t}^{-1}) \mid t \in l^\times\}.$$

Similarly, if we let B be the subgroup of upper-triangular matrices in SU_3, then B is a k-rational Borel subgroup of G. So, using these subgroups, can we consider open compact subgroups of $G(k)$ in the same manner as SL_2? In fact, NOT. This is because the root system of (G, T) has a nontrivial Galois action (equivalently, G is not split). Accordingly, each root subgroup U_a is not defined over k. To remedy this issue, we need to consider the 'relative' root system of G instead of the 'absolute' root system.

We let S be the maximal k-split torus contained in T. Concretely, the group of its k-rational points is given by

$$S(k) = \{\mathrm{diag}(t, 1, t^{-1}) \mid t \in k^\times\} \subset T(k).$$

As in the construction of the absolute root system $\Phi(G, T)$, we consider the adjoint action of S on the Lie algbera of G to get a root system ("relative root system") $\Phi(G, S)$. In this case, the resulting root system $\Phi(G, S)$ is given by

$$\{\pm a, \pm 2a\} \subset X^*(S),$$

where $a\colon \mathrm{diag}(t, 1, t^{-1}) \mapsto t$. (Note that Φ is not reduced!) Each relative root b defines a relative root subgroup U_b of G; by definition, this is a unique smooth closed subgroup of G which is normalized by S such that the weights of S in its Lie algbera are positive integer multiples of b. In the current setting, U_b is described concretely as follows. Firstly, U_{2a} is isomorphic to $\mathbb{G}_{\mathrm{a}}$ over l; its l-rational points are given by

$$U_{2a}(l) = \left\{ \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \,\middle|\, x \in l \right\}.$$

The Galois action of $\mathrm{Gal}(l/k)$ on $U_{2a}(l)$ changes the $(1,3)$-entry x into $-\bar{x}$. Hence we have

$$U_{2a}(k) = \left\{ \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \,\middle|\, x \in l^0 \right\},$$

where $l^0 := \mathrm{Ker}(\mathrm{Tr}_{l/k}\colon l \to k)$. Secondly, U_a is equal to the unipotent radical of B. Hence its l-rational points are given by

$$U_a(l) = \left\{ \begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} \,\middle|\, x, y, z \in l \right\}.$$

The Galois action of $\mathrm{Gal}(l/k)$ on $U_{2a}(l)$ changes the coordinates (x, y, z) into $(\bar{z}, \overline{xz} - \bar{y}, \bar{x})$. Thus, by putting

$$u_a(x, y) = \begin{pmatrix} 1 & x & y \\ 0 & 1 & \bar{x} \\ 0 & 0 & 1 \end{pmatrix},$$

we get

$$U_a(k) = \{u_a(x, y) \mid x, y \in l,\, \mathrm{Nr}_{l/k}(x) = \mathrm{Tr}_{l/k}(y)\}.$$

Note that, with this notation, we can also write

$$U_{2a}(k) = \{u_a(0, y) \mid y \in l^0\}.$$

2.5.2. *Filtrations on maximal torus and root subgroups.* Now, using the maximal k-torus T and 'relative' root subgroups U_b, let us imitate the previous definitions of filtrations of these subgroups.

Firstly, recall that $T(k) \cong l^\times$. Thus we have a natural filtration on $T(k)$ given by

$$l^\times \supset \mathfrak{o}_l^\times \supset 1 + \mathfrak{m}_l \supset 1 + \mathfrak{m}_l^2 \supset 1 + \mathfrak{m}_l^3 \supset \cdots.$$

We give indices on this by $T(k)_0 = \mathfrak{o}_l^\times$ and

$$T(k)_r := \{\mathrm{diag}(x, \bar{x}/x, \bar{x}^{-1}) \in T(k)_0 \mid \omega(x - 1) \geq r\}$$

for $r > 0$. Note that, here, ω denotes the valuation of l normalized so that $\omega(k^\times) = \mathbb{Z}$. In particular, the jumps of the filtration $\{T(k)_r\}_r$ are given by $\frac{1}{e}\mathbb{Z}$, where e is the ramification index of l/k. (Since l/k is quadratic, it is 1 if l/k is unramified and 2 if ramified.)

We next consider the filtration on $U_{2a}(k)$. The natural candidate for the r-th filtration $U_{2a}(k)_r$ for $r \in \mathbb{R}$ may be

$$U_{2a}(k)_r := \{u_a(0, y) \in U_{2a}(k) \mid \omega(y) \geq r\}.$$

However, this makes the set of jumps of $\{U_{2a}(k)_r\}$ a bit tricky because its description involves a constant related to the *ramification filtration* of $\mathrm{Gal}(l/k)$.

Recall that, in general, for a finite Galois extension k'/k and $i \in \mathbb{Z}_{\leq 0}$, the i-th (lower) ramification filtration of $\mathrm{Gal}(k'/k)$ is defined as follows:

$$\mathrm{Gal}(k'/k)_i := \{\sigma \in \mathrm{Gal}(k'/k) \mid \omega'(\sigma(\varpi') - \varpi') \geq i + 1\},$$

where ϖ' is any uniformizer of k' and ω' is the normalized valuation of k'. It is known that $\mathrm{Gal}(k'/k)_i$ is trivial for sufficiently large i depending on k'/k (see [Ser79, Chapter IV] for details).

In the present situation where l/k is quadratic, the lower ramification filtraion jumps at exactly one point. We let s be the jump point, i.e., $\mathrm{Gal}(l/k)_s \neq \{1\}$ and $\mathrm{Gal}(l/k)_{s+1} \neq \{1\}$.

Fact 2.14 ([KP23, Lemma 2.8.1])**.** *We put $\mu := \sup\{\omega(x) \mid x \in l^0\}$.*

(1) *When l/k is unramified, $\mu = 0$ and $\omega(l^0) = \mathbb{Z}$.*
(2) *When l/k is ramified, $\mu = -\frac{s}{2}$ and $\omega(l^0) = \mathbb{Z} + \frac{1}{2} + \mu$.*

Keeping this in mind, let us modify the filtration on $U_{2a}(k)$ in the following way:

$$U_{2a}(k)_r := \{u_a(0,y) \in U_{2a}(k) \mid \omega(y) \geq r + \mu\}.$$

Then the set of jumps is given by $\mathbb{Z}$ when l/k is unramified and $\mathbb{Z} + \frac{1}{2}$ when l/k is ramified.

We finally discuss the filtration on $U_a(k)$. Since $U_a(k)$ is a "2-parameter subgroup", it is not very trivial how to put a 1-parameter filtration on it:

$$U_a(k) = \{u_a(x,y) \mid x, y \in l,\, \mathrm{Nr}_{l/k}(x) = \mathrm{Tr}_{l/k}(y)\}.$$

The point is that the condition $\mathrm{Nr}_{l/k}(x) = \mathrm{Tr}_{l/k}(y)$ implies that

$$2\omega(x) = \omega(\mathrm{Nr}_{l/k}(x)) = \omega(\mathrm{Tr}_{l/k}(y)) \geq \omega(y).$$

Thus, a reasonable option is the following:

$$\begin{aligned} U_a(k)_r &:= \{u_a(x,y) \in U_a(k) \mid \omega(x) \geq r,\, \omega(y) \geq 2r\} \\ &:= \{u_a(x,y) \in U_a(k) \mid \omega(y) \geq 2r\}. \end{aligned}$$

As in the case of U_{2a}, we furthermore modify this filtration as follows:

$$U_a(k)_r := \{u_a(x,y) \in U_a(k) \mid \omega(y) \geq 2r + \mu\}.$$

Exercise 2.15. Show that $U_a(k)_r$ is indeed a subgroup of $U_a(k)$.

2.5.3. *Apartments.* Now let us proceed as in the manner above (for SL_2).

We define the standard apartment $\mathcal{A}$ for SU_3 by

$$\mathcal{A} := X_*(S) \otimes_{\mathbb{Z}} \mathbb{R} \cong X_*(T)^{\Gamma} \otimes_{\mathbb{Z}} \mathbb{R},$$

where Γ is the absolute Galois group of k. For $x \in \mathcal{A}$, we define an open compact subgroup $\mathcal{P}_x$ of $G(k)$ by

$$\mathcal{P}_x := \langle T(k)_0, U_b(k)_{\langle -b, x\rangle} \mid b \in \Phi(G,S)\rangle.$$

2.5.4. *Building.* We define the Bruhat–Tits building $\mathcal{B}$ for SU_3 in the same way as SL_2. To be more precise, we consider the equivalence relation on $G(k) \times \mathcal{A}$ given by

$$(g,x) \sim (h,y) \iff \text{There exists } n \in N(k) \text{ s.t. } y = nx \text{ and } g^{-1}hn \in \mathcal{P}_x.$$

Å Then we put $\mathcal{B} := (G(k) \times \mathcal{A})/{\sim}$. The standard apartment $\mathcal{A}$ is regarded as a subset of $\mathcal{B}$ by

$$\mathcal{A} \hookrightarrow \mathcal{B} \colon x \mapsto (1,x)$$

(for any $x \in \mathcal{A}$, the stabizer of x in $G(k)$ equals $\mathcal{P}_x$, which implies that the map is injective). We call a subset of $\mathcal{B}$ of the form $g\mathcal{A}$ for some $g \in G(k)$ an *apartment* of $\mathcal{B}$. The building $\mathcal{B}$ has a simplicial structure.

In these examples, we define open compact subgroups by using the filtrations on the 'standard' maximal torus and root subgroups. However, we must note that these crucially depends on the 'standard' realization of the groups SL_2 and SU_3. In general, we do not have such a nice (canonical) realization of a given connected reductive groups.

3. Week 3: Abstract building and parahoric subgroups

Our aim of this week is to discuss part of axioms of Bruhat–Tits theory ([KP23, §4]), especially,

- existence of a building ([KP23, Axiom 4.1.1]), and
- existence of parahoric subgroups ([KP23, Axiom 4.1.2]).

Let k be a non-archimedean local field. Let $\mathfrak{o}$ and $\mathfrak{m}$ be the ring of integers of k and the maximal ideal of $\mathfrak{o}$, respectively. Let $\omega\colon k \to \mathbb{Z} \cup \{\infty\}$ be the normalized valuation of k. We write $\mathfrak{f} := \mathfrak{o}/\mathfrak{m}$.

Let K be the maximal unramified extension of k. Let $\mathcal{O}$ and $\mathcal{M}$ be the ring of integers of K and the maximal ideal of $\mathcal{O}$, respectively. Thus we have $\overline{\mathfrak{f}} = \mathcal{O}/\mathcal{M}$. We again write ω for the unique valuation of K extending ω.

3.1. Abstract building.

Definition 3.1. (1) A *simplicial complex* is a pair $(V, \mathcal{B})$ of a nonempty set V (called the set of *vertices*) and a nonempty set $\mathcal{B}$ of nonempty finite subsets of V (called the set of *facets*) such that
   (a) we have $\{x\} \in \mathcal{B}$ for any $x \in V$,
   (b) if $B \in \mathcal{B}$, then any nonempty finite subset of B also belongs to $\mathcal{B}$.

   We often simply write $\mathcal{B}$ for a simplicial complex.

(2) A *polysimplicial complex* is a tuple $\mathcal{B} = (\mathcal{B}_1, \ldots, \mathcal{B}_n)$ of simplicial complexes. We also think of $\mathcal{B}$ as the product set $\mathcal{B}_1 \times \cdots \times \mathcal{B}_n$. Similarly, we let $V := V_1 \times \cdots \times V_n$.

We introduce several terminologies related to a simplicial or polysimplicial complex.

Definition 3.2. Let $\mathcal{B}$ be a simplicial or polysimplicial complex.

(1) If $A, B \in \mathcal{B}$ satisfy $A \subset B$, we say that A is a *face* of B.
(2) A *subcomplex* of $\mathcal{B}$ is a subset $\mathcal{B}' \subset \mathcal{B}$ such that if $A \in \mathcal{B}'$ and $B \subset A$, then $B \in \mathcal{B}'$.

Definition 3.3. (1) An isomoprhism $(V_1, \mathcal{B}_1) \to (V_2, \mathcal{B}_2)$ of simplicial complexes is a bijection $f\colon V_1 \to V_2$ which also induces a bijection between $\mathcal{B}_1$ and $\mathcal{B}_2$.

(2) An isomoprhism $(\mathcal{B}_1, \ldots, \mathcal{B}_n) \to (\mathcal{B}'_1, \ldots, \mathcal{B}'_n)$ of polysimplicial complexes is a tuple $(\sigma, f_1, \ldots, f_n)$ consisting of a permutation $\sigma \in \mathfrak{S}_n$ and isomoprhisms of simplicial complexes $f_i\colon \mathcal{B}_i \to \mathcal{B}'_{\sigma(i)}$ for $i = 1, \ldots, n$.

Definition 3.4. Let $\mathcal{B}$ be a simplicial or polysimplicial complex.

(1) When A is a face of B, we define $\operatorname{codim}(A, B)$ to be the largest integer n such that there exists a chain of facets $A = A_0 \subsetneq A_1 \subsetneq \cdots \subsetneq A_n = B$.
(2) For any facet $B \in \mathcal{B}$, we define $\dim(B)$ to be the maximum of $\operatorname{codim}(A, B)$ for a face $A \subset B$.
(3) We say that $\mathcal{B}$ is a *chamber complex* if it satisfies the following:
   (a) any facet is contained in a maximal facet (maximal facets are called *chambers*);
   (b) for any two maximal facets C and C', there exists a sequence $C = C_1, \ldots, C_n = C'$ such that $C_i \cap C_{i+1} \in \mathcal{B}$ and

$$\operatorname{codim}(C_i \cap C_{i+1}, C_i) = \operatorname{codim}(C_i \cap C_{i+1}, C_{i+1}) = 1$$

for any $i = 1, \dots, n-1$ (we call such a sequence a *gallery joining C and C'*).

(4) We say that a chamber complex is *thick* if each facet of codimension 1 (i.e., codimension 1 in a chamber) is the face of at least three chambers.

(5) We say that a chamber complex is *thin* if each facet of codimension 1 (i.e., codimension 1 in a chamber) is the face of exactly two chambers.

Definition 3.5. An isomorphism of chamber complexes is an isomorphism of (poly-)simplicial complexes which maps chambers to chambers.

Definition 3.6. A *building* (or an *abstract building*) is a chamber complex $\mathcal{B}$ equipped with a collection of subcomplexes called *apartments* satisfying the following axioms:

(1) $\mathcal{B}$ is thick;
(2) each apartment is a thin chamber complex;
(3) any two chambers belong to an apartment (i.e., there exists an apartment containing both chambers);
(4) for any apartments $\mathcal{A}_1$, $\mathcal{A}_2$ and two facets $\mathcal{F}_1, \mathcal{F}_2 \in \mathcal{A}_1 \cap \mathcal{A}_2$, there exists an isomorphism $\mathcal{A}_1 \to \mathcal{A}_2$ which fixes $\mathcal{F}_1$, $\mathcal{F}_2$, and all of their faces.

Now we can state the first axiom of Bruhat–Tits theory:

Axiom 1 (existence of building)**.** Let G be a connected reductive group over k.

(1) There exists a building $\mathcal{B}(G)$ in the sense of Definition 3.6 equipped with an action of $G(k)$ as polysimplicial automorphisms.
(2) The apartments (again in the sense of Definition 3.6) of $\mathcal{B}(G)$ are $G(k)$-equivariantly parametrized by maximal k-split tori of G. In particular, the stabilizer (not pointwise) of the apartment $\mathcal{A}(S)$ associated to a maximal k-split torus S of G is given by $N(k)$, where N denotes the normalizer group of S in G.
(3) Any surjective homomorphism $G \to G'$ with central kernel induces a (G, G')-equivariant isomorphism $\mathcal{B}(G) \to \mathcal{B}(G')$. In particular, the buildings of G, G_{ad} (adjoint group), G_{der} (derived group), G_{sc} (simply-connected cover of G_{der}) are canonically identified. Also, the action of $G(k)$ on $\mathcal{B}(G)$ factors through $G(k) \to G_{\mathrm{ad}}(k)$.

3.2. **Parahoric subgroups.** Recall that, in the last week, we claimed that we can define certain open subgroups $G(k)^0 \subset G(k)^0$ for a connected reductive group G over k. The second axiom of Bruhat–Tits theory is stated using these subgroups.

Axiom 2 (open compactness of stabilizer)**.** For any point $x \in \mathcal{B}(G)$, its stabilizer group $G(k)^1_x$ taken in $G(k)^1$ is an open compact subgroup.[8]

Definition 3.7. (1) For a facet $\mathcal{F} \in \mathcal{B}(G)$, we call its stabilizer group $G(k)^0_{\mathcal{F}}$ taken in $G(k)^0$ the *parahoric subgroup associated to $\mathcal{F}$*.
(2) We call a parahoric subgroup associated to a chamber an *Iwahori subgroup*.

The definition of the group $G(k)^1$ was already given last week. Recall that

$$G(k)^1 := \{g \in G(k) \mid \omega(\chi(g)) = 0 \text{ for all } \chi \in X^*(G)\}.$$

[8]In [KP23, Axiom 4.1.2], *boundedness* is required instead of compactness. But these are equivalent by assuming that k is a local field; see [KP23, Fact 2.2.3].

On the other hand, $G(k)^0$ is defined to be the kernel of the *Kottwitz homomorphism*

$$\kappa_G\colon G(k) \to (\pi_1(G)_I)^{\mathrm{Gal}(K/k)},$$

where I is the inertia subgroup $\mathrm{Gal}(k^{\mathrm{sep}}/K)$. Since we have not introduced yet the definition/construction of κ_G, we next try to understand it.

3.3. **Algebraic fundamental groups.** Here, temporarily, we let k be any field and G be a connected reductive group over k.

Let T be a k-rational maximal torus of G. Let $R^\vee$ be the set of coroots of G with respect to T and $Q^\vee(T) \subset X_*(T)$ be the coroot lattice, i.e., the subgroup of $X_*(T)$ generated by $R^\vee$. Since T is k-rational, $Q^\vee(T) \subset X_*(T)$ are free abelian group of finite rank equipped with an action of $\mathrm{Gal}(k^{\mathrm{sep}}/k)$. Thus, taking the quotient, we get a finitely generated abelian group $X_*(T)/Q^\vee(T)$ equipped with an action of $\mathrm{Gal}(k^{\mathrm{sep}}/k)$.

Definition 3.8. We define the *algebraic fundamental group* $\pi_1(G)$ of G by

$$\pi_1(G) := X_*(T)/Q^\vee(T).$$

Note that the structure of the $\mathrm{Gal}(k^{\mathrm{sep}}/k)$-module $X_*(T)$ heavily depend on the choice of T. However, we have the following:

Lemma 3.9. *The algebraic fundamental group $\pi_1(G)$ is independent of the choice of a k-rational maximal torus T.*

Proof. For convenience, let us write $\pi_1(G,T)$ instead of $\pi_1(G)$ in this proof; so our aim is to show that $\pi_1(G,T)$ is actually independent of the choice of T.

Let us take another k-rational maximal torus T' of G. Since any two maximal tori of G are conjugate by an element of $G(k^{\mathrm{sep}})$, we can find $g \in G(k^{\mathrm{sep}})$ such that $T' = gTg^{-1}$. In particular, the g-conjugation induces an isomorphism $[g]\colon T_{k^{\mathrm{sep}}} \to T'_{k^{\mathrm{sep}}}$. (Be careful that this isomorphism may not be defined over k.) Since this isomorphism preserves the root system, it induces an isomorphism $[g]\colon \pi_1(G,T) \to \pi_1(G,T')$ which also preserves the coroot lattice.

The point of the proof is that this isomorphism at the level of the quotient is independent of the choice of $g \in G(k^{\mathrm{sep}})$. Indeed, let us take another $g' \in G(k^{\mathrm{sep}})$ such that $T' = g'Tg'^{-1}$. Then we have $gTg^{-1} = T' = g'Tg'^{-1}$, hence $g^{-1}g' \in N_G(T)$. It suffices to check that the conjugation isomorphism $[g^{-1}g']\colon T \to T$ induces the indentity on $\pi(G,T)$. The point is that $g^{-1}g' \in N_G(T)$, hence its action is a Weyl action (element of the Weyl group, which is generated by reflections alog roots). The reflection of $X_*(T)$ along a root $a \in X^*(T)$ is given by adding an integer multiple of the coroot $a^\vee \in X_*(T)$, hence it must be trivial after taking the quotient by $X_*(T_{\mathrm{sc}})$.

Now let us complete the proof; our goal is to show that $[g]\colon \pi_1(G,T) \to \pi_1(G,T')$ is $\mathrm{Gal}(k^{\mathrm{sep}}/k)$-equivariant. Let us first recall how $\mathrm{Gal}(k^{\mathrm{sep}}/k)$ acts on $X_*(T)$. By definition, we have $X_*(T) = \mathrm{Hom}_{k^{\mathrm{sep}}}(\mathbb{G}_{\mathrm{m}}, T_{k^{\mathrm{sep}}})$. For any $\chi^\vee \in X_*(T)$, we define $\sigma(\chi^\vee) := \sigma \circ \chi^\vee \circ \sigma^{-1}$, where σ and σ^{-1} denote the automorphisms of $\mathbb{G}_{\mathrm{m}}$ and $T_{k^{\mathrm{sep}}}$ obtained by pulling back $\sigma, \sigma^{-1} \in \mathrm{Gal}(k^{\mathrm{sep}}/k)$, respectively.

Let $\chi^\vee$ be an element of $X_*(T)$. Our task is to show that $[g]\circ(\sigma(\chi^\vee)) = \sigma([g]\circ\chi^\vee)$ modulo $Q^\vee(T)$ and $Q^\vee(T')$. By the definition of the Galois action, this is equivalent to that $[g]\circ\sigma\circ\chi^\vee\circ\sigma^{-1} = \sigma\circ[g]\circ\chi^\vee\circ\sigma^{-1}$ modulo $Q^\vee(T)$ and $Q^\vee(T')$. Note that this is furthermore equivalent to that $\sigma^{-1}\circ[g]\circ\sigma = [g]$ on $\pi(G,T)$. Since $\sigma^{-1}\circ[g]\circ\sigma =$

$[\sigma^{-1}(g)]$, the desired equality ($[\sigma^{-1}(g)] = [g]$) is now just a consequence of the fact that $[g]\colon \pi(G,T) \to \pi(G,T)$ is independent of the choice of g. $\square$

Remark 3.10. (1) When $k = \mathbb{C}$, the algebraic fundamental group $\pi_1(G)$ coincides with the topological fundamental group of $G(\mathbb{C})$.
(2) To avoid choosing a k-rational maximal torus, we sometimes (often) think of $\pi_1(G)$ as the limit (or colimit) of $X_*(T)/Q^\vee(T)$.
(3) Let G_{sc} be the simply-connected cover of the derived group G_{der} of G. Thus, we have a natural homomorphism $G_{\mathrm{sc}} \to G$. If we take a k-rational maximal torus $T \subset G$, then its preimage T_{sc} in G_{sc} is a k-raional maximal torus of G_{sc}. Note that $Q^\vee(T)$ is nothing but the image of $X_*(T_{\mathrm{sc}})$ under the homomorphism $X_*(T_{\mathrm{sc}}) \to X_*(T)$.

The following are immediate consequences of the definition of $\pi_1(G)$:

Lemma 3.11. (1) *The assignment $G \mapsto \pi_1(G)$ defines a functor from the category of connected reductive groups over k to the category of finitely generated abelian groups equipped with an action of* $\mathrm{Gal}(k^{\mathrm{sep}}/k)$.
(2) *The algebraic fundamental group $\pi_1(G)$ is trivial if and only if G is semisimple and simply-connected.*
(3) *If $G = T$ is a torus, then $\pi_1(G) = X_*(T)$.*

Exercise 3.12. Compute $\pi_1(G)$ in the following cases:
(1) $G = \mathrm{GL}_n$.
(2) $G = \mathrm{Sp}_{2n}$.
(3) $G = \mathrm{SO}_n$.

(You can refer to any textbook on algebraic groups which explains the root data of these groups.)

3.4. **Group cohomology.** We quickly recall basics of group cohomology (see [Ser79, §VII] and also [Har77, §III.1] for more details.)

Let G be a group. We consider the category $\mathcal{C}$ of abelian groups equipped with an action of G (or, equivalently, the category of $\mathbb{Z}[G]$-modules). Then it can be proved that $\mathcal{C}$ is an abelian category which is enough injective. More precisely, for any $A \in \mathcal{C}$, we can find an exact sequence

$$0 \to A \to I^0 \to I^1 \to I^2 \to \cdots,$$

where each $I^i \in \mathcal{C}$ is an injective object, i.e., the functor $\mathrm{Hom}_{\mathcal{C}}(-, I^i)\colon \mathcal{C} \to \mathrm{Ab}$ is exact (Ab denotes the category of abelian groups). Such an exact sequence is called an injective resolution of A. We consider the functor of G-invariants

$$(-)^G\colon \mathcal{C} \to \mathrm{Ab}; \quad A \mapsto A^G,$$

which is left-exact.

We define the i-th group cohomology $H^i(G, A)$ of $A \in \mathcal{C}$ as follows. We first choose an injective resolution $0 \to A \to I^0 \to I^1 \to \cdots$ of A and apply the functor $(-)^G$ to get a sequence

$$0 \to (I^0)^G \xrightarrow{d_0} (I^1)^G \xrightarrow{d_1} (I^2)^G \xrightarrow{d_2} \cdots.$$

Note that this sequence is not exact in general since $(-)^G$ is not an exact funcor. We define

$$H^i(G, A) := \mathrm{Ker}(d_i)/\mathrm{Im}(d_{i-1})$$

for $i \geq 0$ (we put $d_{-1} := 0$). Note that, as $(-)^G$ is left-exact, the sequence

$$0 \to A^G \to (I^0)^G \xrightarrow{d_0} (I^1)^G$$

is exact. Hence

$$H^0(G, A) = \mathrm{Ker}(d_0) \cong A^G.$$

Although this construction uses a fixed injective resolution of A, which is not unique, the resulting $H^i(G, A)$ is independent of the choice of an injective resolution.

Note that $H^i(G, A)$ is functorial in $A \in \mathcal{C}$, i.e., if we have a G-equivariant homomorphism $A \to B$, then it naturally induces a homomorphism $H^i(G, A) \to H^i(G, B)$ for each $i \geq 0$. Furthermore, $H^i(G, -)$ associates a long exact sequence to any short exact sequence. To be more precise, let

$$0 \to A \to B \to C \to 0$$

be any exact sequence in $\mathcal{C}$. Then we have a natural homomorphism $H^i(G, C) \to H^{i+1}(G, A)$ for each $i \geq 0$ (called the *connecting homomorphism*) such that

$$\begin{aligned} 0 &\to H^0(G, A) \to H^0(G, B) \to H^0(G, C) \\ &\to H^1(G, A) \to H^1(G, B) \to H^1(G, C) \\ &\to H^2(G, A) \to H^2(G, B) \to H^2(G, C) \to \cdots \end{aligned}$$

is exact.

All of these are part of a general theory of *(right) derived functors* for left-exact functors on abelian categories.

3.5. **Interpretation via cochains.** The above definition of the group cohomology is somewhat too abstract. But, in fact, there is a practical way to compute the group cohomology concretely. The idea is that the group cohomology $H^i(G, A)$ can be also thought of as $\mathrm{Ext}^i_{\mathcal{C}}(\mathbb{Z}, A)$, where $\mathrm{Ext}^i_{\mathcal{C}}(-, A)$ is the "left" derived functor of the right-exact functor $\mathrm{Hom}_{\mathcal{C}}(-, A)$. By choosing a "standard" projective resolution of $\mathbb{Z}$, we can identify $H^i(G, A)$ with a more concrete group as follows.

For $i \geq 0$, we let K^i be the set of maps from $G^i = \underbrace{G \times \cdots \times G}_{i}$ to A. Here, we let $G^0 := \{1\}$, hence $K^0 \cong A$. We define a map $d\colon K^i \to K^{i+1}$ by, for any $f \in K^i$,

$$\begin{aligned} df(g_1, \ldots, g_{i+1}) &:= g_1 \cdot f(g_2, \ldots, g_{i+1}) \\ &\quad + \sum_{j=0}^{i+1} (-1)^j f(g_1, \ldots, g_{j-1}, g_j g_{j+1}, g_{j+2}, \ldots g_{i+1}) + (-1)^{i+1} f(g_1, \ldots, g_i). \end{aligned}$$

Then we have

$$H^i(G, A) \cong \mathrm{Ker}(d\colon K^i \to K^{i+1})/\mathrm{Im}(d\colon K^{i-1} \to K^i),$$

where we let the denominator of the righ-hand side be simply zero when $i = 0$. Elements of K^i, $\mathrm{Ker}(d\colon K^i \to K^{i+1})$, and $\mathrm{Im}(d\colon K^{i-1} \to K^i)$ are referred to as *i-cochains*, *i-cocycles*, and *i-cobundaries*, respectively. If i-cochains are in the same class in the quotient by i-cobundaries, we say that they are *cohomologous*.

3.5.1. $H^0(G, A)$. When $i = 0$, as remarked above, any 0-cochain is identified with an element $a \in A$. By definition, $da \in K^1$ is given by

$$da(g_1) = g_1 \cdot a - a.$$

Hence, a is a 0-cocycle if and only if $g_1 \cdot a - a = 0$ for any $g_1 \in G$, i.e., $a \in A^G$. This means that $H^0(G, A) \cong A^G$ as expected.

3.5.2. $H^1(G, A)$. Let us take a 1-cochain $f \in K^1$, i.e., $f\colon G \to A$. By definition, $df \in K^2$ is given by

$$df(g_1, g_2) = g_1 \cdot f(g_2) - f(g_1 \cdot g_2) + f(g_1).$$

Hence, f is a 1-cocylce if and only if $g_1 \cdot f(g_2) - f(g_1 \cdot g_2) + f(g_1)$ for any $g_1, g_2 \in G$. In other words,

$$f(g_1 \cdot g_2) = f(g_1) + g_1 \cdot f(g_2).$$

This condition is often referred to as the 1-*cocycle condition*. We see that $H^1(G, A)$ is the quotient of such cocycles by 1-coboundaries described above.

3.5.3. $H^2(G, A)$. Let us take a 2-cochain $f \in K^2$, i.e., $f\colon G^2 \to A$. By definition, $df \in K^3$ is given by

$$df(g_1, g_2, g_3) = g_1 \cdot f(g_2, g_3) - f(g_1 \cdot g_2, g_3) + f(g_1, g_2 \cdot g_3) - f(g_1, g_2).$$

Hence, f is a 2-cocylce if and only if

$$g_1 \cdot f(g_2, g_3) - f(g_1 \cdot g_2, g_3) + f(g_1, g_2 \cdot g_3) - f(g_1, g_2) = 0$$

for any $g_1, g_2, g_3 \in G$.

3.6. **Non-abelian group cohomology.** We next consider the case where A is a group (not necessarily abelian) equipped with an action of a group G. In this case, since the category of such A's is not abelian, at least we cannot appeal to theory of derived functors to define the group cohomology $H^i(G, A)$. However, at least the interpretation in terms of cochains still makes sense up to the first degree. Keeping this in mind, we put $H^0(G, A) := A^G$ and define $H^1(G, A)$ as follows. Again, we call functions $G \to A$ 1-*cochains*. We say that a 1-cochain $z_{(-)}\colon G \to A$ is a 1-*cocycle* if it satisfies

$$z_{g_1 \cdot g_2} = z_{g_1} \cdot g_2(z_{g_1})$$

for any $g_1, g_2 \in G$. Here note that we write the equality via the multiplicative notation since G is supposed to be non-abelian. We say that two 1-cocycles $z_{(-)}$ and $z'_{(-)}$ are *cohomologous* if there exists an element $a \in A$ satisfying

$$z'_g = a^{-1} \cdot z_g \cdot g(a)$$

for any $g \in G$. We define $H^1(G, A)$ to be the set of cohomologous classes of 1-cocycles $G \to A$.

In contrast to the case where A is abelian, we can define $H^i(G, A)$ only for $i = 0, 1$ in this way. Furthermore, $H^1(G, A)$ does not have a group structure in general (but it at least has a trivial cohomology class, so has a structure of a pointed set). However, $H^i(G, -)$ still associates a partial long exact sequence to any short exact sequence. More precisely, for any exact sequence

$$1 \to A \to B \to C \to 1$$

of groups equipped with G-actions, we get an exact sequence (as pointed sets)

$$1 \to H^0(G, A) \to H^0(G, B) \to H^0(G, C) \to H^1(G, A) \to H^1(G, B) \to H^1(G, C).$$

Remark 3.13. When A is abelian and maps into the center of B, we can furthermore put $\to H^2(G, A)$ at the end of the above sequence.

For the later application, let us also introduce the *(infinite) Galois cohomology*. For any group $A(k^{\mathrm{sep}})$ on which $\mathrm{Gal}(k^{\mathrm{sep}}/k)$ acts smoothly (i.e., continuous with respect to the discrete topology of $A(k^{\mathrm{sep}})$), we define $H^0(k, A(k^{\mathrm{sep}})) := A(k) := A(k^{\mathrm{sep}})^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}$ and

$$H^1(k, A(k^{\mathrm{sep}})) := \varinjlim_{k'/k} H^1(\mathrm{Gal}(k'/k), A(k')),$$

where the injective limit is over all finite Galois extensions k' of k and $A(k') := A(k^{\mathrm{sep}})^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}$. The group $A(k^{\mathrm{sep}})$ is supposed to be taken to be $G = G(k^{\mathrm{sep}})$ for a connected reductive group G over k. (Or, we can also define $H^1(k, A(k^{\mathrm{sep}}))$ directly by using smooth (continuous with respect to the discrete topology on $A(k^{\mathrm{sep}})$) 1-cocycles $\Gamma_k \to A(k^{\mathrm{sep}})$.)

Some important results on the Galois cohomology of connected reductive groups are as follows.

Fact 3.14 (Hilbert 90th theorem)**.** *Let k be a field. Then we have $H^1(k, \mathrm{GL}_n) = \{1\}$*

Fact 3.15 (Lang's theorem)**.** *Let k be a finite field. Let G be a connected algebraic group over k. Then we have $H^1(k, G) = \{1\}$.*

Fact 3.16 (Kneser's theorem)**.** *Let k be a non-archimedean local field. Let G be a connected simply-connected semisimple group. Then we have $H^1(k, G) = \{1\}$.*

Fact 3.17 (Kottwitz isomorphism)**.** *Let k be a non-archimedean local field. Let G be a connected reductive group over k. Let $\hat{G}$ be the Langlands dual group of G over $\mathbb{C}$ (i.e., the root datum of $\hat{G}$ is the dual of that of G) equipped with a Galois action naturally induced from the k-rational structure of G. Then we have a natural isomorphism*

$$H^1(k, G) \cong \pi_0(Z(\hat{G})^{\mathrm{Gal}(k^{\mathrm{sep}}/k)})^{\vee}.$$

3.7. **The group $G(k)^\natural$.** Let G be a connected reductive group over k. We let G_{der} be the derived subgroup (commutator subgroup) of G and G_{sc} its simply-connected cover. Hence we have natural homomorphisms of algebraic groups $G_{\mathrm{sc}} \twoheadrightarrow G_{\mathrm{der}} \hookrightarrow G$. By taking k-rational points, we get a homomorphism $G_{\mathrm{sc}}(k) \to G(k)$.

Definition 3.18. Let $G(k)^\natural \subset G(k)$ be the image of the natural homomorphism $G_{\mathrm{sc}}(k) \to G(k)$.

Remark 3.19. The homomorphism $G_{\mathrm{sc}} \twoheadrightarrow G_{\mathrm{der}}$ is surjective as an algebraic group homomorphism. Note that this does not necessarily implies that $G_{\mathrm{sc}}(k) \to G_{\mathrm{der}}(k)$ is surjective. This discrepancy can be measured by using the Galois cohomology. If we let Z be the kernel of $G_{\mathrm{sc}} \twoheadrightarrow G_{\mathrm{der}}$, then we get a short exact sequence

$$1 \to Z \to G_{\mathrm{sc}} \to G_{\mathrm{der}} \to 1.$$

By taking the associated long exact sequence, we obtain

$$1 \to Z(k) \to G_{\mathrm{sc}}(k) \to G_{\mathrm{der}}(k) \to H^1(k, Z) \to H^1(k, G_{\mathrm{sc}}).$$

In fact, it is known that the Galois cohomology of any simply-connected semisimple group over a non-archimedean local field vanishes (*Kneser's theorem*). Hence $H^1(k, Z)$ coincides with the cokernel of $G_{\mathrm{sc}}(k) \to G_{\mathrm{der}}(k)$.

Exercise 3.20. Compute the order of $G_{\mathrm{der}}(k)/G_{\mathrm{sc}}(k)$ in the following cases:

(1) $G = \mathrm{GL}_n$ (Hint: you can use Hilbert's 90th theorem).
(2) $G = \mathrm{Sp}_{2n}$ (Hint: Sp_{2n} is simply-connected).
(3) $G = \mathrm{SO}_n$ (Hint: the simply-connected cover of SO_n is Spin_n, which is a double cover).

4. Week 4: Kottwitz homomorphisms

Let k be a non-archimedean local field. In the following, we write K for the maximal unramified extension of k (taken in k^{sep}). For any algebraic extension l of k, we let Θ_l denote the absolute Galois group $\mathrm{Gal}(k^{\mathrm{sep}}/l)$ of l. We write I_k $(=: I)$ for the inertia subgroup of Θ_k, hence $I_k = \mathrm{Gal}(k^{\mathrm{sep}}/K)$.

Our aim of this week is to, for a connected reductive group G over k, define the Kottwitz homomorphism

$$\kappa_G \colon G(k) \to (\pi_1(G)_I)^{\mathrm{Gal}(K/k)},$$

which is crucial for the definition of parahoric subgroups.

The idea of the construction of the Kottwitz homomorphism is to reduce to the case of tori. So we will start with investigating tori.

4.1. Norm morphisms for tori.

Definition 4.1. Let T be a k-rational torus, where k denotes temporarily any field. Let l be any finite Galois extension of k. We define the *norm map*

$$\mathrm{Nr}_{l/k} \colon T(l) \to T(k)$$

by

$$\mathrm{Nr}_{l/k}(t) := \prod_{\sigma \in \mathrm{Gal}(L/K)} \sigma(t).$$

The norm map of a k-rational torus can be also defined algebraically as follows. Let T_l be the base change of T to l (hence T_l is an l-rational torus) and $\mathrm{Res}_{l/k} T_l$ the Weil restriction of T_l from l to k. Note that we have

$$(\mathrm{Res}_{l/k} T_l) \times_k k^{\mathrm{sep}} \cong \prod_{\sigma \in \mathrm{Gal}(l/k)} T_{k^{\mathrm{sep}}}.$$

At the level of k^{sep}-rational points, this morphism is described as follows:

$$\begin{aligned}((\mathrm{Res}_{l/k} T_l) \times_k k^{\mathrm{sep}})(k^{\mathrm{sep}}) &\cong (\mathrm{Res}_{l/k} T_l)(k^{\mathrm{sep}}) \\ &\cong T_l(k^{\mathrm{sep}} \otimes_k l) \\ &\cong T_l\Big(\prod_{\sigma \in \mathrm{Gal}(l/k)} k^{\mathrm{sep}}\Big) \cong \prod_{\sigma \in \mathrm{Gal}(l/k)} T_{k^{\mathrm{sep}}}(k^{\mathrm{sep}}).\end{aligned}$$

Then the norm map $\mathrm{Nr}_{l/k}$ can be realized as the morphism of k-tori just by

$$\mathrm{Nr}_{l/k} \colon (\mathrm{Res}_{l/k} T_l) \times_k k^{\mathrm{sep}} \cong \prod_{\sigma \in \mathrm{Gal}(l/k)} T_{k^{\mathrm{sep}}} \to T_{k^{\mathrm{sep}}}; \quad (t_\sigma)_{\sigma \in \mathrm{Gal}(l/k)} \mapsto \prod_{\sigma \in \mathrm{Gal}(l/k)} t_\sigma.$$

(This morphism is $\mathrm{Gal}(k^{\mathrm{sep}}/k)$-invariant, hence descends to a morphism of k-tori $\mathrm{Res}_{l/k} T_l \to T$.)

By this description, we see that the kernel of the norm morphism $\mathrm{Ker}(\mathrm{Nr}_{l/k})$ is a k-rational subtorus of $\mathrm{Res}_{l/k} T_l$. By taking the Galois cohomology of the short exact sequence

$$1 \to \mathrm{Ker}(\mathrm{Nr}_{l/k}) \to \mathrm{Res}_{l/k} T_l \xrightarrow{\mathrm{Nr}_{l/k}} T \to 1,$$

we get a long exact sequence

$$1 \to \mathrm{Ker}(\mathrm{Nr}_{l/k})(k) \to T(l) \xrightarrow{\mathrm{Nr}_{l/k}} T(k) \to H^1(k, \mathrm{Ker}(\mathrm{Nr}_{l/k})).$$

In particular, note that

- the norm map is not necessarily surjective, and
- the cokernel is given by $H^1(k, \mathrm{Ker}(\mathrm{Nr}_{l/k}))$.

To investigate this quantity $H^1(k, \mathrm{Ker}(\mathrm{Nr}_{l/k}))$, we review a bit more about the Galois cohomology.

Definition 4.2. Let k be a field. We write "$\dim(k) \le 1$" if, for any algebraic extension l and its finite Galois extension l', the norm map $\mathrm{Nr}_{l'/l} \colon l'^{\times} \to l^{\times}$ is surjective.

Example 4.3. Any finite field k clrealy satisfies $\dim(k) \le 1$.

Fact 4.4. *The maximal unramified extension K of a non-archimedean local field k satisfies* $\dim(K) \le 1$.

Fact 4.5. *Let k be a field satisfying* $\dim(k) \le 1$.

(1) *For any connected reductive group G over k, we have $H^1(k, G) = \{1\}$.*
(2) *Any connected reductive group over k is quasi-split.*

Lemma 4.6. *If T is a K-rational torus, where K is the maximal unramified extension of a non-archimedean local field k, the norm map $\mathrm{Nr}_{L/K} \colon T(L) \to T(K)$ is continuous and open.*

Proof. By Fact 4.5 (1), we have $H^1(K, \mathrm{Ker}(\mathrm{Nr}_{L/K})) = \{1\}$, which implies that $\mathrm{Nr}_{L/K} \colon T(L) \to T(K)$ is surjective. Continuity and openness of the norm map follows from that the norm morphism is smooth (after applying the implicit function theorem, $\mathrm{Nr}_{l/k}$ can be thought of as a projection locally, which is obviously open and continuous.) $\square$

Definition 4.7. Let T be a k-rational torus. We say that T is an *induced torus* if $X^*(T)$, or equivalently, $X_*(T)$, has a $\mathbb{Z}$-basis which is invariant (stable as a finite set) under the action of $\mathrm{Gal}(k^{\mathrm{sep}}/k)$.

Example 4.8. We let $T := \mathrm{Res}_{l/k} \mathbb{G}_{\mathrm{m}}$, where l/k is a finite separable extension. Then T is an induced torus. More precisely, we have

$$X^*(T) = \mathrm{Ind}_{\mathrm{Gal}(k^{\mathrm{sep}}/l)}^{\mathrm{Gal}(k^{\mathrm{sep}}/k)} X^*(\mathbb{G}_{\mathrm{m}}) \quad \text{and} \quad X_*(T) = \mathrm{Ind}_{\mathrm{Gal}(k^{\mathrm{sep}}/l)}^{\mathrm{Gal}(k^{\mathrm{sep}}/k)} X_*(\mathbb{G}_{\mathrm{m}}),$$

(note that $X^*(\mathbb{G}_{\mathrm{m}})$ and $X_*(\mathbb{G}_{\mathrm{m}})$ is just $\mathbb{Z}$; free of rank 1 with trivial Galois action). It is not difficult to see that we can choose a $\mathbb{Z}$-basis of this module satisfying the condition above.

Fact 4.9. *Any induced torus is the product of tori of the form as in the above example.*

The following fact is a consequence of so-called Shapiro's lemma and Hilbert 90.

Fact 4.10. *Let T be an induced k-torus. Then $H^1(k, T) = \{1\}$.*

Exercise 4.11. Let k be a non-archimedean local field and l a quadratic extension of k. We consider the norm morphism $\mathrm{Nr}_{l/k} \colon \mathrm{Res}_{l/k} \mathbb{G}_{\mathrm{m}} \to \mathbb{G}_{\mathrm{m}}$ and let T be its kernel. Compute $H^1(k, T)$.

4.2. **Kottwitz homomorphism for tori.**

Definition 4.12. For a k-torus T, we define the *valuation homomorphism* ω_T by

$$\omega_T \colon T(k) \to \mathrm{Hom}_{\mathbb{Z}}(X^*(T)^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}, \mathbb{Z}); \quad t \mapsto [\chi \mapsto -\omega(\chi(t))].$$

Note that we have

$$\begin{aligned}\mathrm{Hom}_{\mathbb{Z}}(X^*(T)^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}, \mathbb{Z})_{\mathbb{Q}} &\cong \mathrm{Hom}_{\mathbb{Z}}(X^*(T)^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}, \mathbb{Q})\\ &\cong \mathrm{Hom}_{\mathbb{Z}}(X^*(T), \mathbb{Q})^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}\\ &\cong (X_*(T)_{\mathbb{Q}})^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}.\end{aligned}$$

Proposition 4.13. *For any K-torus T, there exists a unique homomorphism*

$$\kappa_T \colon T(K) \to X_*(T)_I$$

such that, for any finite Galois extension L/K splitting T, the following diagram commutes:

$$\begin{tikzcd} T(L) \arrow[r, "\omega_{T_L}"] \arrow[d] & X_*(T) \arrow[d] \\ T(K) \arrow[r, "\kappa_T"] & X_*(T) \end{tikzcd}$$

Here, the right vertical arrow is the natural projection. (The top horizontal map is define by using the normalized valuation of L.) Furthermore, if T is k-rational, then κ_T is $\mathrm{Gal}(K/k)$*-equivariant.*

Proof. We first show the existence of a homomorphism κ_T making the diagram commutative for one particular finite Galois extension L/K splitting T; so let us fix L/K. Since the homomorphism $T(L) \xrightarrow{\omega_{T_L}} X_*(T) \to X_*(T)_I$ is I-invariant, it factors through the natural quotient map to the I-coinvariant $T(L)_I$. In fact, the homomorphism $T(L) \to T(L)_I$ can be naturally identified with the norm map $T(L) \to T(K)$; this follows from that $\dim(K) \leq 1$.

Now suppose that L'/K is another finite Galois extension splitting T. Our task is to check that the homomorphism κ_T defined in the previous paragraph does not change even if we replace L with L'. To show this, by replacing L' with the composition field of L and L', we may assume that $L' \supset L$. Then it suffices to show that the diagram

$$\begin{tikzcd} T(L') \arrow[r, "\omega_{T_{L'}}"] \arrow[d, "\mathrm{Nr}_{L'/L}"'] & X_*(T)_I \arrow[d] \\ T(L) \arrow[r, "\omega_{T_L}"] & X_*(T)_I \end{tikzcd}$$

Since T splits over L, we may suppose that $T = \mathbb{G}_{\mathrm{m}}$. Let $\chi \in X^*(T)$ be the standard basis $\chi \colon \mathbb{G}_{\mathrm{m}} \to \mathbb{G}_{\mathrm{m}}; t \mapsto t$. Let $t \in T(L') = (L')^\times$. Then we have

$$\omega_{T_{L'}}(t)(\chi) = -\omega_{L'}(\chi(t)) = -\omega_{L'}(t).$$

On the other hand,

$$\omega_{T_L}(\mathrm{Nr}_{L'/L}(t))(\chi) = -\omega_L(\chi(\mathrm{Nr}_{L'/L}(t))) = -\omega_L(\mathrm{Nr}_{L'/L}(t)).$$

These are equal since ω_L and $\omega_{L'}$ are the normalized valuations and K is the maximal unramified extension of k, hence L'/L is necessarily totally ramified. $\square$

Let T be a k-torus. We define the Kottwitz homomorphism κ_T for T by restricting κ_{T_K} from $T(K)$ to $T(k)$:

$$\kappa_T \colon T(k) \to (X_*(T)_I)^{\mathrm{Gal}(K/k)}.$$

4.3. **z-extensions.** Here, again we let k be a general field.

Definition 4.14. Let G be a connected reductive group over k. A *z-extension* of G is an extension

$$1 \to Z \to \tilde{G} \to G \to 1$$

such that

- Z is a k-rational induced torus which is central in $\tilde{G}$, and
- $\tilde{G}$ is a connected reductive group over k whose derived group $\tilde{G}_{\mathrm{der}}$ is simply-connected.

Proposition 4.15. *Any connected reductive group G defined over k admits a z-extension.*

Proof. Let G_{sc} be the simply-connected cover of the derived group G_{der} of G. Let $Z(G)^\circ$ be the identity component of the center $Z(G)$ of G. Then we have a natural product homomorphism $G_{\mathrm{sc}} \times Z(G)^\circ \to G$, which is surjective and has finite kernel. We write μ for the kernel of this homomorphism.

Let l/k be a finite Galois extension splitting G (hence G_{sc} and $Z(G)^\circ$ also split over l). Then $X^*(\mu)$ is a finite $\mathbb{Z}[\mathrm{Gal}(l/k)]$-module. We take a free $\mathbb{Z}[\mathrm{Gal}(l/k)]$-module M of finite rank equipped with a $\mathrm{Gal}(l/k)$-equivariant surjection $M \to X^*(\mu)$. We let Z be a k-rational torus such that $X^*(Z) \cong M$ as $\mathrm{Gal}(l/k)$-modules. Note that such a torus always exists since we have a natural categorical equivalent $T \mapsto X^*(T)$ between the category of k-rational tori (more generally, diagonalizable groups) and the category of free abelian groups of finite rank (more generally, finitely generated abelian groups) equipped with an action of $\mathrm{Gal}(k^{\mathrm{sep}}/k)$. Moreover, since we can choose a $\mathbb{Z}$-basis of M which is stable under $\mathrm{Gal}(k^{\mathrm{sep}}/k)$ (take a $\mathbb{Z}[\mathrm{Gal}(l/k)]$-basis $e_1, \ldots, e_r$ of M and choose $\{\sigma(e_i) \mid \sigma \in \mathrm{Gal}(k^{\mathrm{sep}}/k), i = 1, \ldots, r\}$), Z is an induced torus. Also, we have a k-rational homomorphism $\mu \to Z$ which induces $X^*(Z) \cong M \to X^*(\mu)$.

We define $\tilde{G}$ to be the pushout of

$$1 \to \mu \to G_{\mathrm{sc}} \times Z(G)^\circ \to G \to 1$$

by $\mu \to Z$. More concretely, we put

$$\tilde{G} := (Z \times (G_{\mathrm{sc}} \times Z(G)^\circ))/\mu,$$

where μ is embedded into $Z \times (G_{\mathrm{sc}} \times Z(G)^\circ)$ by $z \mapsto (z^{-1}, z)$. Then the resulting extension

$$1 \to Z \to \tilde{G} \to G \to 1$$

satisfies all the desired properties. $\square$

Note that any z-extension

$$1 \to Z \to \tilde{G} \to G \to 1$$

induces the short exact sequence

$$1 \to Z(k) \to \tilde{G}(k) \to G(k) \to 1$$

since H^1 of an induced torus is trivial.

4.4. **Kottwitz homomorphisms for general G.** We next consider the case where the derived subgroup G_{der} of G is simply-connected. Let $D := G/G_{\mathrm{der}}$ (note that this is a maximal torus quotient of G). Then the functoriality of π_1 induces a Galois-equivariant homomorphism $\pi_1(G) \to \pi_1(D)$.

Here, recall that $\pi(G) = X_*(T)/Q^\vee(T)$ for a k-rational maximal torus T of G. In general, if we let G_{sc} be the simply-connected cover of the derived group G_{der} of G and T_{sc} the preimage of T in G_{sc} (this is a k-raional maximal torus of G_{sc}), then $Q^\vee(T)$ is nothing but the image of $X_*(T_{\mathrm{sc}})$ under the homomorphism $X_*(T_{\mathrm{sc}}) \to X_*(T)$. Hence, when $G_{\mathrm{der}} = G_{\mathrm{sc}}$, we have $\pi_1(G) \cong \pi_1(D)$.

We define

$$\kappa_G \colon G(K) \to \pi_1(G)_I$$

to be the composition

$$G(K) \to D(K) \xrightarrow{\kappa_D} \pi_1(D)_I \cong \pi_1(G)_I,$$

where κ_D is the Kottwitz homomorphism for the torus D.

Now we finally consider the general case where G is a connected reductive group over F. We take a z-extension of G:

$$1 \to Z \to \tilde{G} \to G \to 1.$$

Proposition 4.16. *There exists a unique homomorphism $\kappa_G \colon G(K) \to \pi_1(\tilde{G})_I$ making the following diagram commutative:*

$$\begin{tikzcd} \tilde{G}(K) \arrow[r] \arrow[d, "\kappa_{\tilde{G}}"] & G(K) \arrow[d, dashed, "\kappa_{\tilde{G}}"] \\ \pi(\tilde{G})_I \arrow[r] & \pi(G)_I \end{tikzcd}$$

Moreover, κ_G is independent of the choice of a z-extension, surjective, $\mathrm{Gal}(K/k)$-*equivariant, and functorial in G.*

Proof. Let us fix a k-rational maximal torus T of G. We write $\tilde{T}$ for its preimage in $\tilde{G}$, which is also a k-rational maximal torus of $\tilde{G}$. Again recalling that $Q^\vee(T)$ and $Q^\vee(\tilde{T})$ are naturally identified with $X_*(G_{\mathrm{sc}})$ (G_{sc} is nothing but $\tilde{G}_{\mathrm{der}}$ since $\tilde{G}$ is a z-extension of G) and noting that $X_*(T) \cap Q^\vee(\tilde{T}) = \{1\}$, we see that

$$1 \to \pi_1(Z) \to \pi(\tilde{G}) \to \pi_1(G) \to 1$$

is exact. This short exact sequence fits into the following commutative diagram:

$$\begin{tikzcd} 1 \arrow[r] & Z(K) \arrow[r] \arrow[d, "\kappa_Z"] & \tilde{G}(K) \arrow[r] \arrow[d, "\kappa_{\tilde{G}}"] & G(K) \arrow[r] \arrow[d, dashed] & 1 \\ & \pi_1(Z)_I \arrow[r] & \pi_1(\tilde{G})_I \arrow[r] & \pi_1(G)_I \arrow[r] & 1 \end{tikzcd}$$

(First row is exact since $H^1(K, -)$ is trivial.) Here, we can check that the left square commutes by applying the functoriality of the Kottwitz homomorphism for tori to $Z \to D := \tilde{G}/\tilde{G}_{\mathrm{sc}}$.

This formally implies that we have a homomorphism from $G(K) \to \pi_1(G)_I$ making the right square of the above diagram commutative. Since $\tilde{G}(K) \to G(K)$ is surjective, obviously such a homomorphism must be unique. Let κ_G denote this homomorphism. Since $\kappa_{\tilde{G}}$ is surjective (this also follows from that $\tilde{G}(K) \to$

$\tilde{G}/\tilde{G}_{\mathrm{der}}(K)$ is surjective), κ_G is surjective. As κ_Z and $\kappa_{\tilde{G}}$ are $\mathrm{Gal}(K/k)$-equivariant, the uniqueness of κ_G implies that κ_G is also $\mathrm{Gal}(K/k)$-equivariant.

Finally, the functoriality and the independence of a z-extension follows from the following lemmas (see also the exercise below). □

Lemma 4.17 ([KP23, Fact 11.4.2])**.** *Suppose that $\tilde{G}_1 \to G$ and $\tilde{G}_2 \to G$ are z-extensions. Then $\tilde{G}_1 \times_G \tilde{G}_2 \to G$ is also a z-extension.*

Lemma 4.18 ([KP23, Lemma 11.4.3])**.** *Suppose that $\tilde{G} \to G$ and $\tilde{H} \to H$ are z-extensions. For any homomorphism $G \to H$, we can find a z-extension $\tilde{G}' \to G$ which makes the following diagram commutative:*

$$\begin{tikzcd} \tilde{G} \arrow[d] & \tilde{G}' \arrow[l] \arrow[r] \arrow[d] & \tilde{H} \arrow[d] \\ G \arrow[r, equal] & G \arrow[r] & H \end{tikzcd}$$

Exercise 4.19. Prove Lemma 4.17 and complete the proof of Proposition 4.16 (i.e., show the independence of the choice of a z-extension and also the functoriality).

We define

$$\kappa_G \colon G(k) \to (\pi(G)_I)^{\mathrm{Gal}(K/k)}$$

to be the homomorphism induced from κ_G defined above.

Definition 4.20. We let $G(k)^0 \subset G(k)$ denote the kernel of κ_G.

5. Week 5: Apartments and affine root systems

Let k be a non-archimedean local field. Let $\mathfrak{o}$ and $\mathfrak{m}$ be the ring of integers of k and the maximal ideal of $\mathfrak{o}$, respectively. Let $\omega\colon k \to \mathbb{Z} \cup \{\infty\}$ be the normalized valuation of k. We write $\mathfrak{f} := \mathfrak{o}/\mathfrak{m}$.

Let K be the maximal unramified extension of k. Let $\mathcal{O}$ and $\mathcal{M}$ be the ring of integers of K and the maximal ideal of $\mathcal{O}$, respectively. Thus we have $\overline{\mathfrak{f}} = \mathcal{O}/\mathcal{M}$. We again write ω for the unique valuation of K extending ω.

5.1. **Affine structure on apartments.** Let G be a connected reductive group over k.

Before we state the next axiom on the affine structure on the apartments, we make some observations. Let S be a maximal k-split torus of G. Let $Z := Z_G(S)$ and $N := N_G(S)$ be the centralizer and normalizer of S in G, respectively.

Remark 5.1. Be careful that S is NOT a "k-split maximal torus" (a k-split maximal torus can exist only when G is split over k). This implies that Z may not be a maximal torus of G.

Exercise 5.2. Find examples of a connected reductive group G over k such that

1. S is not a maximal torus of G but Z is a maximal torus;
2. S and Z are not maximal tori.

(Hint: In fact, Z is a maximal torus if and only if G is quasi-split. So, choose a non-split but quasi-split group for (1) and a non-quasi-split group for (2).)

Let S' be the unique maximal subtorus of S contained in G_{der}. If we let A_G be the maximal k-split central torus of G, then the homomorphism $S' \to S \to S/A_G$ is an isogeny. This implies that, if we put $V(-) := X_*(-) \otimes_{\mathbb{Z}} \mathbb{R}$, then we have a canonical decomposition $V(S) = V(S') \oplus V(A_G)$. Especially, we have a canonical retraction $V(S) \twoheadrightarrow V(S')$.

Recall that, for a k-torus T, valuation homomorphism ω_T is defined by

$$\omega_T\colon T(k) \to \mathrm{Hom}_{\mathbb{Z}}(X^*(T)^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}, \mathbb{Z}); \quad t \mapsto [\chi \mapsto -\omega(\chi(t))].$$

We extend this definition to any connected reductive group H over k simply by

$$\omega_H\colon H(k) \to \mathrm{Hom}_{\mathbb{Z}}(X^*(H)^{\mathrm{Gal}(k^{\mathrm{sep}}/k)}, \mathbb{Z}); \quad t \mapsto [\chi \mapsto -\omega(\chi(t))].$$

Using ω_Z, we define the *translation action* ν of $Z(k)$ on $V_*(S')$ by

$$\nu(z)(v) := v - \omega_Z(z)$$

for any $v \in V_*(S')$ and $z \in Z(k)$, where $\omega_Z(z) \in \mathrm{Hom}_{\mathbb{Z}}(X^*(Z), \mathbb{Z})$ is regarded as an element of $V_*(S')$ through

$$\mathrm{Hom}_{\mathbb{Z}}(X^*(Z), \mathbb{Z}) \subset \mathrm{Hom}_{\mathbb{Z}}(X^*(Z), \mathbb{Z})_{\mathbb{R}} \cong X_*(Z)_{\mathbb{R}} \cong X_*(S)_{\mathbb{R}} = V(S) \twoheadrightarrow V(S').$$

Now recall that the first axiom of Bruhat–Tits theory says that there exists a building $\mathcal{B}(G)$ equipped with a $G(k)$-action and a set of apartments which bijectively and $G(k)$-equivariantly corresponds to the set of maximal k-split tori of G ($\mathcal{A}(S) \leftrightarrow S$).

Axiom 3 (affine structure on apartment)**.** Let S be a maximal k-split torus of G. Then the apartment $\mathcal{A}(S)$ associated to S satisfies the following:

1. $\mathcal{A}(S)$ is an affine space under $V(S')$ (i.e., a set with simply-transitive $V(S')$-action), where S' is as above.

(2) The action of $N(k)$ on $\mathcal{A}(S)$ is given by affine transformations. If we write $f\colon N(k) \to \mathrm{Aff}(\mathcal{A}(S))$ for the resulting homomorphism, then
   (a) its derivative $df(n) \in N(k) \to \mathrm{Aut}_{\mathbb{R}}(V'(S))$ coincides with the natural action of $N(k) \twoheadrightarrow N(k)/Z(k)$ on $V(S') = X_*(S')_{\mathbb{R}}$, and
   (b) the action of $Z(k)$ on $\mathcal{A}(S)$ is given by the translation ν, i.e., $z \in Z(k)$ acts via $-\omega_Z(z)$-translation.
(3) For any $g \in G(k)$, the isomoprhism $g\colon \mathcal{A}(S) \to g\mathcal{A}(S) = \mathcal{A}(gSg^{-1})$ is affine such that its derivative is the natural isomorphism $g\colon V(S') \to V(gS'g^{-1})$.

5.2. **Affine root systems and apartments.** Let us first recall some basic notions related to affine root systems. Let V be a finite-dimensional $\mathbb{R}$-vector space. Let A be an affine space over V. We write A^* for the set of affine functionals on V; note that this has a natural $\mathbb{R}$-vector space structure. An affine root system is a pair (A, Ψ) of an affine space A over V and a set of affine roots $\Psi \subset A^*$ satisfying several conditions. Recall that the derivative $\Phi := \dot{\Psi} \subset V^*$ is a finite root system. In the following, by fixing a $W(\Phi)$-invariant inner product on V^*, we loosely identify V with V^*.

For each affine root $\psi \in \Psi$, we can consider the associated reflection $r_\psi \in \mathrm{Aut}_k(A^*)$. (This was previously written by $r_{\psi,\dot{\psi}^\vee}$.) Recall that the group $W(\Psi) \subset \mathrm{Aut}_k(A^*)$ generated by all such reflections $\{r_\psi \mid \psi \in \Psi\}$ is called the affine Weyl group of Ψ. The affine Weyl group can be also regarded as a subgroup of $\mathrm{Aff}(A)$ in a way such that derivative of r_ψ is $r_{\dot{\psi}} \in W(\Phi)$ and the hyperplane $H_\psi := \{x \in A \mid \psi(x) = 0\}$ is fixed.

Definition 5.3. We let $W(\Psi)^{\mathrm{ext}}$ be the subgroup of $\mathrm{Aff}(A)$ consisting of affine automorphisms of A which preserves $\Psi \subset A^*$ and whose derivative is an element of $W(\Phi)$. We call $W(\Psi)^{\mathrm{ext}}$ the *extended affine Weyl group* of Ψ.

Note that the affine Weyl group $W(\Psi)$ is contained in the extended affine Weyl group $W(\Psi)^{\mathrm{ext}}$.

We next recall a fact on relative root subgroups. Let S be a maximal k-split torus of a connected reductive group G over k. Then we have the associated relative root system $\Phi := \Phi(G, S)$. Each relative root $a \in \Phi$ defines a relative root subgroup U_a of G; by definition, this is a unique smooth closed subgroup of G which is normalized by S such that the weights of S in its Lie algbera are positive integer multiples of a.

Fact 5.4. *For any $u \in U_a(k) \smallsetminus \{1\}$, there exist unique elements $u', u'' \in U_{-a}(k) \smallsetminus \{1\}$ such that $m(u) := u'uu'' \in G(k)$ normalizes S. Furthermore, the action of $m(u)$ on $V(S) = X_*(S)_{\mathbb{R}}$ is given by the reflection along a.*

Axiom 4 (affine root system)**.** Let S be a maximal k-split torus of G. Let $\mathcal{A}(S)$ be the associated apartment and Φ the relative root system of S in G. For any $a \in \Phi$ and $u \in U_a(k) \smallsetminus \{1\}$, we let u', u'', and $m(u) \in N(k)$ be as in Fact 5.4.

(1) The action of $m(u)$ on $\mathcal{A}(S)$ fixes a hyperplane $\mathcal{H}_u$ which is an affine subspace of $\mathcal{A}(S)$ under the vector subspace $\mathrm{Ker}(a) \subset V(S)$, where a is also regarded as a natural homomorphism $V(S) \twoheadrightarrow V(\mathbb{G}_{\mathrm{m}}) \cong \mathbb{R}$ induced by the relative root $a\colon S \to \mathbb{G}_{\mathrm{m}}$.
(2) Let $\psi_a^u\colon \mathcal{A}(S) \to \mathbb{R}$ be the unique affine function whose derivative is a and vanishing locus is $\mathcal{H}_u$. Then $\Psi' := \{\psi_a^u \mid a \in \Phi,\, u \in U_a(k) \smallsetminus \{1\}\}$ forms an affine root system with derivative Φ.

(3) For affine function $\psi \in \mathcal{A}(S)^*$ with derivative $a = \dot{\psi} \in \Phi$, we put

$$U_\psi := \{u \in U_a(k) \smallsetminus \{1\} \mid \psi_a^u \geq \psi\} \cup \{1\},$$

$$U_{\psi+} := \{u \in U_a(k) \smallsetminus \{1\} \mid \psi_a^u > \psi\} \cup \{1\}.$$

Then $\Psi := \{\psi \in \Psi' \mid U_{\psi+} \cdot U_{2a}(k) \subsetneq U_\psi \cdot U_{2a}(k)\}$ forms an affine root system with derivative Φ, where U_{2a} is defined to be $\{1\}$ when $2a \notin \Phi$.

(4) For any $\psi \in \Psi'$, at least one of ψ or 2ψ is in Ψ.

(5) The image of $f\colon N(k) \to \mathrm{Aff}(\mathcal{A}(S))$ contains $W(\Psi)$ and is contained in $W(\Psi)^{\mathrm{ext}}$.
- If G is simply-connected, then the image equals $W(\Psi)$.
- If G is quasi-split adjoint, then the image equals $W(\Psi)^{\mathrm{ext}}$.

(6) The polysimplicial structure on $\mathcal{A}$ given by the affine root hyperplanes of Ψ (see the paragraph below) coincides with the polysimplicial structure induced by $\mathcal{B}(G)$.

We explain how a polysimplicial structure on $\mathcal{A}$ is induced by the affine root system. In general, let us suppose that (A, V) is an affine root system. For each $\psi \in \Psi$, we have the corresponding hyperplane $H_\psi \subset A$. We define the chambers of $\mathcal{A}$ to be the connected components of

$$A \smallsetminus \bigcup_{\psi \in \Psi} H_\psi.$$

Then we define the codimension 1 facets to be the connected components of

$$\bigcup_{\psi \in \Psi} H_\psi \smallsetminus \bigcup_{\substack{\psi, \psi' \in \Psi \\ H_\psi \neq H_{\psi'}}} (H_\psi \cap H_{\psi'}).$$

Repeating this procedure, eventually we arrive at the facets of the smallest dimension (vertices).

5.3. **Tits systems.** We first recall the following basic fact.

Fact 5.5 (Bruhat decomposition)**.** *Let k be an algebraically closed field. Let G be a connected reductive group over k. Let T be a maximal torus of G contained in a Borel subgroup B. Then we have*

$$G = \bigsqcup_{w \in W} BwB,$$

where $W := N_G(T)/T$ is the Weyl group.

On the other hand, we also have the following:

Fact 5.6. *Let $G := \mathrm{GL}_2$ over a non-archimedean local field k. Let T be the diagonal maximal torus of G and N its normalizer gruop in G. Let I be the standard Iwahori subgroup of $G(k)$. We put $\tilde{W} := N(k)/T(\mathfrak{o})$. Then we have*

$$G(k) = \bigsqcup_{w \in \tilde{W}} IwI.$$

Exercise 5.7. Prove Fact 5.6. (Hint: Combine the Cartan decomposition of $\mathrm{GL}_2(k)$ and the Bruhat decomposition of $\mathrm{GL}_2(\mathfrak{f})$.)

Observe that the statements of these two facts are quite parallel although the groups are in different contexts. In fact, these can be treated simultaneously via the framework of "Tits system".

Definition 5.8. A *Tits system* is a tuple (G, B, N, S) consisting of

- a group G,
- subgroups $B \subset G$ and $N \subset G$, and
- a subset S of $N/(B \cap N)$

satisfying the following conditions

(1) The set $B \cup N$ generates G and $T := B \cap N$ is a normal subgroup of N.
(2) All elements of S are of order 2 and S generates the group $W := N/T$ (we call W the *Weyl group* of the Tits system (G, B, N, S)).
(3) For any $s \in S$ and $w \in W$, we have $sBw \subset BwB \cup BswB$.
(4) For any $s \in S$, we have $sBs \neq B$.

When a Tits system (G, B, N, S) furthermore satisfies

(5) $\bigcap_{n \in N} nBn^{-1} = B \cap N$,

we say (G, B, N, S) is *saturated*.

Remark 5.9. A Tits system is also called a *BN-pair*.

Definition 5.10. Let (G, B, N, S) be a Tits system.

(1) For any subset $X \subset S$, let $W_X := \langle X \rangle \subset W$ and $G_X := BW_XB$. We call G_X the *standard parabolic subgroup* (associated to $X \subset S$).
(2) We call any subgroup of G containing a conjugate of a standard parabolic subgroup a *parabolic subgroup* of G.

Proposition 5.11. *Let (G, B, N, S) be a Tits system.*

(1) *We have $G = BWB$. Moreover, the map $w \mapsto BwB$ induces a bijection $W \xrightarrow{1:1} B\backslash G/B$. (We call the resulting equality $G = \bigsqcup_{w \in W} BwB$ the Bruhat decomposition.)*
(2) *Any parabolic subgroup is conjugate to a unique standard parabolic subgroup.*
(3) *For any parabolic subgroup $P \subset G$, we have $N_G(P) = P$.*

Recall that an *abstract building* is a (poly-)simplicial complex $(V, \mathcal{B})$ consisting of a set of vertices V and a set of facets $\mathcal{B}$ satisfying several conditions. We may associate an abstract building to a Tits system (G, B, N, S) in the following manner.

- We define V to be the set of all maximal proper parabolic subgroups of G.
- We define $\mathcal{B}$ to be the set of finite sets $\{P_0, \dots, P_n\}$ of maximal proper parabolic subgroups such that $\bigcap_{i=0}^{n} P_i$ is a parabolic subgroup.

Proposition 5.12 ([KP23, Proposition 1.5.6 (1)])**.** *The pair $(V, \mathcal{B})$ forms a simplicial complex.*

We furthermore introduce the following additional structure on $(V, \mathcal{B})$:

- We define an action of G on the set $\mathcal{B}$ by
$$g \cdot \{P_0, \dots, P_n\} := \{gP_0g^{-1}, \dots, gP_ng^{-1}\}.$$
- Let $\mathcal{C} \in \mathcal{B}$ be the set of all standard maximal proper parabolic subgroups.
- Let $\mathcal{A} \subset \mathcal{B}$ be the union of all N-conjugate of $\mathcal{C}$.

Proposition 5.13 ([KP23, Proposition 1.5.6 (2)–(7)]). *The simplicial complex $(V, \mathcal{B})$ is an abstract building with set of apartments given by $\{g\mathcal{A} \mid g \in G\}$. (We call $\mathcal{A}$ the "standard apartment" of $\mathcal{B}$.) Moreover, the following hold:*

(1) *If we define $P_{\mathcal{F}} := \bigcap_{i=0}^{n} P_i$ for a facet $\mathcal{F} = \{P_0, \ldots, P_n\}$, then the association $\mathcal{F} \mapsto P_{\mathcal{F}}$ gives a G-equivariant, inclusion-reversing, and bijective map between $\mathcal{B}$ and the set of proper parabolic subgroups of G.*
(2) *The set $\mathcal{C}$ is a chamber (we call $\mathcal{C}$ the standard chamber).*
(3) *If an element $g \in G$ fixes a facet, then g fixes all its vertices.*
(4) *For any faces $\mathcal{F}$ and $\mathcal{F}'$ of the standard chamber $\mathcal{C}$ and elements $n \in N$ and $g \in G$ such that $g\mathcal{A} \supset \mathcal{F}$ and $g\mathcal{A} \supset \mathcal{F}'$, there exists an element $g' \in G$ such that $g'\mathcal{A} = g\mathcal{A}$ and g' fixes all vertices of $\mathcal{F}$ and $n\mathcal{F}'$.*
(5) *The group G acts on the set of pairs $(\mathcal{A}', \mathcal{C}')$ of an apartment $\mathcal{A}'$ and its chamber $\mathcal{C}' \subset \mathcal{A}'$ transitively.*

Axiom 5 (Tits system). Let $\mathcal{A} \subset \mathcal{B}(G)$ be an apartment corresponding to a maximal k-split torus $S \subset G$. Let $\mathcal{C} \subset \mathcal{A}$ be a chamber. If we put $\mathcal{G} := G(k)^0$, $\mathcal{J} := G(k)^0_{\mathcal{C}}$, and $\mathcal{N} := N(k) \cap \mathcal{G}$, then $(\mathcal{G}, \mathcal{J}, \mathcal{N})$ forms a saturated Tits system. Moreover, the associated restricted building is equal to $\mathcal{B}(G)$ in the sense that the stabilizer group $G(k)^0_{\mathcal{F}}$ of any facet $\mathcal{F} \subset \mathcal{B}(G)$ is equal to the parabolic subgroup in the sense of Tits system.

6. Week 6: Further properties of parahoric subgroups

We use the usual notation. Let G be a connected reductive group over a non-archimedean local field k.

6.1. **Action of U_ψ on apartments.** Let $S \subset G$ be a maximal k-split torus of G. As usual, we let Z and N denote the centralizer and the normalizer of S in G, respectively. Let $\Phi := \Phi(G, S)$ be the realtive root system of S in G. Let $\mathcal{A}(S) \subset \mathcal{B}(G)$ be the apartment corresponding to S.

Recall that (Axiom 4), the poly-simplicial structure on $\mathcal{A}(S)$ is realized by an affine root system Ψ on the affine space $\mathcal{A}(S)$ such that $\dot{\Psi} = \Phi$. To be more precise, we first have defined an affine root system Ψ' to be the set of affine functionals $\{\psi_a^u \mid a \in \Phi, u \in U_a(k) \smallsetminus \{1\}\}$, where $\psi_a^u \colon \mathcal{A}(S) \to \mathbb{R}$ is the affine functional whose derivative is a and which vanishes on the hyperplane $\mathcal{H}_u$ fixed by the action of $m(u) \in N(k)$. For any affine any affine functional on $\mathcal{A}(S)$ whose derivative lies in Φ, we have defined subgroups $U_\psi, U_{\psi+}$ of $U_a(k)$, where U_a denotes the root subgroup of the root $a := \dot{\psi} \in \Phi$:

$$U_\psi := \{u \in U_a(k) \smallsetminus \{1\} \mid \psi_a^u \geq \psi\} \cup \{1\},$$

$$U_{\psi+} := \{u \in U_a(k) \smallsetminus \{1\} \mid \psi_a^u > \psi\} \cup \{1\}.$$

Then we defined $\Psi := \{\psi \in \Psi' \mid U_\psi \subsetneq U_{\psi+} \cdot U_{2a}(k)\}$.

Let us try to understand the subgroup U_ψ concretely in the case of SL_2. In the case where $G = \mathrm{SL}_2$, we can choose S to be the diagonal maximal torus. The set of roots Φ consists of just two elements a and $-a$, where a denotes $T \to \mathbb{G}_{\mathrm{m}} \colon \mathrm{diag}(z, z^{-1}) \mapsto z^2$. The coroot $a^\vee$ of a is given by $\mathbb{G}_{\mathrm{m}} \to T \colon z \mapsto \mathrm{diag}(z, z^{-1})$. The root subgroups corresponding to a and $-a$ are upper-triangular and lower-triangular unipotent subgroups, respectively.

As claimed in Week 2, the standard apartment in this case can be realized as

$$\mathcal{A}(T) := X_*(T)_{\mathbb{R}} \cong \mathbb{R}a^\vee,$$

where the affine structure on $\mathcal{A}(T)$ over $X_*(T)_{\mathbb{R}} (=: V(T))$ is given in an obvious way. Then any affine functional $\psi \colon \mathcal{A}(T) \to \mathbb{R}$ is of the form

$$\psi \colon ra^\vee \mapsto b(ra^\vee) + c,$$

where $b \in V(T)^* \cong X^*(T)_{\mathbb{R}}$ and $c \in \mathbb{R}$.

We first determine ψ_a^u. For this, let us compute $m(u) \in N(k)$ for $u \in U_a(k) \smallsetminus \{1\}$. Recall that $m(u)$ is defined to be the unique product $m(u) := u'uu''$ satisfying $m(u) \in N(k)$, where $u', u'' \in U_{-a}(k) \smallsetminus \{1\}$. It can be checked that, if $u = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$, then u' and u'' are given by $\left(\begin{smallmatrix} 1 & 0 \\ -x^{-1} & 1 \end{smallmatrix}\right)$; then the product $m(u) := u'uu''$ is given by

$$\begin{pmatrix} 0 & x \\ -x^{-1} & 0 \end{pmatrix} = a^\vee(x) \cdot w_0,$$

where $w_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. As claimed in Week 2, the action of w_0 on $\mathcal{A}(T) = V(T)$ is given by inversion. On the other hand, the action of $T(k)$ on $\mathcal{A}(T)$ is given by translation, i.e., $\left(\begin{smallmatrix} x & 0 \\ 0 & x^{-1} \end{smallmatrix}\right) \cdot (ra^\vee) := (r - \omega(x))a^\vee$. Therefore, we get

$$m(u) \colon \mathcal{A}(T) \to \mathcal{A}(T); \quad ra^\vee \mapsto (-r - \omega(x))a^\vee.$$

This implies that the hyperplane $\mathcal{H}_u$ fixed by the action of $m(u) \in N(k)$ is given by $\mathcal{H}_u = \{-\frac{\omega(x)}{2}a^\vee\}$. Hence, the affine functional ψ_a^u is given by

$$\psi_a^u\colon ra^\vee \mapsto a(ra^\vee) + \omega(x).$$

(Indeed, the derivative of ψ_a^u is a and $\psi_a^u(-\frac{\omega(x)}{2}a^\vee) = 0$; recall that $a(a^\vee) = \langle a, a^\vee\rangle = 2$.)

Now we compute U_ψ for any affine functional $\psi\colon ra^\vee \mapsto a(ra^\vee)+c$. (Let us write "$a+c$" for this.) The condition $\psi_a^u \geq \psi$ is equivalent to that $\omega(x) \geq c$. Hence we obtain

$$\begin{aligned} U_\psi &:= \{u \in U_a(k) \smallsetminus \{1\} \mid \psi_a^u \geq \psi\} \cup \{1\} \\ &= \{u = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right) \mid \omega(x) \geq c\} = \begin{pmatrix} 1 & \mathfrak{m}^{\lceil c \rceil} \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Similarly,

$$U_{\psi+} := \{u \in U_a(k) \smallsetminus \{1\} \mid \psi_a^u > \psi\} \cup \{1\} = \begin{pmatrix} 1 & \mathfrak{m}^{\lceil c+\varepsilon \rceil} \\ 0 & 1 \end{pmatrix},$$

where ε is any sufficiently small positive number. Therefore we see that Ψ' and Ψ in this case are given by

$$\Psi' = \Psi = \{b + c \mid b \in \{\pm a\}, c \in \mathbb{Z}\}.$$

Now let us go back to the general situation to state the next axiom on how U_ψ acts on the Bruhat–Tits building $\mathcal{B}(G)$.

Axiom 6. Let $\psi \in \Psi$ be any affine root on the apartment $\mathcal{A}(S)$.

(1) The action of U_ψ on $\mathcal{B}(G)$ fixes the "half-apartment" $\mathcal{A}(S)^{\psi \geq 0} := \{x \in \mathcal{A}(S) \mid \psi(x) \geq 0\}$.
(2) For any non-empty bounded subset $\Omega \subset \mathcal{A}(S)$, the stabilizer group $G(k)^0_\Omega$ in $G(k)^0$ is generated by $Z(k)^0$ and U_ψ for $\psi \in \{\psi \in \Psi \mid \psi(\Omega) \geq 0\}$.
(3) For $a \in \Phi$, if we let $\psi \in \Psi'$ be the smallest affine functional such that $\dot{\psi} = a$ and $\psi(\Omega) \geq 0$, then $U_a(k) \cap G(k)^0_\Omega = U_\psi$.

6.2. **Iwahori decomposition.** Let $\mathcal{A}(S) \subset \mathcal{B}(G)$ be the apartment corresponding to S. Recall that, for a facet $\mathcal{F} \in \mathcal{B}(G)$, we call its stabilizer group $G(k)^0_\mathcal{F}$ taken in $G(k)^0$ the *parahoric subgroup* associated to $\mathcal{F}$. When $\mathcal{F} \in \mathcal{B}(G)$ is a chamber, we expecially call the parahoric subgroup $G(k)^0_\mathcal{F}$ an *Iwahori subgroup*. We fix a chamber $\mathcal{C} \subset \mathcal{A}(S)$ and consider the corresponding Iwahori subgroup $\mathcal{I} := G(k)^0_\mathcal{C}$. Note that Axiom 3 (2) (b) and the definition of $Z(k)^0$ (i.e., the kernel of the Kottwitz homomorphism) implies that $Z(k)^0$ is contained in $\mathcal{I}$.

Axiom 7. We fix any set $\Phi^+ \subset \Phi$ of positive roots; note that then this uniquely defines a minimal parabolic subgroup P of G over k with Levi decomposition $P = ZU^+$, where U^+ is the unipotent radical of P. Let U^- be the unipotent radical of the opposite parabolic to P.

(1) The product map

$$(U^+(k) \cap \mathcal{I}) \times Z(k)^0 \times (U^-(k) \cap \mathcal{I}) \to \mathcal{I}$$

is bijective.

(2) For each $a \in \Phi^{+,\mathrm{nd}}$ ("nd" means "non-divisible", i.e., $\frac{1}{2} \notin \Phi^{+}$), we let ψ_a be the smallest affine functional on $\mathcal{A}(S)$ such that the derivative of ψ_a is a and that ψ_a is non-negative on the fixed chamber $\mathcal{C}$. Then the product map (where the product can be taken in any order)

$$\prod_{a \in \Phi^{+,\mathrm{nd}}} U_{\psi_a} \to U^{+}(k)$$

is injective and its image is given by $U^{+}(k) \cap \mathcal{I}$. The same statement holds also for U^{-}.

6.3. **Compatibility with isomorphisms.** We next state an axiom on the compatibility of the Bruhat–Tits buildings with isomorphisms of base fields and groups.

Axiom 8. Let $f_k \colon k_1 \xrightarrow{\cong} k_2$ be an isomoprhism of non-archimedean local fields k_1 and k_2. Let G_1 and G_2 be connected reductive groups over k_1 and k_2, respectively, equipped with an isomoprhism $f_G \colon G_1 \times_{k_1} k_2 \xrightarrow{\cong} G_2$ as algebraic groups over k_2. Then (f_k, f_G) induces a natural identification $f_{\mathcal{B}} \colon \mathcal{B}(G_1, k_1) \cong \mathcal{B}(G_2, k_2)$ satisfying $f_{\mathcal{B}}(g \cdot x) = f_G(g) \cdot f_{\mathcal{B}}(x)$ for any $x \in \mathcal{B}(G_1, k_2)$ and $g \in G_1(k_1)$.

Remark 6.1. So far, we have only considered non-archimedean local fields. However, we also want to consider the buildings and its related objects for connected reductive groups over the maximal unramified extension K of a non-archimedean local field k. For this reason, we actually need to state all of the axioms for any connected reductive group over any Henselian discretely valuation field with perfect residue field.

Remark 6.2. The above axiom can be especially applied to the following. Let G be a connected reductive group over k. Let l be any finite Galois extension of k or the maximal unramified extension of k. Then the Galois group $\mathrm{Gal}(l/k)$ acts on $G_l := G \times_k l$, hence induces an action on $\mathcal{B}(G_l)$.

6.4. **Bruhat–Tits group schemes.**

Definition 6.3. Let X be an affine variety (reduced affine scheme of finite type) over k. An *integral model of* X *(over* $\mathfrak{o}$*)* is a flat affine scheme $\mathcal{X}$ of finite tyep over $\mathfrak{o}$ equipped with an isomorphism $\mathcal{X} \times_{\mathfrak{o}} k \,(:= \mathcal{X} \times_{\mathrm{Spec}\,\mathfrak{o}} \mathrm{Spec}\, k) \cong X$ of schemes over k.

Let $\mathcal{X}$ be an integral model of an affine variety X over k. Then any morphism $\mathrm{Spec}\, R \to \mathcal{X}$ over $\mathfrak{o}$, where R is an $\mathfrak{o}$-algebra, induces a morphism $\mathrm{Spec}(R \otimes_{\mathfrak{o}} k) \to \mathcal{X} \times_{\mathfrak{o}} k$ over k by the base change along $\mathrm{Spec}\, k \to \mathrm{Spec}\, \mathfrak{o}$. In other words, we have a natural map $\mathcal{X}(R) \to (\mathcal{X} \times_{\mathfrak{o}} k)(R \otimes_{\mathfrak{o}} k)$. Thus, by composing this map with the bijection $(\mathcal{X} \times_{\mathfrak{o}} k)(R \otimes_{\mathfrak{o}} k) \cong X(R \otimes_{\mathfrak{o}} k)$ induced by the isomorphism $\mathcal{X} \times_{\mathfrak{o}} k \cong X$ over k, we obtain a map $\mathcal{X}(R) \to X(R \otimes_{\mathfrak{o}} k)$.

Lemma 6.4. *If R is flat over $\mathfrak{o}$, then $\mathcal{X}(R) \to X(R \otimes_{\mathfrak{o}} k)$ is injective.*

Proof. Let $\mathfrak{o}[\mathcal{X}]$ be the coordinate ring of $\mathcal{X}$. In terms of commutative rings, we need to check that the map

$$\mathrm{Hom}_{\mathfrak{o}\text{-alg}}(\mathfrak{o}[\mathcal{X}], R) \to \mathrm{Hom}_{k\text{-alg}}(\mathfrak{o}[\mathcal{X}] \otimes_{\mathfrak{o}} k, R \otimes_{\mathfrak{o}} k)$$

is injective. This immediately follows from that $R \to R \otimes_{\mathfrak{o}} k$ is injective, which is a consequence of the flatness of R over $\mathfrak{o}$. $\square$

Especially, by choosing R to be $\mathfrak{o}$, we may identify $\mathcal{X}(\mathfrak{o})$ as a subset of $X(k)$.

On the other hand, we also have a natural map $\mathcal{X}(R) \to (\mathcal{X} \times_{\mathfrak{o}} \mathfrak{f})(R \otimes_{\mathfrak{o}} \mathfrak{f})$ by the base change along $\operatorname{Spec} \mathfrak{f} \to \operatorname{Spec} \mathfrak{o}$, for any $\mathfrak{o}$-algebra R. Especially, by choosing R to be $\mathfrak{o}$, we have a map $\mathcal{X}(\mathfrak{o}) \to (\mathcal{X} \times_{\mathfrak{o}} \mathfrak{f})(\mathfrak{f})$. We have the following fact, which is the geometric version of Hensel's lemma (so it is important here is that k is a non-archimedean local field, especially, henselian).

Fact 6.5. *If an integral model $\mathcal{X}$ of an affine variety X over k is smooth over $\mathfrak{o}$, then the map $\mathcal{X}(\mathfrak{o}) \to (\mathcal{X} \times_{\mathfrak{o}} \mathfrak{f})(\mathfrak{f})$ is surjective.*

Exercise 6.6. Prove this fact. (Hint: Combine the Jacobian criterion for smoothness and Hensel's lemma for multi-variable polynomials.)

Fact 6.7. *Let $\mathcal{G}$ be a smooth affine group scheme over $\mathfrak{o}$. Then there exists a unique largest open affine subgroup scheme $\mathcal{G}^0$ of $\mathcal{G}$ such that every geometric fiber of $\mathcal{G}^0$ is connected (i.e., $\mathcal{G}^0_{k^{\mathrm{sep}}}$ and $\mathcal{G}^0_{\bar{\mathfrak{f}}}$ are connected). We call $\mathcal{G}^0$ the "relative identity component" of $\mathcal{G}$.*

Axiom 9 (Bruhat–Tits group scheme)**.** For any non-empty bounded subset Ω of an apartment $\mathcal{A}$ of $\mathcal{B}(G)$, there exists a smooth affine group scheme $\mathcal{G}^1_\Omega$ oever $\mathfrak{o}$ which is an integral model of G and satisfies $\mathcal{G}^1_\Omega(\mathfrak{o}) = G(k)^1_\Omega$ ("Bruhat–Tits group scheme"). Furthermore, if we let $\mathcal{G}^0_\Omega$ be the relative identity component of $\mathcal{G}^1_\Omega$ (called the "parahoric group scheme"), then we have $\mathcal{G}^0_\Omega(\mathfrak{o}) = G(k)^0_\Omega$.

6.5. **Reductive quotient of parahoric subgroups.** We first look at the following example. Let $G := \mathrm{SL}_2$. We consider the standard maximal parahoric subgroup $\mathcal{P} := \mathrm{SL}_2(\mathfrak{o})$ and its Iwahori subgroup $\mathcal{I} := \left(\begin{smallmatrix} \mathfrak{o}^\times & \mathfrak{o} \\ \mathfrak{m} & \mathfrak{o}^\times \end{smallmatrix}\right)$. By taking the "modulo-$\mathfrak{m}$ reduction", we obtain $\mathrm{SL}_2(\mathfrak{f})$ from $\mathcal{P}$. Moreover, the image of $\mathcal{I}$ in $\mathrm{SL}_2(\mathfrak{f})$ defines a parabolic (Borel) subgroup $\left(\begin{smallmatrix} \mathfrak{f}^\times & \mathfrak{f} \\ 0 & \mathfrak{f}^\times \end{smallmatrix}\right)$ of $\mathrm{SL}_2(\mathfrak{f})$. What we want to do next is to generalize this to parahoric subgroups of general p-adic reductive groups.

Suppose that Ω and Ω' are non-empty bounded subsets of $\mathcal{B}(G)$ satisfying $\Omega \preceq \Omega'$, i.e., $\Omega \subset \overline{\Omega'}$. We consider the corresponding open compact subgroups $G(k)^1_\Omega, G(k)^1_{\Omega'}$ and also Bruhat–Tits groups schemes $\mathcal{G}^1_\Omega$, $\mathcal{G}^1_{\Omega'}$. Note that the relation $\Omega \preceq \Omega'$ implies that $G(k)^1_\Omega \supset G(k)^1_{\Omega'}$; in other words, we have $\mathcal{G}^1_{\Omega'}(\mathfrak{o}) \subset \mathcal{G}^1_\Omega(\mathfrak{o})$.

Let us suppose that there exists a homomorphism of group schemes

$$\rho_{\Omega,\Omega'} \colon \mathcal{G}^1_{\Omega'} \to \mathcal{G}^1_\Omega$$

such that $\rho_{\Omega,\Omega'}$ induces the identity morphism of G on the generic fiber and induces an injection on $\mathfrak{o}$-rational points. Note that such $\rho_{\Omega,\Omega'}$ induces also a homomorphism $\mathcal{G}^0_{\Omega'} \to \mathcal{G}^0_\Omega$ of Bruhat–Tits parahoric group schemes over $\mathfrak{o}$. We let

$$\bar{\rho}_{\Omega,\Omega'} \colon \bar{\mathcal{G}}^0_{\Omega'} \to \bar{\mathcal{G}}^0_\Omega$$

be the homomorphism induced on the special fibers (hence over $\mathfrak{f}$). We put

$$\mathbb{G}_\Omega := \bar{\mathcal{G}}^0_\Omega / R_u(\bar{\mathcal{G}}^0_\Omega),$$

i.e., $\mathbb{G}_\Omega$ is the maximal reductve quotient of the special fiber of the parahoric group scheme.

Axiom 10. There uniquely exists a homomorphism $\rho_{\Omega,\Omega'}$. The kernel of $\bar{\rho}_{\Omega,\Omega'}$ is a smooth conencted unipotent subgroup of $\bar{\mathcal{G}}^0_{\Omega'}$. When $\Omega \preceq \Omega'$ are facets $\mathcal{F} \preceq \mathcal{F}'$ of an apartment $\mathcal{A}(S)$, we furthermore have the following:

(1) If we let $\mathcal{S}$ be the maximal closed split subtorus of $\mathcal{G}^0_{\mathcal{F}'}$ with generic fiber S, then $\bar{\mathcal{S}}$ and $\bar{\rho}_{\mathcal{F},\mathcal{F}'}(\bar{\mathcal{S}})$ are maximal $\mathfrak{f}$-split tori of $\bar{\mathcal{G}}^0_{\mathcal{F}'}$ and $\bar{\mathcal{G}}^0_{\mathcal{F}}$, respectively. Moreover, the image of $\bar{\rho}_{\mathcal{F},\mathcal{F}'}(\bar{\mathcal{S}})$ in $\mathbb{G}_{\mathcal{F}}$ defines a maximal $\mathfrak{f}$-split torus, for which we write $\mathbb{S}$.

(2) The image $\mathfrak{p}_{\mathcal{F}}(\mathcal{F}') := \bar{\rho}_{\mathcal{F},\mathcal{F}'}(\bar{\mathcal{G}}^0_{\mathcal{F}'})$ is a parabolic subgroup of $\bar{\mathcal{G}}^0_{\mathcal{F}}$. We write $\mathbb{P}_{\mathcal{F}}(\mathcal{F}')$ for the image of $\mathfrak{p}_{\mathcal{F}}(\mathcal{F}')$ in $\mathbb{G}_{\mathcal{F}}$, which is a parabolic subgroup of $\mathbb{G}_{\mathcal{F}}$.

(3) Let $x \in \mathcal{F}$ be a point and $v \in V(S)$ be a vector such that $x + v \in \mathcal{F}'$. Then the image of $\mathbb{P}_{\mathcal{F}}(\mathcal{F}')$ in $\mathbb{G}_{\mathcal{F}}$ is the parabolic subgroup corresponding to the set of roots $\{a \in \Phi(\mathbb{G}_{\mathcal{F}}, \mathbb{S}) \mid a(v) \geq 0\}$.

(4) Let
$$\pi_{\mathcal{F}} \colon \mathcal{G}^0_{\mathcal{F}}(\mathfrak{o}) \twoheadrightarrow \bar{\mathcal{G}}^0_{\mathcal{F}}(\mathfrak{f})$$
be the natural reduction homomorphism. Then we have $\pi_{\mathcal{F}}^{-1}(\mathfrak{p}_{\mathcal{F}}(\mathcal{F}')(\mathfrak{f})) = \mathcal{G}^0_{\mathcal{F}'}(\mathfrak{o})$.

(5) The association $\mathcal{F}' \mapsto \mathbb{P}_{\mathcal{F}}(\mathcal{F}')$ induces an order-reversing bijection between the set of facets $\mathcal{F}' \subset \mathcal{A}(S)$ such that $\mathcal{F} \preceq \mathcal{F}'$ and the set of parabolic subgroups of $\mathbb{G}_{\mathcal{F}}$.

(6) The root datum of $\mathbb{G}_{\mathcal{F}}$ with respect to $\mathbb{S}$ is given by $(X^*(S), \Phi_{\mathcal{F}}, X_*(S), \Phi^\vee_{\mathcal{F}})$, where $\Phi_{\mathcal{F}} \subset \Phi(G, S)$ is the derivative of $\Psi_{\mathcal{F}} \subset \Psi$, which is deifned by
$$\Psi_{\mathcal{F}} := \{\psi \in \Psi \mid \psi(\mathcal{F}) = 0\},$$
and $\Phi^\vee_{\mathcal{F}} := \{a^\vee \in \Phi(G, S)^\vee \mid a \in \Phi_{\mathcal{F}}\}$.

(7) Let $G(k)^{0+}_{\mathcal{F}}$ be the preimage of $R_u(\bar{\mathcal{G}}^0_{\mathcal{F}})(\mathfrak{f})$ under $\pi_{\mathcal{F}}$. Then $G(k)^{0+}_{\mathcal{F}} \subset G(k)^{0+}_{\mathcal{F}'}$.

6.6. **Definition of "Bruhat–Tits theory".** Now we can state the definition of Bruhat–Tits theory.

Definition 6.8. We say that *Bruhat–Tits theory is available for G over k* if the following axioms hold:

- Axiom 1 (existence of a bulding),
- Axiom 2 (open compactness of parahoric subgroups),
- Axiom 3 (affine structure of apartments),
- Axiom 4 (relation to affine root systems),
- Axiom 5 (relation to Tits systems),
- Axiom 6 (action of U_ψ),
- Axiom 7 (Iwahori decomposition),
- Axiom 8 (compatibility with isomorphisms),
- Axiom 9 (existence of Bruhat–Tits group schemes),
- Axiom 10 (structure of reductive quotients of parahoric subgroups).

7. Week 7: Construction of apartments: split case

Let k be a non-archimedean local field. Let $\mathfrak{o}$ and $\mathfrak{m}$ be the ring of integers of k and the maximal ideal of $\mathfrak{o}$, respectively. Let $\omega\colon k \to \mathbb{Z} \cup \{\infty\}$ be the normalized valuation of k.

Let G be a split connected reductive group over k. The aim of this week is to construct the apartment $\mathcal{A}(S)$ associated to any maximal k-split torus S of G.

7.1. **Valuation of root data.** Here we temporarily drop the splitness assumption on G. That is, the following definition makes sense for any connected reductive group G over k (with usual S, Z, and N):

Definition 7.1. A *valuation of the root datum* of (G, S) is a family

$$\varphi = \{\varphi_a\colon U_a(k) \to \mathbb{R} \cup \{\infty\}\}_{a\in\Phi}$$

of maps satisfying the following:

(1) For any $a \in \Phi$, the image $\varphi_a(U_a(k))$ has at least three elements.
(2) For any $a \in \Phi$ and $r \in \mathbb{R} \cup \{\infty\}$, if we let

$$U_{a,\varphi,r} := \varphi_a^{-1}([r, \infty]) = \{u \in U_a(k) \mid \varphi_a(u) \geq r\} \subset U_a(k),$$

then $U_{a,\varphi,r}$ is a subgroup of $U_a(k)$ and satisfies $U_{a,\varphi,\infty} = \{1\}$.
(3) For any $n \in N(k)$ which acts on $V(S) = X_*(S)_{\mathbb{R}}$ as the reflection along $a \in \Phi$, the function

$$U_a(k)^* \to \mathbb{R}\colon u \mapsto \varphi_a(u) - \varphi_{-a}(nun^{-1})$$

is constant.
(4) For any $a, b \in \Phi$ and $i, j \in \mathbb{R}$ such that $b \notin -\mathbb{R}_{>0}a$, we have

$$[U_{a,\varphi,i}, U_{b,\varphi,j}] \subset \langle U_{pa+qb,\varphi,pi+qj} \mid p, q \in \mathbb{Z}_{>0} \text{ such that } pa + qb \in \Phi\rangle.$$

(In particular, if $p = q = 1$ are only p, q satisfying $pa + qb \in \Phi$, we have $[U_{a,\varphi,i}, U_{b,\varphi,j}] \subset U_{a+b,\varphi,i+j}$.
(5) If $a, 2a \in \Phi$, then we have $\varphi_{2a} = 2 \cdot \varphi_a|_{U_{2a}(k)}$. Thus, in particular, we have $U_{a,\varphi,r} \cap U_{2a}(k) = U_{2a,\varphi,2r}$.
(6) For any $a \in \Phi$ and $u \in U_a(k)$, if we let $u', u'' \in U_{-a}(k)$ be elements as in Fact 5.4, then we have $\varphi_{-a}(u') = -\varphi_a(u)$.
(7) For any $a \in \Phi$ and $z \in Z(k)$, we have

$$\varphi_a(z^{-1}uz) = \varphi_a(u) + a(\nu(z)),$$

where $\nu\colon Z(k) \to V(S')$ is the "translation" homomorphism.
(8) For any $a \in \Phi$, the subgroups $\{U_{a,\varphi,r}\}_{r\in\mathbb{R}}$ are open compact subgroups of $U_a(k)$ which consistitute a basis of open neighborhood of 1.

We let $\mathrm{VRD}(G, S)$ be the set of valuations of root datum of (G, S).

Exercise 7.2. Show that, in condition (6), we also have $\varphi_{-a}(u'') = -\varphi_a(u)$ by assuming the other conditions.

Definition 7.3. For each valuation of root datum $\varphi \in \mathrm{VRD}(G, S)$, we define an open compact subgroup $\mathcal{P}_\varphi$ of $G(k)$ by

$$\mathcal{P}_\varphi := \langle S(k)^0, U_{a,\varphi,0} \mid a \in \Phi\rangle.$$

In the future, we will define the Bruhat–Tits building $\mathcal{B}(G)$ containing $\mathcal{A}(S)$ and having an action of $G(k)$ such that the stabilizer group of $\varphi \in \mathcal{A}(S)$ taken in $G(k)^0$ coincides with $\mathcal{P}_\varphi$.

7.2. **Equipollence classes of VRD.** (In this subsection, G still can be non-split connected reductive group over k.)

Let $\varphi = \{\varphi_a\}_{a\in\Phi}$ be a valuation of root datum of (G,S). For any $v \in V(S)$, we define a family

$$\varphi_v = \{\varphi_{v,a} \colon U_a(k) \to \mathbb{R} \cup \{\infty\}\}_{a\in\Phi}$$

of maps by

$$\varphi_{v,a}(u) := \varphi_a(u) + a(v)$$

for any $a \in \Phi$ and $u \in U_a(k)$ (a is naturally regarded as an element of $V(S)^*$; in other words, $a(v) = \langle a, v\rangle$).

Lemma 7.4. *For any $v \in V(S)$, the family φ_v is also a valuation of root datum of (G,S). In particular, this procedure defines an action of $V(S)$ on* $\mathrm{VRD}(G,S)$. *Furthermore, for any $a \in \Phi$ and $r \in \mathbb{R}$, we have*

$$U_{a,\varphi,r} = U_{a,\varphi_v,r+a(v)}.$$

In the following, we use the additive symbol "+" for the action of $V(S)$ on $\mathrm{VRD}(G,S)$, i.e., we write $\varphi + v$ for φ_v.

Definition 7.5. We say that two valuations of root datum φ and φ' of (G,S) are *equipollent* if there exists $v \in V(S)$ such that $\varphi' = \varphi + v$. Note that then the equipollence relation defines an equivalence relation on $\mathrm{VRD}(G,S)$.

We write G_{ad} for the adjoint group of G. If we let S_{ad} be the image of S in G_{ad}, then S_{ad} is a maximal k-split torus of G_{ad}. Moreover, the images N_{ad} and Z_{ad} of N and Z in G_{ad} are nothing but the normalizer and centralizer groups of S_{ad} in G_{ad}, respectively. Then $\Phi := \Phi(G,S)$ is naturally identified with $\Phi(G_{\mathrm{ad}}, S_{\mathrm{ad}})$.

Here recall that $G_{\mathrm{ad}}(k)$ acts on $G(k)$. Indeed, the conjugate action of G on G itself, which is k-rational, is trivial on the center of G, hence factors through the quotient to G_{ad}. The resulting action of G_{ad} on G is again k-rational, hence we have an action of $G_{\mathrm{ad}}(k)$ on $G(k)$.

Let us define an action of $N_{\mathrm{ad}}(k)$ on $\mathrm{VRD}(G,S)$. Suppose that $\varphi \in \mathrm{VRD}(G,S)$, i.e., $\varphi = (\varphi_a)_{a\in\Phi}$ is a valuation of root datum of (G,S). For any $n \in N_{\mathrm{ad}}(k)$ and $a \in \Phi$, we define a map

$$n \cdot \varphi_a \colon U_a(k) \to \mathbb{R} \cup \{\infty\}$$

by $(n \cdot \varphi_a)(u) := \varphi_{n^{-1}\cdot a}(n^{-1}un)$ for any $u \in U_a(k)$. Here, $n^{-1} \cdot a$ denotes the character $n^{-1}Sn = S \to \mathbb{G}_{\mathrm{m}} \colon t \mapsto a(ntn^{-1})$, which is again a root of S in G. We let $n \cdot \varphi := \{n \cdot \varphi_a\}_{a\in\Phi}$. Then it is a routine task to check that $n \cdot \varphi$ is a valuation of root datum of (G,S) and that this procedure defines an action of $N_{\mathrm{ad}}(k)$ on $\mathrm{VRD}(G,S)$.

We remark that the actions of $N_{\mathrm{ad}}(k)$ and $V(S)$ on $\mathrm{VRD}(G,S)$ are compatible in the following sense (see Lemma 7.15).

> Let $\varphi = \{\varphi_a\}_{a\in\Phi} \in \mathrm{VRD}(G,S)$. Let $n \in N_{\mathrm{ad}}(k)$ and $v \in V(S)$. Then we have $n \cdot (\varphi + v) = n \cdot \varphi + n \cdot v$.

This implies that these actions give rise to an action of the semidirect product $V(S) \rtimes N_{\mathrm{ad}}(k)$ on $\mathrm{VRD}(G,S)$.

7.3. **Chevalley systems.** Let G be a simply-connected split semisimple group over k. Let T be a split maximal torus of G over k. We write $\Phi := \Phi(G,T)$. We let $\mathfrak{g}$ be the Lie algebra of G and $\mathfrak{g}_a$ the root space for a root $a \in \Phi$.

Fact 7.6. *Let $X_a \in \mathfrak{g}_a(k)$ be a non-zero element for $a \in \Phi$. Then there exists a unique homomorphism $\eta_{X_a}\colon \mathrm{SL}_2 \to G$ such that its differential $d\eta_{X_a}\colon \mathfrak{sl}_2 \to \mathfrak{g}$ satisfies*

$$d\eta_{X_a}\left(\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right) = H_a := da^\vee(1), \quad d\eta_{X_a}\left(\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}\right) = X_a.$$

Definition 7.7. Let $X_a \in \mathfrak{g}_a(k)$ be a non-zero element for $a \in \Phi$. Let η_{X_a} be the homomorphism as in the previous fact. We define $w(X_a) \in N$ to be $\eta_{X_a}\left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)$.

Definition 7.8. (1) A *weak Chevalley system* is a family $X' := \{\pm X_a \in \mathfrak{g}_a(k) \mid a \in \Phi\}$ of pairs $\pm X_a$ of nonzero root vectors such that X' is preserved by the action of $w(X_a)$ for any $a \in \Phi$.

(2) A *Chevalley system* (or *Chevalley basis*) is a family $X := \{X_a \in \mathfrak{g}_a(k) \mid a \in \Phi\}$ of nonzero root vectors such that $[X_a, X_{-a}] = H_a$ and $X' := \{\pm X_a \mid X_a \in X\}$ is a weak Chevalley system. (In this situation, we say that X refines X'.)

Example 7.9. Suppose that $G = \mathrm{SL}_n$. Let T be the diagonal maximal torus of G. Then the set of roots Φ of T in G is given by

$$\Phi = \{\pm(e_i - e_j) \mid 1 \leq i < j \leq n\},$$

where e_i denotes the i-th projection $T \to \mathbb{G}_\mathrm{m}\colon \mathrm{diag}(t_1, \ldots, t_n) \mapsto t_i$. Then the root subgroup corresponding to a positive root $a := e_i - e_j$ is given by

$$U_a(k) = \{I_n + xX_{ij} \mid x \in k\},$$

where I_n is the identity matrix and X_{ij} is the matrix whose (i,j)-entry is 1 and all other entries are zero. Similarly, for $-a = -(e_i - e_j)$, we have

$$U_{-a}(k) = \{I_n + xX_{ji} \mid x \in k\}.$$

Then $\{X_{ij} \mid 1 \leq i \neq j \leq n\}$ forms a Chevalley basis.

Example 7.10. We next consider the case of $G = \mathrm{Sp}_{2n}$, where the defining symplectic matrix is chosen to be the anti-diagonal matrix J_{2n} with anti-diaognal entries $(+1, -1, +1, -1, \ldots)$, i.e.,

$$\mathrm{Sp}_{2n}(k) = \{g \in \mathrm{GL}_{2n} \mid {}^t g J_{2n} g = J_{2n}\}.$$

Let T be the diagonal maximal torus of G; explicitly, this is given by

$$T(k) = \{\mathrm{diag}(t_1, \ldots, t_n, t_n^{-1}, \ldots, t_1^{-1}) \mid t_1, \ldots, t_n \in k^\times\}.$$

Then the set of roots Φ of T in G is given by

$$\Phi = \{\pm e_i \pm e_j \mid 1 \leq i < j \leq n\} \cup \{\pm 2e_i \mid 1 \leq i \leq n\}.$$

Let us again use the notation as in the previous example, i.e., X_{ij} is the matrix whose (i,j)-entry is 1 and all other entries are zero. Then the root subgroup corresponding to a positive root $a \in \Phi$ is given by

$$U_a(k) = \{I_n + xY_a \mid x \in k\},$$

where Y_a is as follows (only for positive a; Y_{-a} is given by the transpose of Y_a):

- When $a = e_i - e_j$ $(1 \leq i < j \leq n)$, we put
$$Y_a := X_{ij} + (-1)^{i-j+1} X_{2n+1-j,2n+1-i}.$$
- When $a = e_i + e_j$ $(1 \leq i < j \leq n)$, we put
$$Y_a := X_{i,2n+1-j} + (-1)^{i-j} X_{j,2n+1-i}.$$
- When $a = 2e_i$, we put
$$Y_a := X_{ii}.$$

Then $\{X_a \mid a \in \Phi\}$ forms a Chevalley basis.

As these examples show, for a classical group, we can find a Chevalley system by its explicit realization as a matrix group. However, in general, a connected reductive group does not have a canonical realization as a matrix group (i.e., a faithful embedding into GL_n). In this sense, we may think of a Chevalley basis as a tool which enables us to treat an abstract connected reductive group as if it were a concrete matrix group.

Recall that a *k-pinning* of G is a tuple $(T, B, \{X_a\}_{a\in\Delta})$ of

- a k-split maximal torus T,
- a k-rational Borel subgroup containing T, and
- a family of non-zero k-rational root vectors $\{X_a\}_{a\in\Delta}$ for simple roots $a \in \Delta$.

Fact 7.11. *For any k-pinning $(T, B, \{X_a\}_{a\in\Delta})$, there exists a unique weak Chevalley system extending it. Conversely, any weak Chevalley system can be obtained in this way.*

7.4. **Chevalley valuation and apartments.** Let us assume that G is split over k. We fix a weak Chevalley system $o = \{\pm X_a \in \mathfrak{g}_a(k)\}_{a\in\Phi}$ of G. Recall that, for each $a \in \Phi$, o determines a k-rational isomorphism $u_{o,a} \colon \mathbb{G}_\mathrm{a} \xrightarrow{\cong} U_a$, which is characterized by the condition that $du_{o,a}(1) = X_a$.

We define a family
$$\varphi_o = \{\varphi_{o,a} \colon U_a(k) \to \mathbb{R} \cup \{\infty\}\}_{a\in\Phi}$$
of maps by
$$\varphi_{o,a}(u_{o,a}(t)) := \begin{cases} \omega(t) & \text{if } t \neq 0, \\ \infty & \text{if } t = 0, \end{cases}$$
for any $a \in \Phi$ and $t \in k$. Then it is a routine task to check that φ_o is a valuation of root datum of (G, S). We call φ_o the *Chevalley valuation* associated to o.

Lemma 7.12. *Let φ be a Chevalley valuation of (G, S) (i.e., the Chevalley valuation associated to some weak Chevalley system o). Then, for any $\varphi' \in \mathrm{VRD}(G, S)$, φ' is a Chevalley valuation if and only if there exists a (necessarily unique) $v \in X_*(S_\mathrm{ad}) \subset V(S)$ such that $\varphi' = \varphi + v$. In particular, all Chevalley valuations of (G, S) are equipollent to each other.*

Proof. The "if" direction is easy, so let us consider the "only if" direction. For this, we may suppose that φ and φ' arise from k-pinnings $(S, B, \{X_a\})$ and $(S, B, \{X'_a\})$ of G sharing the same B. Then we may find $t \in S_\mathrm{ad}(k)$ such that $a(t) \cdot X_a = X'_a$ for all $a \in \Phi$. Using this t, we can check that $\varphi' = \varphi - \omega(t)$. □

Definition 7.13. Let $\mathcal{A}(S)$ be the set of valuation of root datum of (G, S) which are equipollent to some (hence all by the previous lemma) Chevalley valuation. We call $\mathcal{A}(S)$ the *apartment* of S.

Let S' be the unique maximal subtorus of S contained in G_{der} and A_G the maximal central split torus of G. Recall that we have a canonical decomposition $V(S) = V(S') \oplus V(A_G)$ induced by the natural homomorphism $S' \to S \to S/A_G$, which is an isogeny. Note that, for each $\varphi \in \mathrm{VRD}(G, S)$, the stabilizer of φ with respect to the action of $V(S)$ is given by $V(A_G)$. Indeed, for any $v \in V(S)$, we have $\varphi = \varphi + v$ if and only if $\varphi_a(u) = (\varphi + v)_a(u) = \varphi_a(u) + a(v)$ for any $a \in \Phi$ and $u \in U_a(k)$, which is equivalent to $a(v) = 0$ for any $a \in \Phi$. This is furthermore equivalent to $v \in V(A_G)$. Therefore, by the definition of $\mathcal{A}(S)$, we see that $V(S')$ acts on $\mathcal{A}(S)$ simply transitively. In other words, $\mathcal{A}(S)$ is an affine space under $V(S')$.

Lemma 7.14. *The action of $N_{\mathrm{ad}}(k)$ on $\mathrm{VRD}(G, S)$ preserves $\mathcal{A}(S)$.*

Proof. Let $\varphi \in \mathcal{A}(S)$ be the Chevalley valuation associated to a weak Chevalley system o. Let $n \in N_{\mathrm{ad}}(k)$. If we let $(S, B, \{X_a\})$ be the k-pinning corresponding to o, then $(S, nBn^{-1}, \{X_{n^{-1}\cdot a}\})$ is also a k-pinning. It is straightforward to check that $n \cdot \varphi$ is the Chevalley valuation associated to the weak Chevalley system corresponding to $(S, nBn^{-1}, \{X_{n^{-1}\cdot a}\})$. □

Now let us verify that the apartment $\mathcal{A}(S)$ satisfies Axiom 3, which consists of the following conditions.

(1) $\mathcal{A}(S)$ is an affine space under $V(S')$.
(2) The action of $N(k)$ on $\mathcal{A}(S)$ is given by affine transformations. If we write $f\colon N(k) \to \mathrm{Aff}(\mathcal{A}(S))$ for the resulting homomorphism, then
  (a) its derivative $df(n) \in N(k) \to \mathrm{Aut}_{\mathbb{R}}(V'(S))$ coincides with the natural action of $N(k) \twoheadrightarrow N(k)/Z(k)$ on $V(S') = X_*(S')_{\mathbb{R}}$, and
  (b) the action of $Z(k)$ on $\mathcal{A}(S)$ is given by the translation ν, i.e., $z \in Z(k)$ acts via $-\omega_Z(z)$-translation.
(3) For any $g \in G(k)$, the isomoprhism $g\colon \mathcal{A}(S) \to g\mathcal{A}(S) = \mathcal{A}(gSg^{-1})$ is affine such that its derivative is the natural isomorphism $g\colon V(S') \to V(gS'g^{-1})$.

The condition (1) is already verified. The condition (3) is clear. We consider the condition (2).

Lemma 7.15. *Let $\varphi \in \mathrm{VRD}(G, S)$.*

(1) *For any $n \in N_{\mathrm{ad}}(k)$ with image w in W and $v \in V(S')$, we have $n\cdot(\varphi+v) = n\cdot\varphi + w\cdot v$. In other words, the action of $N_{\mathrm{ad}}(k)$ on $\mathrm{VRD}(G, S)$ is given by affine transformations with linear part given by the natural action of $N_{\mathrm{ad}}(k)$ on $V(S')$.*
(2) *If $\varphi \in \mathcal{A}(S)$, then, for any $t \in S_{\mathrm{ad}}(k)$, we have $t \cdot \varphi = \varphi - \omega(t)$. In other words, the action of $S(k)$ on $\mathcal{A}(S)$ is given by the $-\omega$-translation.*

Proof. Let us first check (1). By definition, for any $u \in U_a(k)$, we have

$$\begin{aligned}(n \cdot (\varphi + v))_a(u) &= (\varphi + v)_{n^{-1}\cdot a}(n^{-1}un) \\ &= \varphi_{n^{-1}\cdot a}(n^{-1}un) + (n^{-1} \cdot a)(v) \\ &= (n \cdot \varphi)_a(u) + a(w \cdot v) = (n \cdot \varphi + w \cdot v)_a(u).\end{aligned}$$

We next check (2). Let $\varphi = \varphi_o + v$ for some weak Chevalley system o and $v \in V(S')$. By (1), we have $t \cdot \varphi = t \cdot (\varphi_o + v) = t \cdot \varphi_o + t \cdot v = t \cdot \varphi_o + v$. Thus it suffices to check the claim only in the case where $\varphi = \varphi_o$. Then, for any $a \in \Phi$,

$t \in S_{\mathrm{ad}}(k)$ and $u \in U_a(k)$,

$$(t \cdot \varphi_o)_a(u) = \varphi_{o,a}(t^{-1}ut) = \varphi_{o,a}(u) + a(\nu(t)),$$

where $\nu\colon Z(k) \to V(S')$ is the translation homomorphism. Noting that S is maximal (G is split), hence $Z = S$, we have $a(\nu(t)) = -\omega(a(t))$. $\square$

8. Week 8: Construction of apartments: quasi-split case

In the last week, we constructed the apartment $\mathcal{A}(S)$ associated to a maximal k-split torus S of G and verified that $\mathcal{A}(S)$ satisfies Axiom 3 under the assumption that G is split. This week we consider the case where a connected reductive group G is quasi-split over k. We furtheremore assume that G is semisimple for a moment. Let S be a maximal k-split torus of G and $Z =: T$ the centralizer of S in G (in this case, T is a maximal torus of G). We put $\Theta := \mathrm{Gal}(k^{\mathrm{sep}}/k)$.

8.1. **Absolute root system vs relative root system.** We write $\tilde{\Phi} := \Phi(G, T)$ and $\Phi := \Phi(G, S)$. Since T is k-rational, $X^*(T)$ has a natural action of the Galois group Θ given by $\sigma(a) := \sigma \circ a \circ \sigma^{-1}$ (for $\sigma \in \Theta$ and $a \in X^*(T)$) and the set of absolute roots $\Phi(G, T)$ is preserved by this action. Also note that Φ may not be a reduced root system while $\tilde{\Phi}$ is a reduced root system. In particular, for any $a \in \Phi$, we have the following three possibilities:

- *a is of type R1*, which means $\frac{1}{2}a, 2a \notin \Phi$.
- *a is of type R2*, which means $2a \in \Phi$,
- *a is of type R3*, which means $\frac{1}{2}a \in \Phi$.

(Note that these three possibilities are mutually exclusive since Φ is a root system.)

We summarize basic facts on the relationship between $\tilde{\Phi}$ and Φ in the following. Since S is contained in T, we have the natural restriction map $X^*(T) \to X^*(S)$. In fact, this restriction map induces a surjection

$$\tilde{\Phi} \to \Phi \colon \tilde{a} \mapsto \tilde{a}|_S.$$

Moreover, the fiber of each relative root $a \in \Phi$ is a single Θ-orbit in $\tilde{\Phi}$. For any element $\tilde{a} \in \tilde{\Phi}$, we write $\Theta_{\tilde{a}}$ for the stabilizer of $\tilde{a}$ in Θ and $k_{\tilde{a}} := (k^{\mathrm{sep}})^{\Theta_{\tilde{a}}}$:

$$k \subset k_{\tilde{a}} \subset k^{\mathrm{sep}} \longleftrightarrow \Theta \supset \Theta_{\tilde{a}} = \mathrm{Gal}(k^{\mathrm{sep}}/k_{\tilde{a}}) \supset \{1\}.$$

For $\tilde{a} \in \tilde{\Phi}$, we write $U_{\tilde{a}}$ for the corresponding absolute root subgroups, respectively. Recall that $U_{\tilde{a}}$ is a 1-dimensional unipotent subgroup of G which is defined over $k_{\tilde{a}}$. On the other hand, for $a \in \Phi$, we also have the corresponding relative root subgroup U_a. By definition, this is a maximal closed unipotent subgroup of G which is stable under the conjugate action of S such that all weights of S in its Lie algebra are $\mathbb{Z}_{>0}$-multiples of a. In contrast to $U_{\tilde{a}}$, the relative root subgroup U_a is defined over k but not necessarily 1-dimensional.

type R1: We first consider the case where $a \in \Phi$ is of type R1. In this case, the fiber of $a \in \Phi$ under the restriction map $\tilde{\Phi} \to \Phi$ consists of pairwise orthogonal roots. If $\tilde{a} \in \tilde{\Phi}$ is in the fiber, then we have an

$$\mathrm{Res}_{k_{\tilde{a}}/k} U_{\tilde{a}} \cong U_a$$

defined over k. By noting that $U_{\tilde{a}}$ is contained in U_a over $k_{\tilde{a}}$, this isomorphism can be described at the level of k-rational points as follows:

$$\mathrm{Res}_{k_{\tilde{a}}/k} U_{\tilde{a}}(k) \cong U_{\tilde{a}}(k_{\tilde{a}}) \xrightarrow{\cong} U_a(k) \colon u \mapsto \prod_{\sigma \in \Theta/\Theta_{\tilde{a}}} \sigma(u).$$

type R2 and R3: We next consider the case where $a \in \Phi$ is of type R2, hence $2a \in \Phi$ is of type R3. Then the preimage of a under the restriction map $\tilde{\Phi} \to \Phi$ necessarily consists of even number, say $2m$, of roots $\{\tilde{a}_1, \tilde{a}'_1, \ldots, \tilde{a}_m, \tilde{a}'_m\}$. We can arrange the $2m$ roots such that

- each $\tilde{a}_i$ is orthogonal to any other root in this fiber except for $\tilde{a}'_i$,

- $\tilde{b}_i := \tilde{a}_i + \tilde{a}'_i \in \tilde{\Phi}$ for each i ($\tilde{b}_i$ restricts to $2a$).

Let us fix i and write $\tilde{a} = \tilde{a}_i$, $\tilde{a}' = \tilde{a}'_i$, and $\tilde{b} := \tilde{b}_i$ for simplicity. In this situation, we have that

- $k_{\tilde{a}}$ contains $k_{\tilde{a}}$ and $k_{\tilde{b}}/k_{\tilde{a}}$ is a quadratic extension, and
- the Galois group of $k_{\tilde{b}}/k_{\tilde{a}}$ acts on $\{\tilde{a}, \tilde{a}'\}$ by interchanging $\tilde{a}$ and $\tilde{a}'$.

If we put $U_{[\tilde{a}]} := U_{\tilde{a}} \cdot U_{\tilde{a}'} \cdot U_{\tilde{b}}$ (this is a 3-dimensional unipotent subgroup of G defined over $k_{\tilde{b}}$), then we have an isomorphism

$$\mathrm{Res}_{k_{\tilde{b}}/k} U_{[\tilde{a}]} \cong U_a.$$

Noting that $U_{[\tilde{a}]}$ is contained in U_a, at the level of k-rational points, this isomorphism can be described as follows:

$$\mathrm{Res}_{k_{\tilde{b}}/k} U_{[\tilde{a}]}(k) \cong U_{[\tilde{a}]}(k_{\tilde{b}}) \xrightarrow{\cong} U_a(k) \colon u \mapsto \prod_{\sigma \in \Theta/\Theta_{\tilde{b}}} \sigma(u).$$

In fact, the group $U_{[\tilde{a}]}$ is isomorphic to the group $U_{k_{\tilde{a}}/k_{\tilde{b}}}$ over $k_{\tilde{b}}$, which is defined by

$$U_{k_{\tilde{a}}/k_{\tilde{b}}}(R) := \{(u, v) \in (R \otimes_{k_{\tilde{b}}} k_{\tilde{a}}) \mid v + \overline{v} = u\overline{u}\},$$

where $\overline{(-)}$ denotes the automorphism of $R \otimes_{k_{\tilde{b}}} k_{\tilde{a}}$ induced by the nontrivial Galois action on $k_{\tilde{a}}$ (the group structure is given by $(u, v) \cdot (u', v') = (u + u', v + v' + u\overline{u'})$).

Furthermore, if we let $U^0_{k_{\tilde{a}}/k_{\tilde{b}}}$ be the subgroup of $U_{k_{\tilde{a}}/k_{\tilde{b}}}$ defined by

$$U^0_{k_{\tilde{a}}/k_{\tilde{b}}}(R) := \{(0, v) \in U_{k_{\tilde{a}}/k_{\tilde{b}}}(R)\},$$

then U_{2a} is isomorphic to $\mathrm{Res}_{k_{\tilde{b}}/k} U^0_{k_{\tilde{a}}/k_{\tilde{b}}}$ over k.

Remark 8.1. The relative root system Φ of a quasi-split connected reductive group G contains a relative root of type R2 or R3 only when the absolute root system $\tilde{\Phi}$ contains an irreducible factor of type A_{2n} and some element of Θ acts on the A_{2n}-factor nontrivially. (The simplest example is thus $\mathrm{SU}_{l/k}(3)$.)

8.2. **Chevalley–Steinberg systems.** Recall that, in the split case, the construction of the apartment $\mathcal{A}(S)$ was given as follows:

(1) Define the notion of a valuation of root datum and the set of valuations of root datum $\mathrm{VRD}(G, S)$.
(2) Define the notion of the equipollence relation on $\mathrm{VRD}(G, S)$.
(3) Define Chevalley valuations using the notion of a weak Chevalley system.
(4) Define $\mathcal{A}(S)$ as the set of valuations of root datum which are equipollent to some Chevalley valuation.
(5) Prove the well-definedness of $\mathcal{A}(S)$, i.e., independence of the choice of a Chevalley valuation (in other words, all Chevalley valuations are equipollent).

Among these steps, (1) and (2) work also for non-split connected reductive groups; in the definitions, we did not assume that G is split. However, the notion of a weak Chevalley system was given in terms of the absolute root system $\tilde{\Phi}$. So our task is to modify the notion of a weak Chevalley system in the quasi-split case so that all (3)–(5) work as well.

Recall that a weak Chevalley system of $G_{k^{\mathrm{sep}}}$ is a family $\{\pm X_a \in \mathfrak{g}_a(k) \mid a \in \Phi(G, T)\}$ of pairs $\pm X_a$ of nonzero root vectors such that $\{\pm X_a \in \mathfrak{g}_a(k) \mid a \in$

$\Phi(G,T)\}$ is preserved by the Weyl action of $w(X_a)$ for any $a \in \Phi(G,T)$. Here note that, by definition, $\Phi(G,T) \subset X^*(T)$ is the set of weights of $T_{k^{\mathrm{sep}}}$ appearing in the adjoint action on $\mathfrak{g}_{k^{\mathrm{sep}}}$. Also note that, $X^*(T)$ is the group of absolute characters $\mathrm{Hom}_{k^{\mathrm{sep}}}(T_{k^{\mathrm{sep}}}, \mathbb{G}_m)$.

Definition 8.2. A weak Chevalley–Steinberg system for G is a Θ-invariant weak Chevalley system for $G_{k^{\mathrm{sep}}}$.

As in the split case, choosing a weak Chevalley–Steinberg system is in fact equivalent to choosing a k-pinning of G.

Fact 8.3. *Let X' be a weak Chevally–Steinberg system for G. Then there exists a Chevalley system X for $G_{k^{\mathrm{sep}}}$ which refines X' and satisfies the following Θ-invariance properties: Suppose that $\tilde{a} \in \tilde{\Phi}$ restricts to $a \in \Phi$.*

1. *If a is of type R1 or R2, then $\sigma(X_{\tilde{a}}) = X_{\sigma(\tilde{a})}$ for any $\sigma \in \Theta$.*
2. *If a is of type R2 and $\tilde{a}' \in \Theta \cdot \tilde{a}$ is the unique root such that $\tilde{b} = \tilde{a} + \tilde{a}' \in \tilde{\Phi}$, then $\sigma(X_{\tilde{b}}) = \epsilon(\sigma) \cdot X_{\tilde{b}}$ for any $\sigma \in \Theta_{\tilde{b}}$, where $\epsilon\colon \Theta_{\tilde{b}}/\Theta_{\tilde{a}} \to \{\pm 1\}$ is the unique nontrivial character.*

Definition 8.4. A Chevalley system of $G_{k^{\mathrm{sep}}}$ satisfying the property of the above fact is called a *Chevalley–Steinberg system*.

Exercise 8.5. Find a Chevalley–Steinberg system for SU_3 with respect to a quadratic extension l/k. (Look at Section 2.5.)

8.3. **Chevalley valuation in the quasi-split case.** Let G be a quasi-split connected reductive group over k. Let l be a finite Galois extension of k over which G splits.

We fix a Chevalley–Steinberg system $o = \{X_{\tilde{a}} \in \mathfrak{g}_{\tilde{a}}(l) \mid \tilde{a} \in \tilde{\Phi}\}$ of G. (By the definition of a Chevalley–Steinberg system, this is a Chevally system for $G_{k^{\mathrm{sep}}}$. We may choose it so that it is defined over l.) For each $a \in \Phi$, define a function

$$\varphi_a \colon U_a(k) \to \mathbb{R} \cup \{\infty\}$$

using o in the following way.

1. Suppose that a is of type R1, i.e., $2a, \frac{1}{2}a \notin \Phi$. We choose any lift $\tilde{a} \in \tilde{\Phi}$ of a. Then, as discussed above, we have the isomorphism $U_a(k) \cong U_{\tilde{a}}(k_{\tilde{a}})$. By using the fixed Chevalley–Steinberg system o, we define an isomorphism $\mathbb{G}_{\mathrm{a}} \cong U_{\tilde{a}}$ over $k_{\tilde{a}}$ such that its differential maps 1 to $X_{\tilde{a}}$. We define φ_a to be the composition of these isomorphisms with the valuation $\omega\colon k_{\tilde{a}} \to \mathbb{R} \cup \{\infty\}$:
$$\varphi_a \colon U_a(k) \cong U_{\tilde{a}}(k_{\tilde{a}}) \cong k_{\tilde{a}} \xrightarrow{\omega} \mathbb{R} \cup \{\infty\}.$$
2. Suppose that a is of type R2, i.e., $2a \in \Phi$. We choose liftings $\tilde{a}, \tilde{a}' \in \tilde{\Phi}$ of a such that $\tilde{b} := \tilde{a} + \tilde{a} \in \tilde{\Phi}$. Then we have an isomorphism $U_{[\tilde{a}]}(k_{\tilde{b}}) \cong U_a(k)$. By using the fixed Chevalley–Steinberg system o, we define an isomorphism $U_{k_{\tilde{a}}/k_{\tilde{b}}} \cong U_{[\tilde{a}]}$ over $k_{\tilde{a}}$ such that its differential maps 1 to $X_{\tilde{a}}$. We define φ_a using these isomorphisms as follows:
$$\varphi_a \colon U_a(k) \cong U_{[\tilde{a}]}(k_{\tilde{b}}) \cong U_{k_{\tilde{a}}/k_{\tilde{b}}}(k_{\tilde{b}}) \to \mathbb{R} \cup \infty \colon (u,v) \mapsto \frac{1}{2}\omega(v).$$
3. Suppose that a is of type R3, i.e., $\frac{1}{2}a \in \Phi$. In this case, we define $\varphi_a := 2 \cdot \varphi_{\frac{1}{2}a}|_{U_a(k)}$.

We call the valuation of root datum of (G, S) associated to a Chevalley–Steinberg system of G a *Chevalley valuation*. The following lemma can be proved in the exactly same manner as in the split case.

Lemma 8.6. *Let φ be a Chevalley valuation of (G, S). Then, for any $\varphi' \in \mathrm{VRD}(G, S)$, φ' is a Chevalley valuation if and only if there exists a (necessarily unique) $v \in \omega(T_{\mathrm{ad}}) \subset V(S)$ such that $\varphi' = \varphi + v$. In particular, all Chevalley valuations of (G, S) are equipollent to each other.*

Definition 8.7. We let $\mathcal{A}(S)$ be the equipollence class of a Chevalley valuation. We call $\mathcal{A}(S)$ the *apartment of G with respect to S*.

Recall that, in the last week, we have defined an action of $V(S) \rtimes N_{\mathrm{ad}}(k)$ on $\mathrm{VRD}(G, S)$. By the same argument as in the split case, we can verify that $\mathcal{A}(S) \subset \mathrm{VRD}(G, S)$ is preserved under the action of $N_{\mathrm{ad}}(k)$. Furthermore, we can also generalize Lemma 7.15 so that we can verify that Axiom 3 holds for $\mathcal{A}(S)$.

Also recall that we have assumed that G is semisimple so far. Now let G be a general quasi-split connected reductive group with maximal split central torus A_G. Let S be a split maximal torus of G. Let S' be the maximal split subtorus of $S \cap G_{\mathrm{der}}$. Then, since $S' \hookrightarrow S \twoheadrightarrow S/A_G$ is an isogeny, we have that $V(S') \hookrightarrow V(S) \twoheadrightarrow V(S/A_G)$ is an isomorphism (as $\mathbb{R}$-vector spaces). This enables us to obtain a retraction $V(S) \twoheadrightarrow V(S')$ to the inclusion $V(S') \hookrightarrow V(S)$.

By noting that G and G_{der} have the same root systems (and also root subgroups, Weyl groups, and so on), we define the apartment $\mathcal{A}(S)$ of G to be the apartment $\mathcal{A}(S')$ of S'. We let $V(S)$ act on $\mathcal{A}(S)$ through the surjection $V(S) \twoheadrightarrow V(S')$; then $\mathcal{A}(S)$ is an affine space under $V(S')$ and we can check that Axiom 3 holds.

9. Week 9: Affine root systems on apartments

9.1. **Affine root systems.** We begin with defining an affine root system structure on $\mathcal{A}(S)$; so our next aim is to show that Axiom 4 is satisfied by $\mathcal{A}(S)$. By fixing a k-split maximal torus S of G, we shortly write $\mathcal{A}$ for $\mathcal{A}(S)$ in the following.

Recall that $\mathcal{A}$ is defined to be the equipollence of a(ny) Chevalley valuation o of G, i.e., $\mathcal{A} = o + V(S)$. Hence, each point $x \in \mathcal{A}$ defines, for each root $a \in \Phi = \Phi(G,S)$, a descending filtration $\{U_{a,x,r}\}_{r\in\mathbb{R}}$ of the root subgroup $U_a(k)$. (So far, the symbol "φ" had been used for a valuation of root datum; from now on, we use "x" to emphasize our geometric viewpoint, i.e., x is a "point".) To be more precise, recall that $x \in \mathcal{A}$ is a family of maps

$$x = \{x_a \colon U_a(k) \to \mathbb{R} \cup \{\infty\}\}_{a\in\Phi}$$

and $U_{a,x,r}$ is defined by

$$U_{a,x,r} := x_a^{-1}([r,\infty]).$$

Definition 9.1. For any affine functional $\psi \in \mathcal{A}^*$ whose derivative is $\dot{\psi} = a \in \Phi$ and $x \in \mathcal{A}$, we put

$$U_\psi := U_{a,x,\psi(x)} \subset U_a(k)$$

and

$$U_{\psi+} := \bigcup_{\psi'>\psi} U_{\psi'} = U_{a,x,\psi(x)+\varepsilon} \subset U_a(k),$$

where ψ' runs over affine functionals with derivative a which are greater than ψ and ε is any sufficiently small positive real number.

Lemma 9.2. *The subgroup U_ψ is well-defined, i.e., independent of the choice of $x \in \mathcal{A}$.*

Proof. Let $y \in \mathcal{A}$ be another point. Since $\mathcal{A}$ is an affine space under $V(S)$, there uniquely exists an element $v \in \mathcal{A}$ satisfying $y = x + v$. Then we have

$$U_{a,y,\psi(y)} = U_{a,x+v,\psi(x+v)} = U_{a,x+v,\psi(x)+a(v)} = (x+v)_a^{-1}([\psi(x)+a(v),\infty]).$$

By the definition of "$x+v$" (i.e., how $V(S)$ acts on $\mathrm{VRD}(G,S)$), we have

$$(x+v)_a(u) = x_a(u) + a(v)$$

for any $u \in U_a(k)$, which implies that

$$(x+v)_a^{-1}([\psi(x)+a(v),\infty]) = x_a^{-1}([\psi(x),\infty])$$

(this is nothing but the proof of the second statement of Lemma 7.4). In other words, we have $U_{a,y,\psi(y)} = U_{a,x,\psi(x)}$. □

Definition 9.3. We define subsets $\Psi \subset \Psi' \subset \mathcal{A}^*$ by

$$\Psi' := \{\psi \in \mathcal{A}^* \mid \dot{\psi} \in \Phi,\, U_{\psi+} \subsetneq U_\psi\},$$

$$\Psi := \{\psi \in \mathcal{A}^* \mid \dot{\psi} \in \Phi,\, U_{\psi+} \cdot U_{2a}(k) \subsetneq U_\psi \cdot U_{2a}(k)\}.$$

We call Ψ the *affine root system of G (with respect to S)* and an element of Ψ an *affine root.*

Remark 9.4. In the above definition, we think of $U_{2a}(k)$ as $\{1\}$ if $2a$ is not a root. Hence, if Φ is a reduced root system (this always holds when G is split), we have $\Psi = \Psi'$.

Proposition 9.5 ([KP23, Proposition 6.3.13])**.** *The sets Ψ and Ψ' are an affine root system in $\mathcal{A}^*$ with derivative Φ in the abstract sense.*

Recall that the affine root system Ψ in $\mathcal{A}$ induces a polysimplicial structure on $\mathcal{A}$. To be more precise, for each $\psi \in \Psi$, we set

$$H_\psi := \{x \in \mathcal{A}(S) \mid \psi(x) = 0\}.$$

We define the chambers of $\mathcal{A}$ to be the connected components of

$$\mathcal{A} \smallsetminus \bigcup_{\psi \in \Psi} H_\psi.$$

Then we define the codimension 1 facets to be the connected components of

$$\bigcup_{\psi \in \Psi} H_\psi \smallsetminus \bigcup_{\substack{\psi, \psi' \in \Psi \\ H_\psi \neq H_{\psi'}}} (H_\psi \cap H_{\psi'}).$$

Repeating this procedure, eventually we arrive at the facets of the smallest dimension (vertices).

9.2. **Affine root systems in the split case.** We suppose that G is split. We fix a Chevalley valuation $o \in \mathcal{A}(S)$.

Proposition 9.6. *The set of affine roots Ψ is isomorphic to the affine root system $\Phi \times \mathbb{Z}$ (see the proof for the definition of "$\Phi \times \mathbb{Z}$"). Moreover, $\Psi = \Psi'$.*

Proof. We first recall that the sets Ψ and Ψ' are defined by

$$\Psi' := \{\psi \in \mathcal{A}^* \mid \dot{\psi} \in \Phi,\, U_{\psi+} \subsetneq U_\psi\},$$

$$\Psi := \{\psi \in \mathcal{A}^* \mid \dot{\psi} \in \Phi,\, U_{\psi+} \cdot U_{2a}(k) \subsetneq U_\psi \cdot U_{2a}(k)\}.$$

As noted in a Remark, when Φ is reduced (this is true when G is split), these sets are the same.

To compute Ψ, we next recall the definitino of U_ψ. By choosing any Chevalley valuation $x = \{x_a\}_{a \in \Phi} \in \mathcal{A}(S)$, we have defined

$$U_\psi := U_{a,x,\psi(x)} := x_a^{-1}([\psi(x), \infty]),$$

$$U_{\psi+} := U_{a,x,\psi(x)+\varepsilon} := x_a^{-1}([\psi(x) + \varepsilon, \infty]),$$

where $a := \dot{\psi}$. Also recall that, if we let $X = \{X_a \in \mathfrak{g}_a(k)\}_{a \in \Phi}$ be a weak Chevalley system which gives rise to x, then $x_a \colon U_a(k) \to \mathbb{R} \cup \{\infty\}$ is defined by composing $U_a(k) \cong \mathbb{G}_{\mathrm{a}}(k) = k$ which is determined by X_a and the valuation map $\omega \colon k \to \mathbb{Z} \cup \{\infty\}$. In particular, the set of jumps of $\{U_{a,x,r}\}_{r \in \mathbb{R}}$ (i.e., the numbers r where $U_{a,x,r+} \subsetneq U_{a,x,r}$) is exactly given by $\mathbb{Z}$. Therefore, we obtain

$$\Psi = \{\psi_{a,r} \in \mathcal{A}(S)^* \mid a \in \Phi, r \in \mathbb{Z}\},$$

where $\psi_{a,r} \colon \mathcal{A}(S) \to \mathbb{R}$ is the affine functional such that $\psi_{a,r}(x + v) = a(x) + r$ for any $v \in V(S)$. On the other hand, by definition, $\Phi \times \mathbb{Z}$ is the affine root system realized in $V(S)^*$ (we choose the underlying affine space to be $V(S)$), i.e., each $(a, r) \in \Phi \times \mathbb{Z}$ is regarded as the affine functional $v \mapsto a(v) + r$ on $V(S)$. Obviously these two affine root systems are isomorphic by $\psi_{a,r} \leftrightarrow (a, r)$. $\square$

9.3. **Special points and extra special points.** For any $x \in \mathcal{A}(S)$, we set

$$\Psi_x := \{\psi \in \Psi \mid \psi(x) = 0\}$$

and $\Phi_x := \dot{\Psi}_x$.

Definition 9.7. Let $x \in \mathcal{A}(S)$ be a point.

(1) We say that x is a *special point* if, for any $\psi \in \Psi$, there exists $\psi' \in \Psi_x$ such that H_ψ and $H_{\psi'}$ are parallel.
(2) We say that x is an *extra special point* if there exists $\psi_1, \ldots, \psi_l \in \Psi_x$ such that $\{\dot{\psi}_1, \ldots, \dot{\psi}_l\}$ is a basis of $\Phi := \dot{\Psi}$.

Example 9.8. We let $G := \mathrm{SL}_2$. We choose S to be the diagonal maximal split torus, i.e.,

$$S(R) = \{\mathrm{diag}(t, t^{-1}) \mid t \in R^\times\}$$

for any k-algebra R. Then the set Φ of roots is given by

$$\Phi = \{\pm 2e_1\},$$

where $e_1 \colon S \mapsto \mathbb{G}_{\mathrm{m}} \colon \mathrm{diag}(t, t^{-1}) \mapsto t$. The coroot corresponding to $2e_1$ is $e_1^\vee \colon t \mapsto \mathrm{diag}(t, t^{-1})$. As G is split, the affine root system of G is given by $\Phi \times \mathbb{Z}$. Therefore, after choosing a Chevalley valuation of G with respect to S, we may identify the apartment $\mathcal{A}(S)$ with

$$V(S) = X_*(S)_{\mathbb{R}} = \mathbb{R}e_1^\vee.$$

For each affine root $2e_1 + r$ with $r \in \mathbb{Z}$, its hyperplane H_{2e_1+r} is given by $\{x \in \mathcal{A}(S) \mid 2e_1(x) + r = 0\} = \{-\frac{r}{2}e_1^\vee\}$. Thus any vertex is obviously special and extra special.

Example 9.9. We let $G := \mathrm{SL}_3$. We choose S to be the diagonal maximal split torus, i.e.,

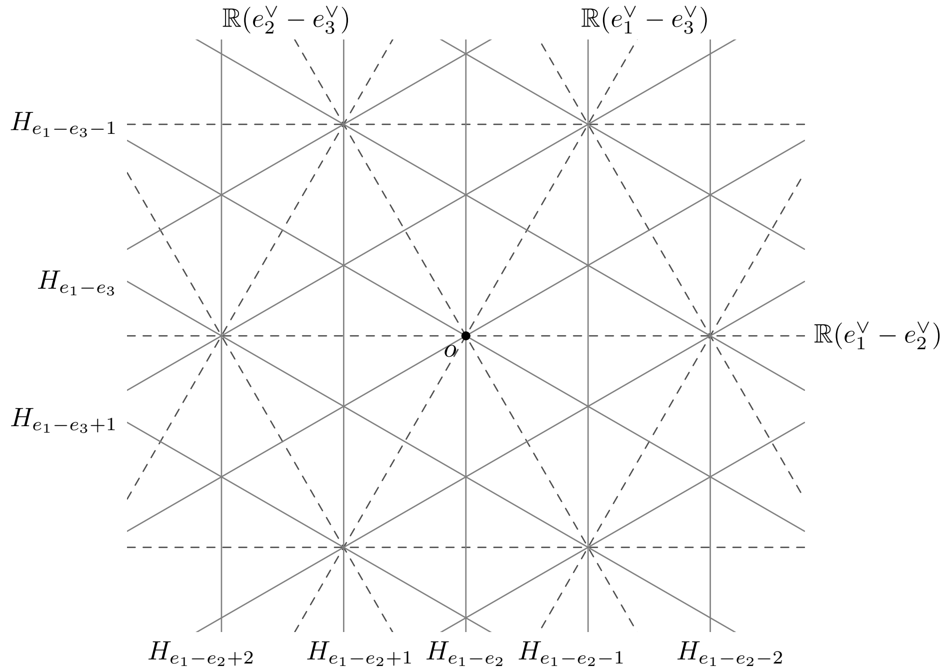
$$S(R) = \{\mathrm{diag}(t_1, t_2, t_3) \mid t_i \in R^\times,\ t_1t_2t_3 = 1\}$$

for any k-algebra R. Then the set Φ of roots is given by

$$\Phi = \{\pm(e_i - e_j) \mid i \neq j,\ 1 \leq i < j \leq 3\},$$

where $e_i \colon S \mapsto \mathbb{G}_{\mathrm{m}} \colon \mathrm{diag}(t_1, t_2, t_3) \mapsto t_i$. With the standard choice of a Borel subgroup (i.e., upper-triangular), its root basis Δ can be taken to be $\{e_1 - e_2,\ e_2 - e_3\}$. As G is split, the affine root system of G is given by $\Phi \times \mathbb{Z}$. Therefore, after choosing a Chevalley valuation of G with respect to S, we may identify the apartment $\mathcal{A}(S)$ with

$$V(S) = X_*(S)_{\mathbb{R}} = \mathbb{R}(e_1^\vee - e_2^\vee) \oplus \mathbb{R}(e_2^\vee - e_3^\vee).$$



By looking at this picture, we can easily see that all vertices are completely symmetric to each other. Each trigangle is a chamber (facet of dimension 2), each edge of a triangle is a dimension 1 facet, and each vertex is a facet of diension 0. At any vertex, there exist exactly three hyperplanes (lines), corresponding to affine roots whose derivatives are $e_1 - e_2$, $e_2 - e_3$, and $e_1 - e_3$. Hence every vertex is extra special.

Example 9.10. We let $G := \mathrm{SO}_5$;

$$G(R) = \{g \in \mathrm{SL}_5(R) \mid {}^t g J_5 g = J_5\},$$

where J_5 is the antidiagonal matrix with antidiagonal entries $(1, -1, 1, -1, 1)$, for any k-algebra R. We choose S to be the diagonal maximal torus, so

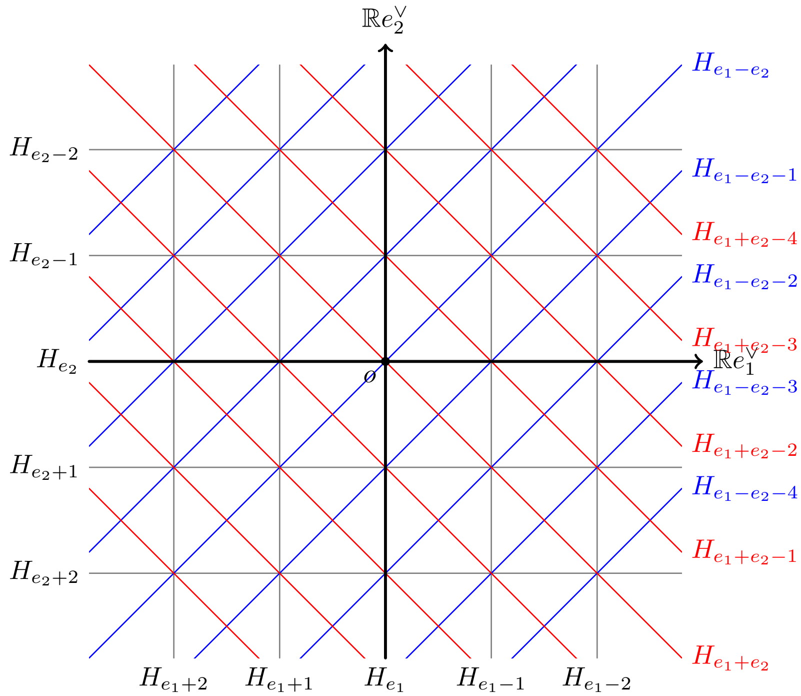
$$S(R) = \{\mathrm{diag}(t_1, t_2, 1, t_2^{-1}, t_1^{-1}) \mid t_1, t_2 \in R^\times\}.$$

Then the set Φ of roots is given by

$$\Phi = \{\pm e_1 \pm e_2\} \sqcup \{\pm e_1, \pm e_2\},$$

where $e_i \colon S \to \mathbb{G}_{\mathrm{m}} \colon \mathrm{diag}(t_1, t_2, 1, t_2^{-1}, t_1^{-1}) \mapsto t_i$. With the standard choice of a Borel subgroup (i.e., upper-triangular), its root basis Δ can be taken to be $\{e_1 - e_2, e_2\}$. As G is split, the affine root system of G is given by $\Phi \times \mathbb{Z}$. Therefore, after choosing a Chevalley valuation of G with respect to S, we may identify $\mathcal{A}(S)$ with

$$V(S) = X_*(S)_{\mathbb{R}} = \mathbb{R}e_1^\vee \oplus \mathbb{R}e_2^\vee.$$



By looking at this picture, we see that a chamber is a right triangle with three edges given by a black line, a red line, and a blue line. Hence there are two kinds of vertices;

(1) corners of the squares, where red, blue, and two black lines cross,
(2) centers of the squares, where only red and blue lines cross.

The vertices of the former type are extra special while the vertices of the latter type are not even special.

Exercise 9.11. Determine and classify (i.e., do the same computation as in the above examples) all the vertices of the apartment of G_2.

Note that the definion of a special point and an extra special point make sense for any affine root system, in general. The following is a general lemma which is true for any affine root system (in the book of Kaletha–Prasad, Bourbaki is cited for the latter assertion.)

Lemma 9.12 ([KP23, Lemmas 1.3.3, 1.3.42])**.** *The derivation map of Weyl groups $W(\Psi) \to W(\Phi)$ is surjective. Moreover, for $x \in \mathcal{A}(S)$, the restriction of the derivation map to the stabilizer group $W(\Psi)_x$ is still surjective if and only if x is a special point.*

Proposition 9.13. (1) *There exists an extra special point in $\mathcal{A}(S)$.*
(2) *Any extra special point is a special point.*
(3) *Any special point is a vertex of $\mathcal{A}(S)$.*
(4) *If $\Phi = \dot{\Psi}$ is reduced, then being special is equivalent to being extra special.*

Proof. We first show (1). Let $\Delta = \{a_1, \ldots, a_l\}$ be a basis (set of simple roots) of Φ. By choosing any $\psi_i \in \Psi$ for each i such that $\dot{\psi}_i = a_i$, we see that the intersection of the hyperplanes $H_{\psi_1}, \ldots, H_{\psi_l}$ consists of exactly one point, which is an extra special point.

We next show (2). Let $x \in \mathcal{A}(S)$ be an extra special point. By definition, we can choose $\psi_1, \ldots, \psi_l \in \Psi_x$ such that $\{\dot{\psi}_1, \ldots, \dot{\psi}_l\}$ is a basis of $\Phi := \dot{\Psi}$. We note that the derivation map $W(\Psi) \to W(\Phi)$ sends the reflection r_ψ with respect to an affine root $\psi \in \Psi$ to the reflection $r_{\dot{\psi}}$ with respect to the derivative $\dot{\psi} \in \Phi$ of ψ. Since $W(\Phi)$ is generated by $r_{\dot{\psi}_i}$ for $i = 1, \ldots, l$, we see that the derivation map $W(\Psi) \to W(\Phi)$ remains surjective when restricted to the stabilizer $W(\Psi)_x$ of x in $W(\Psi)$ (which contains r_{ψ_i} for $i = 1, \ldots, l$). Thus Lemma 9.12 implies that x is special.

We next show (3). Let $x \in \mathcal{A}(S)$ be a special point. Then, for any $a_i \in \Delta$, there exits $\psi_i' \in \Psi_x$ whose derivative is proportional to a_i. In particular, we can find a subset $\{\psi_1', \ldots, \psi_l'\}$ of Ψ_x whose derivatives $\{\dot{\psi}_1', \ldots, \dot{\psi}_l'\}$ form a basis of $V(S)^*$. This implies that

$$\{x\} \subset \bigcap_{\psi \in \Psi_x} H_\psi \subset \bigcap_{i=1}^{l} H_{\psi_i'} = \{x\}.$$

Hence x is a vertex of $\mathcal{A}(S)$.

Finally, we show (4). In the discussion of the previous paragraph, if it is given that Φ is reduced, then the part "there exits $\psi_i' \in \Psi_x$ whose derivative is proportional to a_i" can be strengthened to "there exits $\psi_i' \in \Psi_x$ whose derivative is equal to a_i". This means that x is an extra special point. □

Proposition 9.14. *When G is split, the following are equivalent for a point $x \in \mathcal{A}(S)$:*

(1) *x is a special point.*
(2) *x is an extra special point.*
(3) *x is a Chevalley valuation.*

Proof. By Proposition 9.13 (3), the condition (1) is equivalent to the condition (2). By our discussion in Proposition 9.6, it is obvious that (3) implies (1) or (2). Thus our task is to show that (1) implies (3).

Suppose that o is a Chevalley valuation and $x = o + v \in \mathcal{A}(S)$ is a special point. By definition, for any $\psi \in \Psi$, there exists $\psi' \in \Psi_x$ such that H_ψ and $H_{\psi'}$ are parallel, or equivalently, $\dot{\psi} = \dot{\psi}'$. Since we have $\Psi = \Phi \times \mathbb{Z}$, this means that for any $a \in \Phi$, we have $a(v) \in \mathbb{Z}$. In other words, $v \in V(S) = X_*(S)_{\mathbb{R}}$ in fact belongs to the coweight lattice. Recall that the valuation homomorphism $\omega_{T_{\mathrm{ad}}}$ for T_{ad} (the image of T in the adjoint quotient of G) is a surjective homomorphism from $T_{\mathrm{ad}}(k)$ to the coweight lattice in $V(S)$. If we let $t \in T_{\mathrm{ad}}(k)$ be an element such that $\omega_{T_{\mathrm{ad}}}(t) = v$, then we have $t \cdot o = x$. This means that x is the Chevalley valuation associated to the weak Chevalley system $t \cdot X$, where X is the weak Chevalley system which gives rise to the Chevalley valuation o. □

10. Week 10: More about affine root systems

10.1. **Affine reflection on apartments.** In the following, we write $\mathcal{A} = \mathcal{A}(S)$ and $V = V(S') = X_*(S')_{\mathbb{R}}$ in short.

Let $a \in \Phi(G,S)$. For $u \in U_a(k)^* := U_a(k) \smallsetminus \{1\}$, we define a subset $H_u \subset \mathcal{A}$ by

$$H_u := \{x \in \mathcal{A} \mid x_a(u) = 0\}.$$

Note that this is a hyperplane in $\mathcal{A}$ with derivative $a^{\perp} := \{v \in V \mid a(v) = 0\}$ (i.e., an affine subspace of $\mathcal{A}$ under the $\mathbb{R}$-subspace $a^{\perp} \subset V$). Indeed, by fixing a Chevalley valuation o of G and writing $x = o + v$ with $v \in V$, we have

$$x_a(u) = (o+v)_a(u) = o_a(u) + a(v).$$

Hence, if we choose any $v_0 \in V$ satisiying $a(v_0) = -o_a(u)$, then we have $x_a(u) = 0$ if and only if $v \in v_0 + a^{\perp}$. In other words, we have $H_u = o + v_0 + a^{\perp}$. (Note that v_0 can be chosen more explicitly; since $a(a^{\vee}) = 2$, we may choose $v_0 := -\frac{1}{2}o_a(u)a^{\vee}$.)

Definition 10.1. We let $r_u\colon \mathcal{A} \to \mathcal{A}$ be the unique affine endomorphism such that

(1) the derivative of r_u is the reflection $r_a \in \mathrm{End}_{\mathbb{R}}(V)$ with respect to the root $a \in \Phi$, and
(2) r_u fixes the hyperplane H_u pointwise.

Explicitly, r_u is given by

$$x \mapsto x - x_a(u) \cdot a^{\vee}.$$

We call r_u the *affine reflection along H_u*.

Recall that, for any $u \in U_a(k)^*$, there exist unique $u', u'' \in U_{-a}(k)^*$ satisfying $u'uu'' \in N(k)$, which project to the reflection with respect to the root a in the Weyl group. We put $m(u) := u'uu''$.

Proposition 10.2. *The action of $m(u) \in N(k)$ on $\mathcal{A}$ is the affine reflection along H_u.*

Proof. As noted above, the affine reflection along H_u is explicitly given by $x \mapsto x - x_a(u) \cdot a^{\vee}$. Thus our task is to show that $m(u) \cdot x = x - x_a(u) \cdot a^{\vee}$ for any $x \in \mathcal{A}$. By fixing $x \in \mathcal{A}$, we put $y := x - x_a(u) \cdot a^{\vee}$. Recall that each $x \in \mathcal{A}$ is a family of maps $\{x_b\colon U_b(k) \to \mathbb{R} \cup \{\infty\}\}_{b \in \Phi}$. This can be recovered uniquely by the associated filtration $\{U_{b,x,r}\}_{b \in \Phi, r \in \mathbb{R}}$ defined by $U_{b,x,r} := x_b^{-1}([r,\infty])$. Hence, it is enough to check that $U_{b,m(u)\cdot x,r} = U_{b,y,r}$ for all $b \in \Phi$ and $r \in \mathbb{R}$.

Since $m(u) \in N(k)$ acts on $x \in \mathcal{A}$ by $(m(u) \cdot x)_b := m(u) \cdot x_b$, where $(m(u) \cdot x_b)(v) := x_{m(u)^{-1}\cdot b}(m(u)^{-1} \cdot v \cdot m(u))$ for $v \in U_b(k)$, we can easily check that $U_{b,m(u)\cdot x,r} = m(u) \cdot U_{m(u)^{-1}\cdot b,x,r} \cdot m(u)^{-1}$. On the other hand, we have $U_{b,y,r} = U_{b,x-x_a(u)\cdot a^{\vee},r} = U_{b,x,r+x_a(u)\cdot b(a^{\vee})}$. Therefore, it suffices to show that

$$m(u) \cdot U_{m(u)^{-1}\cdot b,x,r} \cdot m(u)^{-1} = U_{b,x,r+x_a(u)\cdot b(a^{\vee})}$$

for all $b \in \Phi$ and $r \in \mathbb{R}$. Or equivalently, it suffices to show that

$$m(u)^{-1} \cdot U_{b,x,r} \cdot m(u) = U_{m(u)^{-1}\cdot b,x,r-x_a(u)\cdot b(a^{\vee})}.$$

We first consider the case where $b \notin \mathbb{Q} \cdot a$. In the following, we write $s := x_a(u)$; note that hence $u \in U_{a,x,s}$. We put

$$\mathcal{U} := \langle U_{ma+nb,x,ms+nr} \mid m \in \mathbb{Z}_{\geq 0}, n \in \mathbb{Z}_{\geq 1} \text{ such that } ma+nb \in \Phi \rangle,$$

$$\mathcal{V} := \langle U_{ma+nb,x,ms+nr} \mid m \in \mathbb{Z}_{\geq 0}, n \in \mathbb{Z}_{>1} \text{ such that } ma+nb \in \Phi \rangle.$$

By the condition (4) of the definition of a valuation of root datum (Definition 7.1), we have $[\mathcal{U}, \mathcal{U}] \subset \mathcal{V}$. In particular, $\mathcal{V}$ is a normal subgroup of $\mathcal{U}$ such that $\mathcal{U}/\mathcal{V}$ is abelian. Again by the condition (4), we see that $[U_{a,x,s}, \mathcal{U}] \subset \mathcal{U}$ and $[U_{a,x,s}, \mathcal{V}] \subset \mathcal{V}$. In particular, this implies that the conjugate action of u stabilizes $\mathcal{U}$ and $\mathcal{V}$. Similarly, by also using the condition (6) (and Exercise below Definition 7.1), we see that the conjugate actions of u' and u'' stabilizes $\mathcal{U}$ and $\mathcal{V}$. Accordingly, the conjugate action of $m(u) = u'uu''$ also stabilizes $\mathcal{U}$ and $\mathcal{V}$.

We write $\pi \colon \mathcal{U} \twoheadrightarrow \mathcal{U}/\mathcal{V}$ for the natural surjection. Then, as $\mathcal{U}/\mathcal{V}$ is abelian, we have

$$\mathcal{U}/\mathcal{V} \cong \prod_{\substack{m \in \mathbb{Z}_{\geq 1} \\ ma+b \in \Phi}} \pi(U_{ma+b,x,ms+r}),$$

where the product on the right-hand side can be taken in any order. Moreover, under this surjection each $U_{ma+b,x,ms+r}$ is mapped to the product injectively. Note that, as $m(u) \in N(k)$, the action of $m(u)$ (hence also $m(u)^{-1}$) on $\mathcal{U}/\mathcal{V}$ is given by permuting the factors $\pi(U_{ma+b,x,ms+r})$ in the product expression above. Since we have $m(u)^{-1} \cdot b = m(u) \cdot b = -b(a^\vee) \cdot a + b$, we conclude that

$$m(u)^{-1} \cdot \pi(U_{b,x,r}) \cdot m(u) = \pi(U_{m(u)^{-1} \cdot b, x, -b(a^\vee) \cdot s + r}).$$

This implies that we also have

$$m(u)^{-1} \cdot U_{b,x,r} \cdot m(u) = U_{m(u)^{-1} \cdot b, x, -b(a^\vee) \cdot s + r}.$$

We next consider the case where $b \in \mathbb{Q} \cdot a$. Since we have $U_{a,x,r} \cap U_{2a}(k) = U_{2a,x,2r}$ (for any $a \in \Phi$ and $r \in \mathbb{R}$) by the definition of a valuation of root datum, it is enough to treat only the cases where $b = \pm a$.

Suppose that $b = -a$. Then what we have to show is

$$m(u)^{-1} \cdot U_{a,x,r} \cdot m(u) = U_{-a,x,-2s+r}.$$

Since $m(u) = u'uu''$, we have

$$m(u) = u'uu'' = uu''(m(u)^{-1}u'm(u)).$$

Since u and $m(u)^{-1}u'm(u)$ belong to $U_a(k)$, this identity shows that $m(u) = m(u'')$. Hence, by the definition of a valuation of root datum (condition (6)), we have $s = x_a(u) = -x_{-a}(u'') = x_a(m(u)^{-1}u'm(u))$. We note that the function

$$U_{-a}(k) \smallsetminus \{1\} \to \mathbb{R} \colon v \mapsto x_{-a}(v) - x_a(m(u)^{-1}vm(u))$$

is constant by the definition of a valuation of root datum (condition (2)). From the above observation, we see that this function takes the value $-2s$ at $v = u'$. Thereffore, we get $x_a(m(u)^{-1}vm(u)) = x_{-a}(v) + 2s$ for any $v \in U_{-a}(k) \smallsetminus \{1\}$. This is exactly what we wanted to show.

The case $b = a$ can be treated in the same manner. □

The following lemma can be proved by examining the definitions of a valuation of root datum and the action of $N(k)$ on $\mathrm{VRD}(G, S)$ (but not very obvious):

Lemma 10.3 ([KP23, Fact 6.3.3])**.** *For any $n \in N(k)$ and $\psi \in \mathcal{A}^*$ (with derivative in Φ), we have $U_{n\psi} = nU_\psi n^{-1}$.*

Lemma 10.4. *The action of $N(k)$ on $\mathcal{A}^*$ preserves Ψ and Ψ'.*

Proof. It follows from Lemma 10.3 that Ψ' is preserved by $N(k)$. So let us think about Ψ.

Let $a \in \Phi$, $n \in N(k)$, and $b := n \cdot a \in \Phi$. If $\psi\colon \mathcal{A} \to \mathbb{R}$ is an affine functional with derivative $\dot{\psi} = a$, then $n \cdot \psi$ (defined by $n \cdot \psi(x) := \psi(n^{-1} \cdot x)$) is an affine functional with derivative $n \cdot a$. Indeed, we have

$$(n \cdot \psi)(x + v) = \psi(n^{-1} \cdot (x + v)) = \psi(n^{-1} \cdot x) + a(n^{-1} \cdot v) = (n \cdot \psi)(x) + (n \cdot a)(v).$$

Now suppose that $\psi \in \Psi'$. Then ψ belongs to Ψ if and only if $U_{\psi+} \cdot U_{2a}(k) \subsetneq U_{\psi} \cdot U_{2a}(k)$. By applying n-conjugation, this is equivalent to that $(n \cdot U_{\psi+} \cdot n^{-1}) \cdot U_{2na}(k) \subsetneq (n \cdot U_{\psi} \cdot n^{-1}) \cdot U_{2na}(k)$. By the above observation and Lemma 10.3, this is equivalent to $n \cdot \psi \in \Psi$.

$\square$

10.2. **Proof of Proposition 9.5.**

Lemma 10.5. *Let $\psi \in \Psi'$ and $a := \dot{\psi}$.*

1. *If $2a$ is not a root, then $\psi \in \Psi$.*
2. *If $\psi \notin \Psi$, then $2\psi \in \Psi$.*

Proof. The first assertion (1) follows from the definitions of Ψ and Ψ'. For the second assertion (2), please look at [KP23, Proposition 6.3.8]. $\square$

Now we prove Proposition 9.5.

Recall that, in general, a subset Ψ of $\mathcal{A}^*$ is called an affine root system if the following are satisfied:

1. The subset Ψ spans A^* and does not contain constant functionals. Moreover, $\dot{\Psi} \subset V^*$ is a finite set.
2. For any $\psi \in \Psi$, there exists $\dot{\psi}^\vee \in V$ satisfying $\langle \dot{\psi}, \dot{\psi}^\vee \rangle = 2$. Moreover, the *reflection* $r_{\psi, \dot{\psi}^\vee}\colon A^* \to A^*$ defined by
$$r_{\psi,\dot{\psi}^\vee}(x) := x - \dot{x}(\dot{\psi}^\vee)\psi \in A^*$$
preserves the set Ψ.
3. For any $\psi, \eta \in \Psi$, we have $\langle \dot{\psi}, \dot{\eta}^\vee \rangle \in \mathbb{Z}$.
4. For any $a \in \Phi$, $\{\psi \in \Psi \mid \dot{\psi} = a\}$ does not have an accumulation point.

Lemma 10.6 ([KP23, Lemma 1.3.12])**.** *A subset $\Psi \subset \mathcal{A}^*$ is an affine root system if and only if the following are satisfied:*

1. *$\dot{\Psi} \subset V^*$ is a finite root system.*
2. *For each $\psi \in \Psi$, let r_ψ be the affine endomorphism of $\mathcal{A}$ which fixes the hyperplane $H_\psi \subset \mathcal{A}$ and whose derivative is the reflection $r_{\dot{\psi}}$. Then Ψ is preserved by the action of r_ψ on $\mathcal{A}^*$.*
3. *For each $a \in \dot{\Psi}$, the set $\{\psi \in \Psi \mid \dot{\psi} = a\}$ contains at least 2 elements and does not have an accumulation point in $\mathcal{A}^*$.*

Proof of Proposition 9.5. It is enough to verify the conditions (1)–(3) of the above lemma for Ψ and Ψ' and also check that $\dot{\Psi} = \dot{\Psi}' = \Phi$.

First, by the last condition of the definition of a valuation of root datum, $\{U_{a,x,r}\}_{r \in \mathbb{R}}$ forms a fundamental system of open neighborhoods of 1 in $U_a(k)$. In particular, this filtration jumps at infinitely many points, which are discrete in $\mathbb{R}$. Hence, for any $a \in \Phi$, we can find infinitely many $\psi \in \Psi'$ such that $\dot{\psi}$. By Lemma

10.5, this also implies that the same holds for Ψ (in particular, we get $\Phi = \dot{\Psi} = \dot{\Psi}'$). This verifies the conditions (1) and (3) of Lemma 10.6.

Second, recall that the action of $m(u)$ on $\mathcal{A}$ is the affine reflection along H_u (Proposition 10.2). By choosing any point $x \in \mathcal{A}$, we have $U_\psi = U_{a,x,\psi(x)}$ by definition, where $a := \dot{\psi}$. If we choose $u \in U_a(k)$ to be an element of $U_\psi \smallsetminus U_{\psi+} = U_{a,x,\psi(x)} \smallsetminus U_{a,x,\psi(x)+}$, then we have $x_a(u) = \psi(x)$. Hence,

$$\begin{aligned} H_u &= \{y \in \mathcal{A} \mid y_a(u) = 0\} \\ &= \{x + v \in \mathcal{A} \mid (x+v)_a(u) = 0\} \\ &= \{x + v \in \mathcal{A} \mid x_a(u) + a(v) = 0\} \\ &= \{x + v \in \mathcal{A} \mid \psi(x) + a(v) = 0\} \\ &= \{x + v \in \mathcal{A} \mid \psi(x+v) = 0\} = H_\psi. \end{aligned}$$

Thus, the action of $m(u)$ on $\mathcal{A}$ can be also thought of as the affine reflection along H_ψ. □

10.3. **Axiom 4.** Recall that Axiom 4 of Bruhat–Tits theory is as follows:

(1) The action of $m(u)$ on $\mathcal{A}(S)$ fixes a hyperplane $\mathcal{H}_u$ which is an affine subspace of $\mathcal{A}(S)$ under the vector subspace $\mathrm{Ker}(a) \subset V(S)$, where a is also regarded as a natural homomorphism $V(S) \twoheadrightarrow V(\mathbb{G}_{\mathrm{m}}) \cong \mathbb{R}$ induced by the relative root $a\colon S \to \mathbb{G}_{\mathrm{m}}$.
(2) Let $\psi_a^u\colon \mathcal{A}(S) \to \mathbb{R}$ be the unique affine function whose derivative is a and vanishing locus is $\mathcal{H}_u$. Then $\Psi' := \{\psi_a^u \mid a \in \Phi,\, u \in U_a(k) \smallsetminus \{1\}\}$ forms an affine root system with derivative Φ.
(3) For affine function $\psi \in \mathcal{A}(S)^*$ with derivative $a = \dot{\psi} \in \Phi$, we put

$$U_\psi := \{u \in U_a(k) \smallsetminus \{1\} \mid \psi_a^u \geq \psi\} \cup \{1\},$$

$$U_{\psi+} := \{u \in U_a(k) \smallsetminus \{1\} \mid \psi_a^u > \psi\} \cup \{1\}.$$

Then $\Psi := \{\psi \in \Psi' \mid U_{\psi+} \cdot U_{2a}(k) \subsetneq U_\psi \cdot U_{2a}(k)\}$ forms an affine root system with derivative Φ, where U_{2a} is defined to be $\{1\}$ when $2a \notin \Phi$.
(4) For any $\psi \in \Psi'$, at least one of ψ or 2ψ is in Ψ.
(5) The image of $f\colon N(k) \to \mathrm{Aff}(\mathcal{A}(S))$ contains $W(\Psi)$ and is contained in $W(\Psi)^{\mathrm{ext}}$.
   - If G is simply-connected, then the image equals $W(\Psi)$.
   - If G is quasi-split adjoint, then the image equals $W(\Psi)^{\mathrm{ext}}$.
(6) The polysimplicial structure on $\mathcal{A}$ given by the affine root hyperplanes of Ψ (see the paragraph below) coincides with the polysimplicial structure induced by $\mathcal{B}(G)$.

By our discussions so far, we have verified the conditions (1)–(4). The remaining conditions (5) and (6) will be discussed within a few weeks.

10.4. **Affine root systems in the quasi-split case.** Recall that the affine root system Ψ of (G, S) is particularly simple; we have $\Psi = \Phi \times \mathbb{Z}$. Here, let us consider the more general case where G is quasi-split.

We start with the example of a special unitary group. Let SU_3 be the special unitary group associated with a quadratic extension l/k. Recall from Week 2 the following:

- We can take S to be the maximal k-split torus, so that
$$S(k) = \{\mathrm{diag}(t, 1, t^{-1}) \mid t \in k^\times\}.$$
- The relative root system Φ is given by $\{\pm a, \pm 2a\} \subset X^*(S)$, where $a\colon \mathrm{diag}(t, 1, t^{-1}) \mapsto t$.
- The l-rational points of the relative root subgroup for $2a$ are given by
$$U_{2a}(l) = \left\{ \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \middle| \, x \in l \right\}.$$
The Galois action of $\mathrm{Gal}(l/k)$ on $U_{2a}(l)$ changes the $(1,3)$-entry x into $-\bar{x}$. Hence we have
$$U_{2a}(k) = \left\{ \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \middle| \, x \in l^0 \right\},$$
where $l^0 := \mathrm{Ker}(\mathrm{Tr}_{l/k}\colon l \to k)$.
- The l-rational points of the relative root subgroup for a are given by
$$U_a(l) = \left\{ \begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix} \middle| \, x, y, z \in l \right\}.$$
The Galois action of $\mathrm{Gal}(l/k)$ on $U_{2a}(l)$ changes the coordinates (x, y, z) into $(\bar{z}, \overline{xz} - \bar{y}, \bar{x})$. Thus, by putting
$$u_a(x, y) = \begin{pmatrix} 1 & x & y \\ 0 & 1 & \bar{x} \\ 0 & 0 & 1 \end{pmatrix},$$
we get
$$U_a(k) = \{u_a(x, y) \mid x, y \in l, \, \mathrm{Nr}_{l/k}(x) = \mathrm{Tr}_{l/k}(y)\}.$$
Note that, with this notation, we can also write
$$U_{2a}(k) = \{u_a(0, y) \mid y \in l^0\}.$$

What has been implicitly explained so far is that, if we let
$$o_a\colon U_a(k) \to \mathbb{R} \cup \{\infty\}; \quad u_a(x, y) \mapsto \frac{1}{2}\omega(y)$$
and define $o_{2a}\colon U_{2a}(k) \to \mathbb{R} \cup \{\infty\}$ to be $2 \cdot o_a|_{U_{2a}(k)}$ (and also similarly for $-a, -2a$), then $\{o_{\pm a}, o_{\pm 2a}\}$ gives a Chevalley valuation of SU_3. Then, explicitly, the filtrations on the root subgroups are as follows:

- For $2a$, we have
$$U_{o,2a,r} = \{u_a(0, y) \in U_{2a}(k) \mid \omega(y) \geq r\}.$$
- For a, we have
$$U_{o,a,r} = \{u_a(x, y) \in U_{2a}(k) \mid \omega(y) \geq 2r\}.$$

Now let us think about the set of affine roots Ψ. For this, we first need to determine Ψ'. Recall that we have
$$\Psi' = \{\psi \in \mathcal{A}^* \mid \dot{\psi} \in \Phi, \, U_{\psi+} \subsetneq U_\psi\}.$$

When $\dot{\psi} = a$, we have $U_\psi = U_{o,a,\psi(o)}$. Similarly, when $\dot{\psi} = 2a$, we have $U_\psi = U_{o,2a,\psi(o)}$. Hence, our task is to determine the set of jumps of the filtrations $\{U_{o,a,r}\}_{r\in\mathbb{R}}$ and $\{U_{o,2a,r}\}_{r\in\mathbb{R}}$.

Here, we need to face the subtle point, which was originally planned to be explained in Week 2 but eventually skipped. In fact, the set of jumps of can be a bit tricky depending on the *ramification filtration* of $\mathrm{Gal}(l/k)$.

Recall that, in general, for a finite Galois extension k'/k and $i \in \mathbb{Z}_{\leq 0}$, the i-th (lower) ramification filtration of $\mathrm{Gal}(k'/k)$ is defined as follows:

$$\mathrm{Gal}(k'/k)_i := \{\sigma \in \mathrm{Gal}(k'/k) \mid \omega'(\sigma(\varpi') - \varpi') \geq i+1\},$$

where ϖ' is any uniformizer of k' and ω' is the normalized valuation of k'. It is known that $\mathrm{Gal}(k'/k)_i$ is trivial for sufficiently large i depending on k'/k (see [Ser79, Chapter IV] for details).

In the present situation where l/k is quadratic, the lower ramification filtraion jumps at exactly one point. We let s be the jump point, i.e., $\mathrm{Gal}(l/k)_s \neq \{1\}$ and $\mathrm{Gal}(l/k)_{s+1} = \{1\}$. Note that, $s = 0$ unless l/k is wildly ramified; in particular, since l/k is quadratic, the residual characteristic p must be necessarily 2 so that s can be positive.

Exercise 10.7. Determine s when $k = \mathbb{Q}_2$ and $l = \mathbb{Q}_2(\sqrt{2})$.

Fact 10.8 ([KP23, Lemma 2.8.1])**.** *We put $\mu := \sup\{\omega(x) \mid x \in l^0\}$.*

(1) *When l/k is unramified, $\mu = 0$ and $\omega(l^0) = \mathbb{Z}$.*
(2) *When l/k is ramified, $\mu = -\frac{s}{2}$ and $\omega(l^0) = \mathbb{Z} + \frac{1}{2} + \mu$.*

Therefore, if we describe Ψ based on the Chevalley valuation chosen as above, then it naturally involves the constant μ. So, instead, let us consider modifying the Chevalley valuation as follows. We define o'_a and o'_{2a} by

$$o'_a(u_a(x,y)) := \frac{1}{2}(\omega(y) - \mu), \quad o'_{2a}(u_{2a}(0,y)) := \omega(y) - \mu.$$

If we let $J_a := o'_a(U_a(k) \smallsetminus \{1\}) \subset \mathbb{R}$ and $J_{2a} := o'_{2a}(U_{2a}(k) \smallsetminus \{1\}) \subset \mathbb{R}$ (i.e., the set of jumps), then we have the following:

Fact 10.9 ([KP23, Corollary 3.2.4])**.**
(1) *If l/k is unramified, we have $J_a = \frac{1}{2}\mathbb{Z}$ and $J_{2a} = \mathbb{Z}$.*
(2) *If l/k is ramified, we have $J_a = \frac{1}{4}\mathbb{Z}$ and $J_{2a} = \mathbb{Z} + \frac{1}{2}$.*

In other words, we obtain

$$\Psi' = \begin{cases} \{\pm a + \frac{1}{2}\mathbb{Z}, \pm 2a + \mathbb{Z}\} & \text{if } l/k \text{ is unramified,} \\ \{\pm a + \frac{1}{4}\mathbb{Z}, \pm 2a + \mathbb{Z} + \frac{1}{2}\} & \text{if } l/k \text{ is ramified,} \end{cases}$$

where, for example, $a + \frac{1}{2} \in a + \frac{1}{2}\mathbb{Z}$ denotes the affine functional whose derivative is a and takes $\frac{1}{2}$ at o'.

Recall that $\psi \in \Psi'$ (let us suppose that $\dot{\psi} = a$) belongs to Ψ if and only if we have

$$U_{\psi+} \cdot U_{2a}(k) \subsetneq U_\psi \cdot U_{2a}(k).$$

By a further computation on p-adic fields, we can check that Ψ is given as follows:

Fact 10.10 ([KP23, Corollary 3.2.6])**.** *We have*

$$\Psi = \begin{cases} \{\pm a + \mathbb{Z}, \pm 2a + \mathbb{Z}\} & \textit{if } l/k \textit{ is unramified,} \\ \{\pm a + \frac{1}{2}\mathbb{Z}, \pm 2a + \mathbb{Z} + \frac{1}{2}\} & \textit{if } l/k \textit{ is ramified,} \end{cases}$$

Now let us go back to the situation where G is a general quasi-split connected reductive group. Recall that the construction of a Chevalley valuation associated to a weak Chevalley–Steinberg system is a bit subtle for roots of type R2 or R3. If $a \in \Phi = \Phi(G, S)$ is of type R2, i.e., $2a \in \Phi$, we choose $\tilde{a}, \tilde{a}' \in \tilde{\Phi} = \Phi(G, T)$ which are in the fiber of $\tilde{\Phi} \twoheadrightarrow \Phi$ at a and not orthogonal to each other. If we let $\tilde{b} := \tilde{a} + \tilde{a}'$, then we have $\Theta_{\tilde{a}} \subset \Theta_{\tilde{b}}$, hence obtain a quadratic extension $k_{\tilde{a}}/k_{\tilde{b}}$. Keeping the above example in mind, we modify the notion of the Chevalley valuation associated to a weak Chevalley–Steinberg system as follows:

Definition 10.11. Let o be the Chevalley valuation of G associated to a weak Chevalley–Steinberg system X of (G, S). We define $v_0 \in V(S)$ to be the element characterized by the following conditions:

- For each $a \in \Phi$ of type R1, $a(v_0) = 0$.
- For each $a \in \Phi$ of type R2, $a(v_0) = \frac{\mu_a}{2}$, where μ_a is the quantity as above coming from the ramification of the quadratic extension $k_{\tilde{a}}/k_{\tilde{b}}$.

We define the *adjusted Chevalley valuation* associated to X to be $o - v_0$.

Remark 10.12. As we can see from this definition, the adjusted one o' can be different from the original o (i.e., $v_0 \neq 0$) only when Φ is not reduced and $p = 2$.

As in the case of SU_3, we introduce the sets of jumps of filtrations.

Definition 10.13. We fix any adjusted Chevalley valuation o of G. For each $a \in \Phi$,

(1) we set J_a to be the set of jumps of $\{U_{a,o,r}\}_{r\in\mathbb{R}}$, i.e., $J_a := o_a(U_a(k) \smallsetminus \{1\})$,
(2) we set $J_{a/2a}$ to be the set of jumps of $\{U_{o,a,r}/U_{o,2a,r}\}_{r\in\mathbb{R}}$.

(When $2a \notin \Phi$, we define $J_{a/2a} = J_a$.)

For each $a \in \Phi$, we set

$$\Psi_a := \{\psi \in \Psi \mid \dot{\psi} = a\} \quad \subset \quad \Psi'_a := \{\psi \in \Psi' \mid \dot{\psi} = a\}.$$

Then, by the definitions, we have

$$\Psi'_a = a + J_a \quad \text{and} \quad \Psi_a = a + J_{a/2a}.$$

Here, for example, $a+r \in a+J_a$ denotes the affine functional on $\mathcal{A}$ whose derivative is a and value at o is given by r.

Fact 10.14. *We let e_a be the ramification index of $k_{\tilde{a}}/k$ (this is independent of the choice of $\tilde{a}$ in the fiber of a). We let $e_{a/2a}$ be the ramification index of $k_{\tilde{a}}/k_{\tilde{b}}$.*

(1) *If $a \in \Phi$ is of type R1, we have $J_a = J_{a/2a} = \frac{1}{e_a}\mathbb{Z}$.*
(2) *If $a \in \Phi$ is of type R2 and $e_{a/2a} = 1$, we have*

$$J_a = \frac{1}{2e_a}\mathbb{Z}, \quad J_{a/2a} = \frac{1}{e_a}\mathbb{Z}, \quad J_{2a} = \frac{1}{e_a}\mathbb{Z}.$$

(3) *If $a \in \Phi$ is of type R2 and $e_{a/2a} = 2$, we have*

$$J_a = \frac{1}{2e_a}\mathbb{Z}, \quad J_{a/2a} = \frac{1}{e_a}\mathbb{Z}, \quad J_{2a} = \frac{1}{e_a}(\mathbb{Z}+1).$$

10.5. **Special and extra special points.**

Proposition 10.15. *Let o be an adjusted Chevalley valuation. For any $v \in V(S)$, $x := o + v \in \mathcal{A}$ is special if and only if $a(v) \in J_a$ for all roots $a \in \Phi$ of type R1 or R2.*

Proposition 10.16. *A point of $\mathcal{A}$ is extra special if and only if it is an adjusted Chevalley valuation.*

Exercise 10.17. Find an example of a special point which is not extra special. (Hint: Consider SU_3 for a ramified quadratic extension l/k.)

11. Week 11: Construction of the Bruhat–Tits building

The aim of this week is to construct the Bruhat–Tits building in the case where G is quasi-split.

11.1. **Idea of the construction.** Recall that we have introduced the notion of a Tits system in Week 5. To be more precise, a Tits system is a tuple (G, B, N, S) consisting of a group G,subgroups $B \subset G$ and $N \subset G$, and a subset S of $N/(B \cap N)$ satisfying the following conditions

(1) The set $B \cup N$ generates G and $T := B \cap N$ is a normal subgroup of N.
(2) All elements of S are of order 2 and S generates the group $W := N/T$ (we call W the *Weyl group* of the Tits system (G, B, N, S)).
(3) For any $s \in S$ and $w \in W$, we have $sBw \subset BwB \cup BswB$.
(4) For any $s \in S$, we have $sBs \neq B$.

When a Tits system (G, B, N, S) furthermore satisfies

(5) $\bigcap_{n \in N} nBn^{-1} = B \cap N$,

we say (G, B, N, S) is *saturated*.

If a Tits sytem (G, B, N, S) is given, we can consider the notion of a parabolic subgroup of G. More precisely, for any subset $X \subset S$, we let $W_X := \langle X \rangle \subset W$ and $G_X := BW_XB$. We call G_X the standard parabolic subgroup associated to $X \subset S$. In general, we call any subgroup of G containing a standard parabolic subgroup a *parabolic subgroup* of G.

Also recall that a Tits system (G, B, N, S) gives rise to an abstract building $\mathcal{B}$ equipped with an action of G.

- We define V to be the set of all maximal proper parabolic subgroups of G.
- We define $\mathcal{B}$ to be the set of finite sets $\{P_0, \dots, P_n\}$ of maximal proper parabolic subgroups such that $\bigcap_{i=0}^{n} P_i$ is a parabolic subgroup.
- We define an action of G on the set $\mathcal{B}$ by

$$g \cdot \{P_0, \dots, P_n\} := \{gP_0g^{-1}, \dots, gP_ng^{-1}\}.$$

- Let $\mathcal{C} \in \mathcal{B}$ be the set of all standard maximal proper parabolic subgroups (the standard chamber).
- Let $\mathcal{A} \subset \mathcal{B}$ be the union of all N-conjugate of $\mathcal{C}$ (the standard apartment).

Finally, let us also recall Axiom 5.

> Let $\mathcal{A} \subset \mathcal{B}(G)$ be an apartment corresponding to a maximal k-split torus $S \subset G$. Let $\mathcal{C} \subset \mathcal{A}$ be a chamber. If we put $\mathcal{G} := G(k)^0$, $\mathcal{J} := G(k)^0_{\mathcal{C}}$, and $\mathcal{N} := N(k) \cap \mathcal{G}$, then $(\mathcal{G}, \mathcal{J}, \mathcal{N})$ forms a saturated Tits system. Moreover, the associated restricted building is equal to $\mathcal{B}(G)$ in the sense that the stabilizer group $G(k)^0_{\mathcal{F}}$ of any facet $\mathcal{F} \subset \mathcal{B}(G)$ is equal to the parabolic subgroup in the sense of Tits system.

Now we explain the idea of how to construct $\mathcal{B}(G)$.

Recall that we have already defined the apartment $\mathcal{A}(S)$ equipped with an affine root system Ψ. Since $\mathcal{A} := \mathcal{A}(S)$ has a poly-simplicial structure induced by Ψ, the notion of a chamber makes sense; let $\mathcal{C} \subset \mathcal{A}(S)$ be a chamber. Also recall that $\mathcal{A}$ is the equipollence class of Chevalley valuations, hence each point $x \in \mathcal{A}$ gives rise to a valuation of root datum $x = \{x_a \colon U_a(k) \to \mathbb{R} \cup \{\infty\}\}_{a \in \Phi}$. To each point $x \in \mathcal{A} = \mathcal{A}(S)$, an open compact subgroup $\mathcal{P}_x$ of $G(k)$ is associated by

$\mathcal{P}_x := \langle S(k)^0, U_{a,x,0} \mid a \in \Phi \rangle$ (recall that $U_{a,x,0} := x_a^{-1}([0,\infty])$). The subgroup $\mathcal{P}_x$ depends only on the facet $\mathcal{F}$ containing x, so we also write $\mathcal{P}_{\mathcal{F}}$ instead of $\mathcal{P}_x$.

We first prove that the triple ($\mathcal{G} := G(k)^0, \mathcal{I} := \mathcal{P}_{\mathcal{C}}, \mathcal{N} := N(k)$) forms a Tits system (called the *Iwahori–Tits system*). Then we define $\mathcal{B}(G)$ to be the abstract building associated to the Iwahori–Tits system $(\mathcal{G}, \mathcal{I}, \mathcal{N})$. Therefore, what we have to show in order to verify Axiom 5 are the following:

- The action of $\mathcal{G}$ on $\mathcal{B}(G)$ is extended to an action of G.
- The poly-simplicial structure on $\mathcal{A}(S)$ is equal to the one induced from the poly-simplicial structure of $\mathcal{B}(G)$.
- The subgroup $\mathcal{P}_{\mathcal{F}}$ is equal to the stabilizer group $\mathcal{G}(k)^0_{\mathcal{F}}$ for any facet $\mathcal{F}$.
- The subgroup $\mathcal{P}_{\mathcal{F}}$ is also equal to the stabilizer group $\mathcal{G}(k)^0_{\mathcal{F}}$ for any facet $\mathcal{F}$.

11.2. **Open compact subgroup associated to concave functions.** Recall that, for any point $x \in \mathcal{A}$, we have defined a subgroup $\mathcal{P}_x \subset G(k)$ by $\mathcal{P}_x := \langle S(k)^0, U_{a,x,0} \mid a \in \Phi \rangle$. In practice, it is convenient to consider more general subgroups whose index r of each $U_{a,x,r}$ is not necessarily zero. For this purpose, we introduce the notion of a concave function and a subgroup associated to it.

Let $\hat{\Phi} := \Phi \cup \{0\} \subset X^*(S)$. For convenience, we consider the totally ordered monoid

$$\tilde{\mathbb{R}} := \mathbb{R} \sqcup (\mathbb{R}+) \sqcup \{\infty\}$$

whose order and addition are defined in the natural way, where $\mathbb{R}+$ denotes the set of formal symbols $r+$ for $r \in \mathbb{R}$. (See [KP23, Section 1.6] for the details.)

Definition 11.1. A function $f\colon \hat{\Phi} \to \tilde{\mathbb{R}}$ is said to be *concave* if it satisfies

$$f(a+b) \leq f(a) + f(b)$$

for any $a, b \in \hat{\Phi}$ such that $a + b \in \hat{\Phi}$.

To define the subgroup associated to a concave function, we first need to introduce a filtration on $Z(k)^0$. Since we are assuming that G is quasi-split here, we have $Z = T$, which is a k-rational maximal torus of G. Whenever the ramification of the splitting field of T is tame, we adopt the following as the filtration on $Z(k)^0 = T(k)^0$:

$$T(k)_r := \{t \in T(k) \mid \omega(\chi(t) - 1) \geq r \text{ for all } \chi \in X^*(T)\}.$$

In fact, the definition of a filtration is subtle in general, i.e., the ramification of T is wild. We do not explain it here; please see [KP23, Section 7.2 and B].

Definition 11.2. Let $f\colon \hat{\Phi} \to \tilde{\mathbb{R}}$ be a concave function. Let $x \in \mathcal{A}$. For each $a \in \Phi$, we define the following subgroup of $G(k)$:

(1) $U_{a,x,f} := U_{a,x,f(a)} \cdot U_{2a,x,f(2a)}$ for any $a \in \Phi$ (as usual, $U_{2a,x,f(2a)} = \{1\}$ when $2a \notin \Phi$).
(2) $G(k)^{(a)}_{x,f} := \langle U_{a,x,f}, U_{-a,x,f} \rangle$.
(3) $N^{(a)}_{x,f} := G(k)^{(a)}_{x,f} \cap N(k)$.
(4) $Z^{(a)}_{x,f} := G(k)^{(a)}_{x,f} \cap Z(k)$.

Moreover, we define the following subgroup of $G(k)$:

$$G(k)^{\sharp}_{x,f} := \langle U_{a,x,f} \mid a \in \Phi \rangle,$$

$$\mathcal{P}_{x,f} := G(k)_{x,f} := \langle Z(k)_{f(0)}, G(k)^{\sharp}_{x,f} \rangle.$$

Example 11.3. Note that any constant function $f = r \colon \hat{\Phi} \to \tilde{\mathbb{R}}$ is concave as long as $r \in \tilde{\mathbb{R}}_{\geq 0}$. Especially, when $f = 0$, we have $G(k)_{x,0} = \mathcal{P}_x$. In general, for any $f = r \in \tilde{\mathbb{R}}_{\geq 0}$, the corresponding subgroup $G(k)_{x,r}$ is called the *r-th Moy–Prasad filtration subgroup* of $G(k)$ at x.

We finally introduce notation related to parahoric subgroups.

Definition 11.4. For any nonempty subset $\Omega \subset \mathcal{A}$, we define subgroups of $G(k)$ as follows.

(1) $U_{a,\Omega,0} := \bigcap_{x\in\Omega} U_{a,x,0}$ for any $a \in \Phi$.
(2) $G(k)^{\sharp}_{\Omega} := \langle U_{a,\Omega,0} \mid a \in \Phi \rangle$.
(3) $\mathcal{P}_{\Omega} := G(k)^{0}_{\Omega} := Z(k)^0 \cdot G(k)^{\sharp}_{\Omega}$.
(4) $N^{\sharp}_{\Omega} := G(k)^{\sharp}_{\Omega} \cap N(k)$.
(5) $Z^{\sharp}_{\Omega} := G(k)^{\sharp}_{\Omega} \cap Z(k)$.
(6) $N(k)^{0}_{\Omega} := N(k) \cap G(k)^{0}_{\Omega} = N^{\sharp}_{\Omega} \cdot Z(k)^0$.

11.3. **Affine Weyl group.** Recall that, when G is a connected reductive group over k, we can define the (relative) Weyl group of S in G via two different ways. The one is a concrete realization, that is, we define W to be the quotient group N/Z. The other is more abstract; we consider the $\mathbb{R}$-vector space $X^*(S)_{\mathbb{R}}$ (or $X_*(S)_{\mathbb{R}}$) and define W to be the subgroup of $\mathrm{Aut}_{\mathbb{R}}(X^*(S)_{\mathbb{R}})$ generated by the reflections with respect to the roots $a \in \Phi$. Since N acts on S so that the subgroup Z acts trivially, we have a natural map from N/Z to the latter group; this is an isomorphism.

What we discuss here is the affine version of this story. Recall that we have defined an affine root system Ψ on the apartment $\mathcal{A}$ of G, hence we also have the affine Weyl group $W^{\mathrm{aff}} := W(\Psi)$, which is a subgroup of $\mathrm{Aut}_{\mathrm{aff}}(\mathcal{A})$. On the other hand, we define "concrete" groups as follows:

Definition 11.5. (1) We put $\tilde{W}^0 := N(k)/Z(k)^0$. We call this group the *Iwahori–Weyl group*.
(2) We put $\tilde{W}^1 := N(k)/Z(k)^1$.
(3) We put $\tilde{W}_{\mathrm{sc}}$ to be the group $\tilde{W}^0$ taken in G_{sc}, i.e., $N_{\mathrm{sc}}(k)/Z_{\mathrm{sc}}(k)^0$.
(4) We put $\tilde{W}_{\mathrm{ad}}$ to be the group $\tilde{W}^1$ taken in G_{ad}, i.e., $N_{\mathrm{ad}}(k)/Z_{\mathrm{ad}}(k)^1$.

Let us consider which kind of obvious relations betwee these groups we have. Recall that $Z(k)^0$ is the kernel of the Kottwitz homomorphism while $Z(k)^1 = \{z \in Z(k) \mid \omega(\chi(z)) = 0 \text{ for all } \chi \in X^*(G)\}$. As we have $Z(k)^0 \subset Z(k)^1$, we have a natural surjection $\tilde{W}^0 \twoheadrightarrow \tilde{W}^1$. By definitions, we also have injections $\tilde{W}_{\mathrm{sc}} \hookrightarrow \tilde{W}^0$ and $\tilde{W}^1 \hookrightarrow \tilde{W}_{\mathrm{ad}}$:

$$\tilde{W}_{\mathrm{sc}} \hookrightarrow \tilde{W}^0 \twoheadrightarrow \tilde{W}^1 \hookrightarrow \tilde{W}_{\mathrm{ad}}.$$

Note that, although the middle map is not injective in general, the composition $\tilde{W}_{\mathrm{sc}} \to \tilde{W}_{\mathrm{ad}}$ is injective since it is nothing but the natural map induced by $G_{\mathrm{sc}} \to G_{\mathrm{ad}}$.

Also recall that the extended affine Weyl group $W^{\mathrm{ext}} := W(\Psi)^{\mathrm{ext}}$ is defined to be the subgroup of $\mathrm{Aut}_{\mathrm{aff}}(\mathcal{A})$ consisting of all affine transformations preserving Ψ. By Lemma 10.4, we know that the action of $N_{\mathrm{ad}}(k)$ on $\mathcal{A}^*$ preserves Ψ (although Lemma 10.4 is stated for $N(k)$, the same proof works also for $N_{\mathrm{ad}}(k)$). By going back to the definition of the action of $N_{\mathrm{ad}}(k)$ on $\mathcal{A}$, it is not difficult to see that this action is trivial on $Z_{\mathrm{ad}}(k)^1$, hence factors through $\tilde{W}_{\mathrm{ad}}$. Thus, $\tilde{W}_{\mathrm{ad}}$ can be thought of as a subgroup of W^{ext}.

Proposition 11.6 ([KP23, Proposition 6.6.2])**.** *Suppose that G is quasi-split.*

(1) *The image of $\tilde{W}_{\mathrm{sc}}$ in W^{ext} is equal to W^{aff}, hence we have a natural identification $\tilde{W}_{\mathrm{sc}} \xrightarrow{\cong} W^{\mathrm{aff}}$.*
(2) *The image of $\tilde{W}_{\mathrm{ad}}$ in W^{ext} is equal to W^{ext}, hence we have a natural identification $\tilde{W}_{\mathrm{ad}} \xrightarrow{\cong} W^{\mathrm{ext}}$.*

Exercise 11.7. Let $G = \mathrm{SL}_n$, hence $G_{\mathrm{sc}} = G$ and $G_{\mathrm{ad}} = \mathrm{PGL}_n$. Determine $\tilde{W}_{\mathrm{sc}} = \tilde{W}^0$ and its image in $\tilde{W}_{\mathrm{ad}}$.

11.4. **Iwahori–Tits system.** We first assume that G is simply-connected. Let $\mathcal{C} \subset \mathcal{A}$ be a chamber and $\mathcal{I} := G(k)^0_{\mathcal{C}}$ the corresponding Iwahori subgroup. We put $\mathcal{G} := G(k)^0$, $\mathcal{N} := N(k)^0 := N(k) \cap G(k)^0$, and $\mathcal{Z} := Z(k)^0 = S(k)^0$. We let $R \subset W^{\mathrm{aff}}$ be the subset consisting of affine reflections along hyperplanes containing walls (codimension 1 facets) of $\mathcal{C}$.

Theorem 11.8. *The tuple $(\mathcal{G}, \mathcal{I}, \mathcal{N}, R)$ forms a saturated Tits system.*

Let $\Psi(\mathcal{C})$ be the set of affine roots whose hyperplanes contain walls of $\mathcal{C}$; then $\Psi(\mathcal{C})$ forms a basis of the affine root system Ψ (see [KP23, Section 1.3] for details). For any facet $\mathcal{F}$ contained in the closure of $\mathcal{C}$, we define $W^{\mathrm{aff}}_{\mathcal{F}} \subset W^{\mathrm{aff}}$ to be the subgroup generated by relections along affine roots in $\Psi(\mathcal{C})$ which vanishes on $\mathcal{F}$.

Lemma 11.9 ([KP23, Lemma 7.4.7])**.** *We have $W^{\mathrm{aff}}_{\mathcal{F}} \subset G(k)^0_{\mathcal{F}}$, where any element of the left hand-side is regarded as an element of $N(k)$ by fixing a representative of the image in $\tilde{W}^0$.*

Proposition 11.10. *For any facet $\mathcal{F}$ contained in the closure of $\mathcal{C}$,*

(1) *$W^{\mathrm{aff}}_{\mathcal{F}}$ is equal to the stabilizer of $\mathcal{F}$ with respect to the action of W^{aff} on $\mathcal{A}$,*
(2) *we have $G(k)^0_{\mathcal{F}} = \mathcal{I} \cdot W^{\mathrm{aff}}_{\mathcal{F}} \cdot \mathcal{I}$.*

Proof. The first assertion (1) is purely about affine root system, so we skip it here (see [KP23, Lemma 1.3.17], where Bourbaki is cited).

We consider the second assertion (2). We put $X_{\mathcal{F}} \subset R$ to be the subset consisting of reflections stabilizing $\mathcal{F}$. Then the right-hand side of the equality $\mathcal{I} \cdot W^{\mathrm{aff}}_{\mathcal{F}} \cdot \mathcal{I}$ is the standard parabolic subgroup of $\mathcal{G}$ corresponding to $X_{\mathcal{F}}$ in the sense of an abstract Tits system. Thus, in particular, note that it is a group.

An easy, but important, observation here is that we have

$$U_{a,\Omega,0} = \bigcup_{\substack{\psi \in \Psi \\ \dot{\psi} = a, \psi(\Omega) \geq 0}} U_a(k)_\psi.$$

By noting this, we see that $G(k)^0_{\mathcal{C}} \subset G(k)^0_{\mathcal{F}}$ since $\psi(\mathcal{C}) \geq 0$ implies $\psi(\mathcal{F}) \geq 0$ for any affine root $\psi \in \Psi$. Hence, by combining this with the previous lemma, we get $G(k)^0_{\mathcal{F}} \supset \mathcal{I} \cdot W^{\mathrm{aff}}_{\mathcal{F}} \cdot \mathcal{I}$.

We next consider the opposite inclusion. For each $a \in \Phi$, we let $\psi^{\mathcal{F}}_{a,0}$ be the unique affine root whose derivative is a and which is nonnegative and the smallest on $\mathcal{F}$. We note that, again by the "easy" observation above, the group $G(k)^0_{\mathcal{F}}$ is generated by $Z(k)^0$ and U_ψ for affine roots $\psi \in \Psi$ of the form $\psi^{\mathcal{F}}_{a,0}$. Thus it is enough to check that $U_{\psi^{\mathcal{F}}_{a,0}}$ is contained in the right-hand side for each $a \in \Phi$.

We first suppose that $\psi_{a,0}^{\mathcal{F}}$ is strictly positive on $\mathcal{F}$. Note that, for any affine root $\psi \in \Psi$, we have exactly one of $\psi(\mathcal{C}) > 0$ or $\psi(\mathcal{C}) < 0$ since the value of ψ varies on $\mathcal{C}$ continuously and $\mathcal{C}$ does not meet any hyperplane. Hence the assumption $\psi_{a,0}^{\mathcal{F}}(\mathcal{F}) > 0$ implies that $\psi_{a,0}^{\mathcal{F}}(\mathcal{C}) > 0$. Thus we get $U_{\psi_{a,0}^{\mathcal{F}}} \subset G(k)_{\mathcal{C}}^{0}$.

We next suppose that $\psi_{a,0}^{\mathcal{F}}$ is not strictly positive on $\mathcal{F}$. However, in this case, since $\psi_{a,0}^{\mathcal{F}}$ is non-negative on $\mathcal{F}$, we necessarily have $\psi_{a,0}^{\mathcal{F}}(\mathcal{F}) = 0$ (if $\psi_{a,0}^{\mathcal{F}}$ attains 0 at least one point in $\mathcal{F}$, then it must be 0 at all points in $\mathcal{F}$). Accordingly, we have two possibilities; $\psi_{a,0}^{\mathcal{F}}(\mathcal{C}) > 0$ or $\psi_{a,0}^{\mathcal{F}}(\mathcal{C}) < 0$. If $\psi_{a,0}^{\mathcal{F}}(\mathcal{C}) > 0$, then $U_{\psi_{a,0}^{\mathcal{F}}} \subset G(k)_{\mathcal{C}}^{0}$. If $\psi_{a,0}^{\mathcal{F}}(\mathcal{C}) < 0$, then we have $U_{-\psi_{a,0}^{\mathcal{F}}} \subset G(k)_{\mathcal{C}}^{0}$. Since the reflection r along the hyperplane determined by $\psi_{a,0}^{\mathcal{F}}$ is contained in $W_{\mathcal{F}}^{\mathrm{aff}}$, we have

$$U_{-\psi_{a,0}^{\mathcal{F}}} = r \cdot U_{\psi_{a,0}^{\mathcal{F}}} \cdot r^{-1} \subset \mathcal{I} \cdot W_{\mathcal{F}}^{\mathrm{aff}} \cdot \mathcal{I}.$$

(Here we used Lemma 10.3.) □

11.5. **Bruhat–Tits building.**

Definition 11.11. We let $\mathcal{B}(G)$ be the bulding associated to the Iwahori–Tits system of G. We call $\mathcal{B}(G)$ the *Bruhat–Tits bulding* of G.

Note that an abstract building constructed via a Tits system (G, B, N, S) is equipped with an action of G (here G is taken to be $\mathcal{G} = G(k)^0$; sorry for this conflict of notation!). Recall that we have defined the subgroup $\mathcal{P}_\Omega$ by hand. Let us check that this is indeed the same as the stabilizer group of Ω with respect to the action of $\mathcal{G} = G(k)^0$ on $\mathcal{B} := \mathcal{B}(G)$.

Proposition 11.12. *Let $\Omega \subset \mathcal{A}$ be a nonempty subset contained in a facet $\mathcal{F} \subset \mathcal{A}$. Then the group $G(k)_{\Omega}^{0}$ is equal to the stabilizer of Ω with respect to the action of $G(K)^0$ on $\mathcal{B}$, which is furthermore equal to the pointwise stabilizer.*

Proof. For convenience, in the following, we add a subscript "$\bullet$" and "$\dagger$" for denoting the pointwise stabilizer and the stabilizer, respectively. Thus our aim is to show that $G(k)_{\mathcal{F},\bullet}^{0} = G(k)_{\mathcal{F},\dagger}^{0} = G(k)_{\Omega}^{0}$. We have obvious chains $G(k)_{\Omega,\bullet}^{0} \subset G(k)_{\Omega,\dagger}^{0}$, $G(k)_{\mathcal{F},\bullet}^{0} \subset G(k)_{\mathcal{F},\dagger}^{0}$, and $G(k)_{\mathcal{F},\bullet}^{0} \subset G(k)_{\Omega,\bullet}^{0}$ and On the other hand, by construction, we easily see that the action of $G(K)^0$ on $\mathcal{B}(G)$ preserves the poly-simplicial structure of $\mathcal{B}(G)$. Thus, any element of $G(k)^0$ fixes Ω (as a subset) if and only if it fixes $\mathcal{F}$ (as a subset), i.e., $G(k)_{\Omega,\dagger}^{0} = G(k)_{\mathcal{F},\dagger}^{0}$. Furthermore, we also have $G(k)_{\Omega}^{0} = G(k)_{\mathcal{F}}^{0}$. By putting all of these into together, it suffices to show only $G(k)_{\mathcal{F},\bullet}^{0} = G(k)_{\mathcal{F},\dagger}^{0} = G(k)_{\mathcal{F}}^{0}$. However, this statement is something general, which is true for any abstract building associated to a saturated Tits system (recall that we have verified that $G(k)_{\mathcal{F}}^{0}$ is nothing but the standard parabolic subgroup in the sense of the Iwahori–Tits system). □

12. Week 12: Construction of Bruhat–Tits building (continued)

12.1. **Product decomposition.** We state a key proposition on the subgroups associated to concave functions. We fix a set $\Phi^+ \subset \Phi$ of positive relative roots and put $\Phi^- := -\Phi^+$. Let $\Phi^{+,\mathrm{nd}}$ and $\Phi^{-,\mathrm{nd}}$ denote the sets of non-divisible roots in Φ^+ and Φ^- respectively. Let U^+ and U^- be the unipotent radical of the Borel subgroups of G corresponding to Φ^+ and Φ^- respectively.

For a concave function $f\colon \hat{\Phi} \to \tilde{\mathbb{R}}$ and any point $x \in \mathcal{A}$, we define the following subgroups of $G(k)$:

- $U^+_{x,f} := U^+ \cap G(k)^\sharp_{x,f}$, $U^-_{x,f} := U^- \cap G(k)^\sharp_{x,f}$.
- $N^\sharp_{x,f} := N(k) \cap G(k)^\sharp_{x,f}$.
- $Z^\sharp_{x,f} := Z(k) \cap G(k)^\sharp_{x,f}$.

Lemma 12.1. *With the notation as above, we have the following:*

(1) *Let $a, b \in \Phi$ be non-proportional roots. Then we have*
$$[U_{a,x,f}, U_{b,x,f}] \subset \langle U_{pa+qb,x,f} \mid p, q > 0,\, pa + qb \in \Phi\rangle.$$
(2) *For any subset Φ'^+ of Φ^+ which is closed under addition (i.e., if $a, b \in \Phi'^+$ satisfies $a + b \in \Phi$, then $a + b \in \Phi'^+$), the product*
$$\prod_{a\in\Phi'^+} U_{a,x,f}$$
is a subgroup of $G(k)$ independent of the order of the product.

(3) *For any $a \in \Phi$, we have $G(k)^{(a)}_{x,f} = U_{-a,x,f} \cdot U_{a,x,f} \cdot N^{(a)}_{x,f}$.*

(4) *If $f(a) + f(-a) > 0$ and $f(2a) + f(-2a) > 0$ (when $2a \in \Phi$), then we have $N^{(a)}_{x,f} = Z^{(a)}_{x,f}$.*

This lemma is technically very important, but the proof is basically just a combination of properties of a valuation of root datum and a concave function. So we do not explain it here; see [KP23, Lemma 7.3.11].

Proposition 12.2. *With the notation as above, we have the following:*

(1) *We have $G(k)^\sharp_{x,f} = U^+_{x,f} \cdot U^-_{x,f} \cdot N^\sharp_{x,f}$.*

(2) *The group $N^\sharp_{x,f}$ is generated by $N^{(a)}_{x,f}$ for $a \in \Phi$.*

(3) *The product map*
$$\prod_{a\in\Phi^{\pm,\mathrm{nd}}} U_{a,x,f} \to U^\pm_{x,f}$$
is bijective for any choice of an order on $\Phi^{\pm,\mathrm{nd}}$.

(4) *If $f(0) > 0$, then $N^\sharp_{x,f} = Z^\sharp_{x,f}$.*

(5) *If $N^\sharp_{x,f} = Z^\sharp_{x,f}$, the product map*
$$\prod_{a\in\Phi^{+,\mathrm{nd}}} U_{a,x,f} \times Z^\sharp_{x,f} \times \prod_{a\in\Phi^{-,\mathrm{nd}}} U_{a,x,f} \to G(k)^\sharp_{x,f}$$
is bijective for any choice of an order on $\Phi^{\pm,\mathrm{nd}}$.

Proof. For $a \in \Phi^{\mathrm{nd}}$, we simply write $X_a := U_{a,x,f}$. Similarly, we write X^+ and X^- for the product of X_a for $\Phi^{+,\mathrm{nd}}$ and $\Phi^{-,\mathrm{nd}}$, respectively. Note that $X^\pm$ is a subgroup of $G(k)$ independent of the choice of an order of the product by Lemma 12.1 (2). We furthermore write

$$N' := \langle N_{x,f}^{(a)} \mid a \in \Phi \rangle.$$

We first prove that the set $X^- \cdot X^+ \cdot N'$ is independent of the choice of Φ^+. For this, note that any two choices of a set of positive roots are mapped to each other by the action of the Weyl group. Since any two different Weyl chambers are transfereed to each other by a combination of reflections along a wall, it is enough to check that $X^- \cdot X^+ \cdot N'$ does not change even if we replace Φ^+ with $s_b(\Phi^+)$ for any simple root $b \in \Phi^+$. Let us write $X^{\pm b}$ for the product of X_a for $a \in \Phi^{\pm,\mathrm{nd}} \smallsetminus \{\pm b\}$. By noting that $s_b(\Phi^+) = (\Phi^+ \smallsetminus \{b\}) \sqcup \{-b\}$, our task is to check that

$$X^- \cdot X^+ \cdot N' = (X^{-b} \cdot X_b) \cdot (X_{-b} \cdot X^b) \cdot N'.$$

By Lemma 12.1 (3), we have $X_{-b} \cdot X_b \cdot N_{x,f}^{(b)} = G(k)_{x,f}^{(b)} = X_b \cdot X_{-b} \cdot N_{x,f}^{(b)}$. In particular, we have $X_{-b} \cdot X_b \cdot N' = X_b \cdot X_{-b} \cdot N'$. As X^b and X^{-b} are normalized by X_b and X_{-b} by Lemma 12.1 (1), we have

$$\begin{aligned} X^- \cdot X^+ \cdot N' &= (X^{-b} \cdot X_{-b}) \cdot (X_b \cdot X^b) \cdot N' \\ &= X^{-b} \cdot X^b \cdot (X_{-b} \cdot X_b \cdot N') \\ &= X^{-b} \cdot X^b \cdot (X_b \cdot X_{-b} \cdot N') = (X^{-b} \cdot X_b) \cdot (X_{-b} \cdot X^b) \cdot N'. \end{aligned}$$

Let us show (1) and (2). The claim we have just proved especially implies that $X^- \cdot X^+ \cdot N'$ is normalized by N'. By this and also using that $X^- \cdot X^+ \cdot N' = X^+ \cdot X^- \cdot N'$, we can easily check that $X^- \cdot X^+ \cdot N'$ is a subgroup of $G(k)$. Furthermore, as any $U_{a,x,f} = X_a$ is contained in $X^- \cdot X^+ \cdot N'$, this subgroup is nothing but $G(k)_{x,f}^{\sharp}$. To show (1) and (2), it suffices to show that $X^+ = U_{x,f}^+$, $X^- = U_{x,f}^-$, and $N_{x,f}^{\sharp} = N'$. By definition of $U_{x,f}^-$, we have

$$U_{x,f}^- = U^- \cap G(k)_{x,f}^{\sharp} = U^- \cap X^- \cdot X^+ \cdot N'.$$

Suppose that $x^- x^+ n' \in X^- \cdot X^+ \cdot N' = G(k)_{x,f}^{\sharp}$. If $x^- x^+ n' \in U^-$, then $x^+ n' \in U^-$. By the Bruhat decomposition $G = \bigsqcup_{w \in W} BwU^-$, this implies that $x^+ = 1$ and $n' = 1$. In summary, we have obtained that $U^- \cap X^- \cdot X^+ \cdot N' \subset X^-$. The converse inclusion is clear, hence we get $X^- = U_{x,f}^-$. By similar arguments, we can also show $X^+ = U_{x,f}^+$ and $N_{x,f}^{\sharp} = N'$.

Now, the statement (3) is just a consequence of Lemma 12.1 (2). Also, since we can swap X^- with $N' = N_{x,f}^{\sharp}$ in the product expression $X^+ X^- N' = G(k)_{x,f}^{\sharp}$ provided that $N_{x,f}^{\sharp} = Z_{x,f}^{\sharp}$, the surjectivity part of the statement (5) follows. The injectivity part is something general, which holds at the level of algebraic groups after replacing, e.g., $U_{a,x,f}$ with $U_a(k)$ and so on (see [KP23, Proposition 2.11.3]).

Finally, the statement (4) follows from Lemma 12.1 (4) by noting that $N_{x,f}^{\sharp} = N'$ and that $f(a) + f(-a) \geq f(a - a) = f(0) > 0$ (and also $f(2a) + f(-2a) \geq f(2a - 2a) = f(0) > 0$). $\square$

This proposition have the following important consequence.

Corollary 12.3. *Suppose that a concave function $f \colon \hat{\Phi} \to \tilde{\mathbb{R}}$ does not take the value ∞. Then the group $G(k)_{x,f}$ is open compact.*

Proof. We recall the following general fact: the product morphism

$$\prod_{a\in\Phi^{+,\mathrm{nd}}} U_a \times Z \times \prod_{a\in\Phi^{-,\mathrm{nd}}} U_a \to G$$

is an open immersion (see [KP23, Corollary 8.2.6]). In particular, this implies that the image of the morphism induced on k-points

$$\prod_{a\in\Phi^{+,\mathrm{nd}}} U_a(k) \times Z(k) \times \prod_{a\in\Phi^{-,\mathrm{nd}}} U_a(k) \to G(k)$$

is p-adically open. Since the filtration of each component on the left-hand side forms a fundamental system of open neighborhoods, the image of

$$\prod_{a\in\Phi^{+,\mathrm{nd}}} U_{a,x,f} \times Z(k)_{f(0)} \times \prod_{a\in\Phi^{-,\mathrm{nd}}} U_{a,x,f} \to G(k)$$

is open in $G(k)$. Hence, at least we see that $G(k)_{x,f}$ is open since we have $G(k)_{x,f} = \langle Z(k)_{f(0)}, G(k)^\sharp_{x,f}\rangle = \langle Z(k)_{f(0)}, U_{a,x,f} \mid a\in\Phi^{\mathrm{nd}}\rangle$ by definition.

To show the compactness of $G(k)_{x,f}$, it suffices to show the compactness of $G(k)^\sharp_{x,f}$ since $Z(k)_{f(0)}$ is compact. For this, we use Proposition 12.2 (1), that is, $G(k)^\sharp_{x,f} = U^+_{x,f}\cdot U^-_{x,f}\cdot N^\sharp_{x,f}$. As $U^+_{x,f}$ and $U^-_{x,f}$ are compact by Proposition 12.2 (3), our task is to check that $N^\sharp_{x,f}$ is compact.

Here we appeal to several elementary facts on concave functions.

(1) If f_1 and f_2 are concave functions, then so is f_1+f_2.
(2) For any $v\in V$, the function $\hat\Phi\to\mathbb{R}\colon a\mapsto a(v)$ is concave (linear).
(3) Hence, in particular, $f+v$ is concave. Thus concave function satisfies $G(k)^\sharp_{x,f} = G(k)^\sharp_{x+v,f+v}$.
(4) For any concave function $f\colon\hat\Phi\to\tilde{\mathbb{R}}$ which does not take ∞, we can find $v\in V$ such that $f+v$ is non-negative.

From these, we may assume that our concacve function f is non-negative from the beginning. Then, by the following Lemma 12.5, the action of $N^\sharp_{x,f}$ on $\mathcal{A}$ fixes x. In general, the kernel of the derivative map for the affine Weyl group $W(\Psi)\twoheadrightarrow W(\Phi)$ consists of translations contained in $W(\Psi)$. Therefore, we see that the image of $N^\sharp_{x,f}$ in the extended Weyl group does not contain the translation part. Consequently, this shows that the quotiet of $N^\sharp_{x,f}$ by $Z(k)_{f(0)}$ is finite, hence $N^\sharp_{x,f}$ is compact. □

Exercise 12.4. Prove the "elementary facts" (1)–(2) on concave functions used in the above proof.

The idea of the proof of the following lemma, which is used in the above proof, is to reduce the statement to case of SL_2 or SU_3 and deal with these particular cases by a direct computation. See [KP23, Corollary 7.3.13].

Lemma 12.5. *If f is non-negative, then the action of $N^\sharp_{x,f}$ on $\mathcal{A}$ fixes the point x.*

This proposition is particularly applied to an Iwahori subgroup. If we let $\mathcal{C}$ be a chamber of $\mathcal{A}$ and $\mathcal{I} := G(k)^0_{\mathcal{C}}$, then we may regard $\mathcal{I}$ as a subgroup associated to a concave function. More precisely, by choosing any point $x\in\mathcal{C}$ and $f=0$, we have $\mathcal{I} = G(k)_{x,0}$. Again by the above lemma, the action of $N^\sharp_{x,0}$ on $\mathcal{A}$ fixes the point x. This implies that $N^\sharp_{x,0}$ in the extended affine Weyl group is trivial, i.e.,

$N^{\sharp}_{x,0} = Z^{\sharp}_{x,0}$. (This is obvious by choosing x to be any "generic" point in $\mathcal{C}$.) Hence, especially, the product map

$$\prod_{a\in\Phi^{+,\mathrm{nd}}} U_{a,x,0} \times Z^{\sharp}_{x,0} \times \prod_{a\in\Phi^{-,\mathrm{nd}}} U_{a,x,0} \to G(k)^{\sharp}_{x,0}$$

is bijective for any choice of an order on $\Phi^{\pm,\mathrm{nd}}$.

Exercise 12.6. The group $G(k)^{\sharp}_{x,f}$ is not necessarily equal to $G(k)^{0}_{x,f}$. In other words, $G(k)^{\sharp}_{x,f}$ may not contain $Z(k)^0$. Find such an example (e.g., in the case of SL_2).

12.2. **Proof of Theorem 11.8.** Now let us complete the proof of Theorem 11.8, that is, show that the tuple $(\mathcal{G}, \mathcal{I}, \mathcal{N}, R)$ forms a saturated Tits system. Recall that the conditions we have to check are as follows:

(1) The set $\mathcal{I} \cup \mathcal{N}$ generates $\mathcal{G}$ and $\mathcal{Z} := \mathcal{I} \cap \mathcal{N}$ is a normal subgroup of $\mathcal{N}$.
(2) All elements of R are of order 2 and R generates the group $\mathcal{N}/\mathcal{Z}$.
(3) For any $s \in R$ and $w \in \mathcal{N}/\mathcal{Z}$, we have $s\mathcal{I}w \subset \mathcal{I}w\mathcal{I} \cup \mathcal{I}sw\mathcal{I}$.
(4) For any $s \in R$, we have $s\mathcal{I}s \neq \mathcal{I}$.
(5) $\bigcap_{n\in\mathcal{N}} n\mathcal{I}n^{-1} = \mathcal{Z}$.

We first assume that G is simply-connected (quasi-split). Note that, under this assumption, we have

- $\mathcal{G} := G(k)^0 = G(k)$,
- $\mathcal{N} := N(k) \cap G(k)^0 = G(k) = N(k)$,
- $\mathcal{Z} = \mathcal{I} \cap \mathcal{N} = Z(k)^0$,
- $\mathcal{N}/\mathcal{Z} = N(k)/Z(k)^0 = \tilde{W}^0 \cong W^{\mathrm{aff}}$.

Proof of Theorem 11.8: simply-connected case. Let us first check the condition (1) of a saturated Tits system. We first note that $\mathcal{Z} = \mathcal{I} \cap \mathcal{N}$, hence this is normal in $\mathcal{N}$. In order to check that $\mathcal{G}$ is generated by $\mathcal{I} \cup \mathcal{N}$, it suffices to show that $\mathcal{I} \cup \mathcal{N}$ contains $U_a(k)$ for all $a \in \Phi$ (recall that $G(k)$ is generated by $U_a(k)$'s). Recall that $\mathcal{I} = G(k)^0_{\mathcal{C}} = \langle \mathcal{Z}, U_{a,x,0} \mid a \in \Phi\rangle$ for any $x \in \mathcal{C}$. By definition, we have $U_{a,x,0} = x_a^{-1}([0,\infty])$. By the definition of a valuation of root datum, for any $t \in S(k) \subset \mathcal{N}$, we have $x_a(t \cdot u \cdot t^{-1}) = x_a(u) - a(\nu(t)) = x_a(u) + \omega(a(t))$. This implies that

$$t \cdot U_{a,x,0} \cdot t^{-1} = U_{a,x,\omega(a(t))}.$$

Since the filtration $\{U_{a,x,r}\}_{r\in\mathbb{R}}$ gives a fundamental system of open neighborhoods of $1 \in U_a(k)$ and $a(-)\colon S(k) \to k^{\times}$ is surjective, we get

$$\bigcup_{t\in S(k)} t \cdot U_{a,x,0} \cdot t^{-1} = U_a(k).$$

The condition (2) is a general property of the affine root system for an affine root system. More precisely, under our assumption that G is simply-connected, $\mathcal{N}/\mathcal{Z}$ is idenfied with the affine Weyl group W^{aff} of the affine root system Ψ as mentioned above. As R is the set of reflections of simple affine roots of Ψ, all elements of R are of order 2 and generate W^{aff}.

We next check the condition (4), i.e., for any $s \in R$, we have $s\mathcal{I}s \neq \mathcal{I}$. For $s \in R$, we let $\alpha \in \Psi$ be the corresponding affine root, that is, the unique affine root such that s is the reflection along the hyperplane defined by $\alpha = 0$ and also such that $\alpha(\mathcal{C}) > 0$ ($\alpha(x) > 0$ for any $x \in \mathcal{C}$). Then, by letting $a := \dot{\alpha}$, we see that $U_\alpha = U_{a,x,\alpha(x)} \subset U_{a,x,0} \subset \mathcal{I}$. Recall that the product map

$$\prod_{a \in \Phi^{+,\mathrm{nd}}} U_{a,x,0} \times Z(k)^0 \times \prod_{a \in \Phi^{-,\mathrm{nd}}} U_{a,x,0+} \to \mathcal{I}$$

is bijective, where the set of positive roots Φ^+ is the one determined by the choice of $\mathcal{C}$. If we apply the s-conjugation to this bijective map, we also see that

$$\prod_{a \in \Phi^{+,\mathrm{nd}}} sU_{a,x,0}s \times Z(k)^0 \times \prod_{a \in \Phi^{-,\mathrm{nd}}} sU_{a,x,0+}s \to s\mathcal{I}s$$

is bijective. Since each $sU_{-a,x,0(+)}s$ is a subgroup of $U_{s(-a)} = U_a$, by taking the intersection with $U_\alpha \subset U_a(k)$, we obtain

$$U_\alpha \cap s\mathcal{I}s = U_\alpha \cap (sU_{-a,x,0(+)}s) = U_{\alpha+}.$$

As this is not equal to $U_\alpha \cap \mathcal{I} = U_\alpha$, we obtain $s\mathcal{I}s \neq \mathcal{I}$.

Let us consider the condition (3). We fix $s \in R$ and $\alpha \in \Psi$ as above. We start with proving the inclusion $s\mathcal{I} \subset \mathcal{I}sU_\alpha$. Let Ψ' be the set of smallest indivisible affine roots which are positive on $\mathcal{C}$. Thus, in particular, α is an element of Ψ'. Let $-\alpha + k$ be the element of Ψ' which has the same derivative as $-\alpha$ and is greater than $-\alpha$. By the product decomposition of the Iwahori subgroup $\mathcal{I}$, we have

$$\mathcal{I} = \prod_{\substack{\beta \in \Psi' \smallsetminus \{\alpha, -\alpha+k\} \\ \dot{\beta} \in \Phi^{-,\mathrm{nd}}}} U_\beta \cdot U_{-\alpha+k} \cdot \mathcal{Z} \cdot \prod_{\substack{\beta \in \Psi' \smallsetminus \{\alpha, -\alpha+k\} \\ \dot{\beta} \in \Phi^{+,\mathrm{nd}}}} U_\beta \cdot U_\alpha.$$

It is not difficult to see that the action of s on Ψ preserves the set $\Psi' \smallsetminus \{\alpha, -\alpha + k\}$ and maps α and $-\alpha + k$ to $-\alpha$ and $\alpha + k$, respectively. Hence, we get

$$\begin{aligned} s\mathcal{I}s &= \prod_{\substack{\beta \in \Psi' \smallsetminus \{\alpha, -\alpha+k\} \\ \dot{\beta} \in \Phi^{-,\mathrm{nd}}}} U_\beta \cdot (sU_{-\alpha+k}s) \cdot \mathcal{Z} \cdot \prod_{\substack{\beta \in \Psi' \smallsetminus \{\alpha, -\alpha+k\} \\ \dot{\beta} \in \Phi^{+,\mathrm{nd}}}} U_\beta \cdot (sU_\alpha s) \\ &\subset \mathcal{I} \cdot (sU_\alpha s), \end{aligned}$$

which is equivalent to $s\mathcal{I} \subset \mathcal{I}sU_\alpha$.

Now let us take any $w \in W^{\mathrm{aff}}$, thus our task is to show the inclusion $s\mathcal{I}w \subset \mathcal{I}w\mathcal{I} \cup \mathcal{I}sw\mathcal{I}$. For this, we note that

$$X := (U_\alpha s \mathcal{Z} U_\alpha) \sqcup (U_{s(\alpha)+} \mathcal{Z} U_\alpha)$$

is a subgroup of $G(k)$ (see [KP23, Lemma 7.1.3]). From now on, we will prove the inclusion $Xw\mathcal{I} \subset \mathcal{I}sw\mathcal{I} \cup \mathcal{I}w\mathcal{I}$. Note that, by the inclusion proved in the previous paragraph, this implies the inclusion as desired. Indeed, we have

$$s\mathcal{I}w \subset (\mathcal{I}sU_\alpha)w \subset \mathcal{I}Xw \subset \mathcal{I}w\mathcal{I} \cup \mathcal{I}sw\mathcal{I}.$$

Since s belongs to X, we have $Xw\mathcal{I} = X(sw)\mathcal{I}$. This means that we may freely replace w with sw for our purpose. Here we note we have either $w^{-1}(\alpha)(\mathcal{C}) > 0$ or $w^{-1}(\alpha)(\mathcal{C}) < 0$; hence we may suppose to have the former case by replacing w with swif necessary. Then we have $w^{-1}U_\alpha w = U_{w^{-1}(\alpha)} \subset \mathcal{I}$, which is equivalent to $U_\alpha \subset w\mathcal{I}w^{-1}$. Now we can verify the desired inclusion as follows:

- $(U_\alpha s \mathcal{Z} U_\alpha)w\mathcal{I} \subset (U_\alpha s \mathcal{Z} w\mathcal{I}w^{-1})w\mathcal{I} = U_\alpha s \mathcal{Z} w\mathcal{I} \subset \mathcal{I}sw\mathcal{I} \cup \mathcal{I}w\mathcal{I}$.
- $(U_{s(\alpha)+} \mathcal{Z} U_\alpha)w\mathcal{I} \subset (U_{s(\alpha)+} \mathcal{Z} w\mathcal{I}w^{-1})w\mathcal{I} = U_{s(\alpha)+} \mathcal{Z} w\mathcal{I} \subset \mathcal{I}sw\mathcal{I} \cup \mathcal{I}w\mathcal{I}$.

This completes the proof. $\square$

Now we consider the case where G is a general quasi-split connected reductive group. Recall from Week 3 that a subgroup $G(k)^\natural \subset G(k)$ is defined to be the image of the natural homomorphism $G_{\mathrm{sc}}(k) \to G(k)$. Using this subgroup, we define a tuple $(\mathcal{G}^\natural, \mathcal{I}^\natural, \mathcal{N}^\natural, R)$ as follows:

- $\mathcal{G}^\natural := G(k)^\natural$,
- $\mathcal{N}^\natural := N(k) \cap \mathcal{G}^\natural$,
- $\mathcal{I}^\natural := \mathcal{I}^\sharp \cdot \mathcal{Z}^\natural$, where $\mathcal{Z}^\natural$ is the image of $Z_{\mathrm{sc}}(k)^0$ in $G(k)^\natural$.

Then, by using that the Iwahori–Tits system for G_{sc} is a Tits system, which has been already proved, we can deduce that $(\mathcal{G}^\natural, \mathcal{I}^\natural, \mathcal{N}^\natural, R)$ is also a Tits system. Furthermore, using that

$$(\mathcal{G}, \mathcal{I}, \mathcal{N}, R) = (\mathcal{G}^\natural \cdot \mathcal{Z}, \mathcal{I}^\natural \cdot \mathcal{Z}, \mathcal{N}^\natural \cdot \mathcal{Z}, R)$$

we can furthermore deduce that $(\mathcal{G}, \mathcal{I}, \mathcal{N}, R)$ is a Tits system.

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